

MMÜ 497 - DESIGN PROJECT II

ELECTRIC BICYCLE CONVERSION KIT

FINAL REPORT



Submitted by

GROUP 21

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Abstract

This report explains in detail a modular and portable electric bicycle conversion kit. Our goal is to develop an efficient kit that can electrify existing bicycles. Our kit is designed with a focus on reliability and efficiency and it has features like speed control and regenerative braking.

In the report, each part has been looked at separately. For example, alternative motors, control systems, battery technologies, designs, and simulations that ensure design reliability have been detailed. Also, our designs prioritize expandability, and open-source electrical circuits are used. Additional equipment is also included in the report. This detailed preliminary design report will be a guide for our project.

The project's functional requirements include performance criteria such as maximum speed, carrying capacity and range. Throughout the design process, realistic constraints like environmental sustainability, economic feasibility, and user safety have been considered. The design aims for full compliance with national and international engineering standards such as TS EN 15194+A1 and TS ISO 20653

The mechanical and thermal integrity of the design was verified using ANSYS static and transient analyses. Driving dynamics were simulated using a digital twin model developed using Python and Matlab. The results show that the system theoretically has a maximum speed of 43.54 km/h and a net energy consumption of 4.55 Wh/km (on a flat city road), including regenerative gains. Furthermore, a cost analysis calculated the total cost of the project at 14,955 TL. This is within safe limits compared to the initial production and proves that it meets the targeted modularity requirements.

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List of Symbols and Abbreviations

ADC	Analog to Digital Converter
BLDC	Brushless DC (Direct Current)
CAN	Controller Area Network
FOC	Field-Oriented Control
I ² C	Inter-Integrated Circuit
IMU	Inertial Measurement Unit
MPU	Motion Processing Unit
PWM	Pulse Width Modulation
UART	Universal Asynchronous Receiver Transmitter
VESC	Vedder Electronic Speed Controller [Benjamin Vedder (YODA)]
xESC	Extended Electronic Speed Controller
RC Filter	Resistor-Capacitor Filter
DC	Direct Current
AC	Alternating Current
IDE	Integrated Development Environment
ESP32	Espressif (Wi-Fi & Bluetooth)
STM32	STMicroelectronics (ARM)
SDG	Sustainable Development Goals
EPAC	Electrically Assisted Pedal Cycle
IP	Ingress Protection
ABS	Acrylonitrile Butadiene Styrene
ADC	Analog to Digital Converter
BMS	Battery Management System
CAD	Computer-Aided Design
CAN	Controller Area Network
CFD	Computational Fluid Dynamics

DoF	Degrees of Freedom
FOC	Field-Oriented Control
GPIO	General Purpose Input/Output
HIL	Hardware-in-the-Loop
MIL	Model-in-the-Loop
LiFePO4	Lithium Iron Phosphate
PCB	Printed Circuit Board
PWM	Pulse Width Modulation
RPM	Revolutions Per Minute
SDG	Sustainable Development Goals
SIL	Software-in-the-Loop
SoC	State of Charge
SoH	State of Health
UART	Universal Asynchronous Receiver Transmitter

Symbol	Definition	Unit
A_f	Frontal Area	m ²
C_d	Aerodynamic Drag Coefficient	-
C_r	Rolling Resistance Coefficient	-
F_{drag}	Aerodynamic Drag Force	N
F_{roll}	Rolling Resistance Force	N
g	Gravitational Acceleration	m/s ²
K_{bf}	Brake Force Distribution Factor	-
L_{10}	Basic Rating Life (Bearing)	Revolutions
m	Mass	kg
P_{req}	Required Power	W
T_{load}	Total Resistive Torque	Nm
v_g	Ground Speed	m/s
η_m	Motor Efficiency	%
θ	Slope Angle	Degrees (°)
ρ_{air}	Air Density	kg/m ³
ω	Angular Velocity	rad/s

Introduction

1. Problem Statement and Motivation

Problem Statement

Kits that convert bicycles to electric are becoming widespread with the developing of bicycle technologies. At the same time, many electric transportation vehicles are developing and becoming widespread. Compared to large, big, and environmentally harmful transportation vehicles, modular and open-to-development electric vehicles are becoming widespread both environmentally and socially. The kits not only protect nature with green energy but also integrate old and mechanical bicycles into the developing age. The detachable feature of our product solves the problem of unnecessary weight when not in use. In short, our problem is that constantly spreading electric bicycles can be obtained only with small and modular kits.

Motivation

Increasing traffic in urban areas significantly increases both energy demand and pollution. According to a study conducted in the United States in 2022, the average occupancy per vehicle is 1.5 people. In other words, in most trips, there is only one person in the car. A similar study from 2012 shows that when only commuters are considered, about 80% of car trips are made by a single person. This situation both destroys the world's valuable resource reserves and harms individual economies (according to the results of the 2022 National Household Travel Survey). As societies face such problems, the demand for cleaner and more flexible transportation options increases. Moreover, by using existing bicycles, people contribute to the circular economy and reduce waste. Additionally, I would like to state that using an electric bicycle—intended for urban transportation—regularly in off-road conditions or in long-distance sports beyond its range is not sustainable, because we cannot remove its weight. In other words, whatever sport the user performs, they can make their bicycle suitable for urban use as well by using appropriate kit modules.

2. Technical and Other Constraints

Some protocols and justifications for continuing the project to meet the system's needs are outlined below.

The system description examines the primary engine locations, and engine shapes are illustrated with a few examples. Important points are highlighted for future reports.

Alternative Motor Types

When we consider the project's expandability and operating type, we can easily determine the maximum motor power. This can be demonstrated through comparisons. Our expectations from the motor include high efficiency, support for regenerative features, durability, and the widespread availability of compatible packages. The available motor operation options we can use are examined below:

DC Brushed Motor

It supports simple controls with basic driver circuits. It is generally used in low-cost, short-lifespan projects that carry minimal load. Since brushed motors involve mechanical contact, they are subject to wear, generate noise, and produce sparks. Although providing high torque at startup is one of their advantages, the disadvantages outweigh the benefits.



Figure 1 - Cansin Motors 24V 500W Brushed DC Motor

AC Induction Motor

It is a very robust motor type, widely used in industrial fields. However, its size and driver types pose disadvantages. It has a direct shaft on the output side and falls outside our size expectations because it is much larger compared to other alternatives. It can be controlled with a frequency converter, which makes complex motor controls like regenerative braking more difficult.



Figure 2 - Pilmak 72V 6500W AC Asynchronous Motor ^[3]

Stepper Motor

Data such as ideal position is captured precisely for precise motor controls, but it is not ideal for continuous rotation.



Figure 3 - MERMAK Step Motor Nema 23 2.2 Nm

Servo Motor

It is suitable for expensive and complex projects that require the position of parts, such as robotic arms, but it is not the most optimal product for the structures in our project.



Figure 4 - Tower Pro MG995 High-Torque Servo Motor

DC Brushless Motor (BLDC)

It's frictionless, suitable for regenerative braking, durable, supports Arduino, and offers sufficient precision. Its quiet operation is a positive factor for the project. It can be customized with the control card used, and it becomes more complex as it is customized. Choosing a BLDC motor makes the most sense in terms of performance, efficiency, and sustainability. Our motor doesn't require any extra shafts; the outer wall of the motor's core structure can be directly used as our wheel, providing a weight advantage.



Figure 5 - EMAX XA2212 820KV Brushless DC Motor

All the motor types that we discussed here, operate with electronic commutation and the motor will be placed in the hub of the wheel.



Figure 6 - 1000 Watt BLDC Hub Motor

Alternative Motor Placements

We would like to present 3+1 design ideas for the required mechanism, along with their advantages and disadvantages. In short, our goal is to move the bicycle by applying force to any of its rotating parts. The main varying idea here is the point of contact. The drive can be provided through the chain, through the wheel hub, or by contacting the tire sidewalls.

Motor Placement 1 (Chain Drive)

Providing drive through the chain is very efficient and reliable due to the direct transmission of force. However, if the chain rotates, the pedals also rotate, which is an undesirable situation. Chain drive significantly reduces comfort to an unacceptable level.



Figure 7 - Chain Drive System

Motor Placement 2 (Wheel Hub Drive)

A connection made through the wheel hub simplifies our control systems from the start, is as efficient as the chain drive, is minimally affected by environmental conditions, and only heats up as much as the entire system does. This type of system offers indispensable advantages once installed. However, its disadvantages are opposite to our needs. It cannot be mobile—only the battery remains mobile. The installation process may discourage the user from the product, and any procedures performed during installation can affect the system and efficiency.



Figure 8 - Wheel Hub

Motor Placement 3 (Rear Wheel-Back Side)

This system may not be as efficient as the first two because environmental conditions directly affect our system. Also, since the tire sidewalls vary on every bicycle, it cannot be universal. However, the mobility of the system is an appealing feature for us. The design challenges we face are, inability to absorb environmental impacts and unreliability at the connection point.



Figure 9 - Rear Wheel Kit

Motor Placement 4 (Rear Wheel-Front Side)

The unique aspect of this system is that it is a more original and efficient design compared to other products. Looking at similar designs, there is insufficient pressure between the tire and our kit, resulting in products failing to provide enough torque during startup and being overly affected by environmental conditions. In this project, we simulate the ground design-wise. If sufficient power is transmitted to the ground, it is thanks to the friction coefficient between the tire and the surface and the force applied to the ground. The friction coefficient at the contact point is maximized using materials like rubber or plastic, but there is no system applying enough pressure.

The 4th option is optimal for us because mobility, as stated in the problem definition, is our main goal. The 4th design is the only one that possesses features we cannot compromise on. Our design faces engineering challenges such as noise and contact efficiency, which we will address theoretically in the first phase and practically in the second phase. To explain one by one:

1. The 1st design was eliminated due to comfort issues. If we use chain drive, the chain directly rotates the pedal, reducing our comfort.
2. Although the 2nd design largely meets users' desired plug-and-play feature, it requires installation, which creates an undesired situation.
3. The 3rd design has been explored through similar products and found lacking compared to our current design.

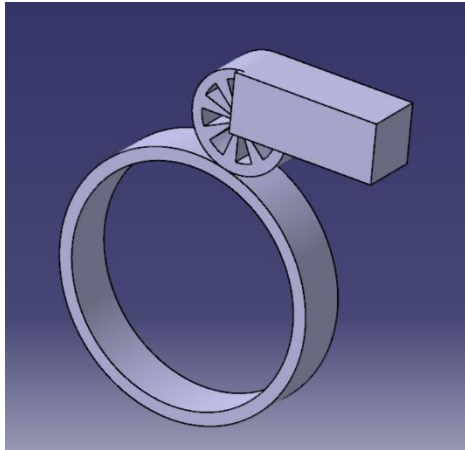


Figure 10 - Representative Design

Functional Requirements

A motor with approximately 75 watts of power is sufficient to hold a 110kg load at 25 km/h, but this calculation should include factors such as environmental conditions and the desired speed on the slope. If we expect the same requirements on a 10% slope, we can reach very high values, such as 890 watts. However, these limitations are not the only constraints, but rather legal regulations. In short, the 250W 36V specification must be met.

Performance Requirements

Maximum Speed

The maximum speed varies depending on the engine selected. However, countries have specific standards, and the criteria for permitting unlicensed use generally include maximum speed adjustments. These criteria were discussed in detail in the previous section, and as can be understood from this, the maximum speed should be 20-25 km/h.

Carrying Capacity

The criteria we consider for weight limits are derived directly from e-bike products. We must not introduce a deficient product into a market where we compete and offer alternatives. Typical products have a loading capacity of 110 kg. Safety factors and lifespan calculations can be made based on this capacity.

Battery Capacity / Range

Our range can be determined by the battery specifications and mAh ratings of our products. We expect a minimum flat road range of 35 km. Factors such as road quality, temperature, and battery quality can directly affect range. Our primary concern is the specific characteristics of the battery; it must be compatible with the engine. Rather than simply choosing the engine type mentioned in the engine selection section, we can calculate the required power output, range/duration, and battery specifications by calculating the engine's power. However, due to legal regulations, we can only increase the battery size.

Mechanisms and Weight

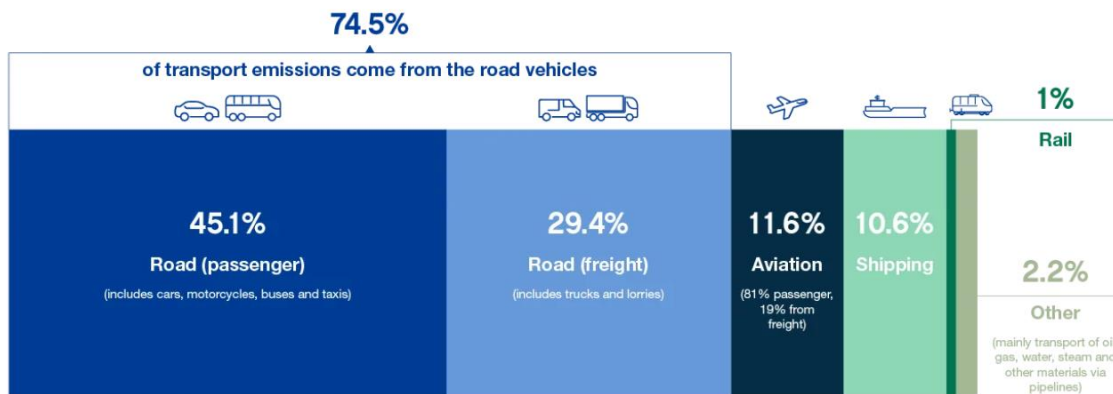
Our product must be mobile and compact, so it will be designed for easy installation and removal. The total weight of the product is evaluated during the Product Development phase and may vary based on user requirements. For example, a 50,000mAh battery is required for a 50 km range, and achieving this would require an increase of approximately 1 kilogram.

3. Engineering Codes/Standards Related to the Project

Environmental and Sustainable Design

Our project places environmental sustainability at the core of its design. Given the significant contribution of the transportation sector to global CO₂ emissions, solutions like electric bicycle conversion kits are of great importance. According to 2018 data, 45.1% of transport emissions from road vehicles carrying passengers, such as cars, buses, and taxis. While there are some efforts being made to reduce these emissions with electric cars, electric buses and shared journeys, only 3.8% of the vehicles in use worldwide are electric vehicles. When it comes to bicycles, the difference in CO₂ emissions is almost incomparable. The CO₂ emission that occurs during the production of the e-bike conversion kit is also much lower.

Our project contributes to various "Sustainable Development Goals". We support good health and well-being (SDG Goal 3) by promoting exercise through bike usage. We encourage the use of affordable and clean energy (SDG Goal 7) by advocating for electric vehicles and bicycles. We adhere to responsible consumption and production (SDG Goal 12) by ensuring our parts are recyclable. Finally, as a green technology, our project directly contributes to climate action (SDG Goal 13).



OurWorldinData.org - Research and data to make progress against the world's largest problems.
Data Source: Our World in Data based on International Energy Agency (IEA) and the International Council on Clean Transportation (ICCT).

Figure 11 - Emission Chart

Economic Considerations

From the consumer's perspective, the equivalents in the market cost much more than the product we intend to design, largely due to customs prices, as they come from abroad. There will be parts that we will also supply from abroad but since the customs cost of individual parts is lower than that of an entire e-bike or an entire e-bike conversion kit, the cost will be lower. The product will have different models for different power needs, such as price performance model or a super powerful model. Our maximum project budget will be 30.000 Turkish Liras. However the price of the full kit will be much lower.

Safety Issues

We want our product to be completely waterproof to prevent liquid damage, which could lead to malfunctions or electrical hazards. We will also provide insulation between the outer case of our product and the inner circuits to prevent electrical leakage. The fact that our product is

removable and portable will eliminate the risk of theft as users can easily detach the kit and carry it with them .

Our product is designed to take a parallel path with national and international engineering standards, particularly Turkish Standards Institution (TSE) and Turkish government regulations related to electric bicycles and road safety

1. TS EN 15194+A1 (2023) - *Cycles - Electrically power assisted cycles - EPAC Bicycles*

This standard applies to EPACs. It aims to cover EPACs with a maximum continuous rated power of 0.25 kW, whose output power reduce as bike reaches 25 km/h or when the user stops pedalling

This European Standard explains the requirements and test methods for engine power management systems and electrical circuits, including the charging system, used in the design and assembly of EPACs. It applies to systems with a rated voltage up to 48V DC, or those with an integrated battery charger with a nominal input of 230V AC.

This Standard also defines the safety and performance requirements related to safety for the design, assembly, and testing of EPACs intended for use on public roads. It also establishes guidelines for providing instructions on the proper use and maintenance of these bicycles

This EPACs can have a maximum saddle height of 63.5 cm or more.

2. TS ISO 20653 (2023) - *Protection of electrical equipment against foreign objects, water and access*

This standard relates to the IP codes. It defines types and levels of protection against liquids, including protection of internal electrical components against the external factors; protection of persons against access to hazardous parts inside the enclosure It also specifies requirements for each degree of protection and tests carried out to confirm that the enclosure complies with the relevant degree of protection requirements.

2. Turkish Traffic Regulations

Currently for individual users in Turkey:

According to the highway traffic law number 2918,

There is no official licensing or type approval process required for e-bike conversion kits. However, using an illegally modified e-bike may result in the vehicle being detained by traffic authorities. Conversion kits sold for commercial use must include a CE certificate, a technical manual, and relevant product safety documentation. E-bikes with a maximum speed of 25 km/h and a motor output not exceeding 250 W are considered non-motor vehicles. Therefore, they do not require a driver's license or license plate but must still follow to standard bicycle safety regulation.

4. Work Packages and Work-Time Schedule

Work Packages

WORK PACKAGES	STATUS	ASSIGNED TO	START DATE	EXPECTED END DATE	BUDGET
Literature Review and Research	In progress	All Members	08/07/2025	15/05/2026	-
Documentation & Reporting	In progress	All Members	15/07/2025	15/05/2026	-
Designing	Completed	Berke Arslan	01/10/2025	01/12/2025	-
Electrical & Control System Development	In progress	Cihan Güldalı	03/11/2025	15/02/2026	3000TL
Battery & Energy Management	In progress	Berke Arslan	15/11/2025	15/02/2026	1500TL
Prototype Assembly	Not started	All Members	10/01/2026	01/02/2026	2500TL

Performance Testing and Calibration	Not started	All Members	05/02/2025	15/05/2026	-
Simulation & Analysis	Completed	Toprak Eren	15/10/2025	10/12/2026	-
Improvements on Design and Systems	In progress	Cihan Güldalı	20/10/2025	01/03/2026	1000TL
Presentation	Completed	Toprak Eren	15/10/2025	19/11/2025	-

Deliverables

Literature Review and Research

Gathering and Reading Articles - **In Progress**

Researching Different Solutions on Market - **In Progress**

Documentation and Reporting

Report Writing - **Completed**

Logbooks - **Completed**

Designing

Creating a 3D CAD Model - **Completed**

Performance Testing and Calibration

Test Results Report - **Not Started**

Calibration Logs - **Not Started**

Simulation & Analysis

Simulation Report - **Completed**

Analysis Report - **Completed**

Improvements on Design and Systems

Redesigning Parts - **In Progress**

Electrical & Control System Development

Circuit Diagrams - [Not Started](#)

Control Software - [Not Started](#)

Battery & Energy Management

Battery Prototype - [Not Started](#)

Battery Tests and Efficiency Report - [Not Started](#)

Prototype Assembly

Assembled Prototype - [Not Started](#)

Presentation

Preparing the Presentation - [Completed](#)

Rehearsal Sessions - [Completed](#)

Work-Time Schedule

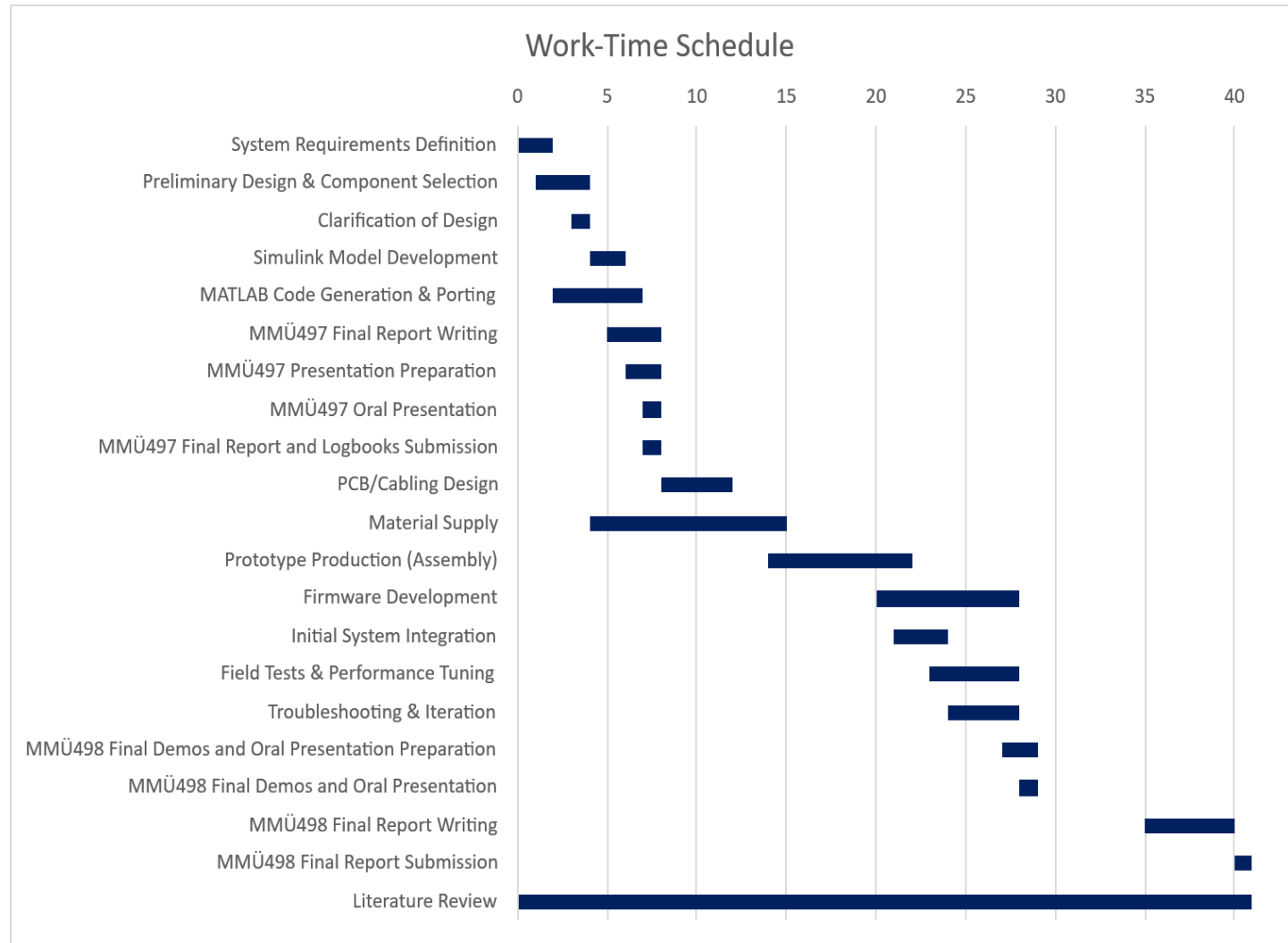


Figure 12 - Gantt Chart for Work-Time Schedule

Literature Review and Preliminary Design

1. Literature Review

Control Mechanism

To begin, let us describe the control system. Control systems are responsible for calculating the energy supplied to the motor based on available data and inputs. In our system, an external target speed is received and transmitted via the throttle. However, simultaneously, we must maintain this speed input constant. In essence, while the target speed can be precisely controlled by the RPM value of our Hall sensor-equipped motor, controlling the speed itself is quite complex. Several parameters oppose our target speed. Dominant external factors include positive or negative forces such as the correct positioning of the kit, pedal assistance, or bicycle brakes. By utilizing appropriate sensors to measure these forces, we can interpret them accurately and ensure stability through the protective code implemented in our control system. Our system will feature several distinct modes: one focused on maintaining speed, and another on determining a target acceleration based on the command received from the throttle. Additionally, certain microcontrollers enable wireless communication via specific protocols. Data can be interpreted and controlled according to the data flow (Advanced communication). Subsequently, this data must be processed.

Primarily, we require a microcontroller that allows us to process and interpret data. This board inherently possesses certain technical specifications, and its tolerance levels are a critical point that must be primarily checked. If excessive voltage is supplied to the chosen board, its components may be damaged. Another crucial consideration is the board's pins; some of these pins are designated for analog inputs, while others are for digital inputs. The input pin types of the microcontrollers must be verified from their respective datasheets, and connections should be made accordingly. All sensors converge and are interpreted here. The command lines to be processed in the microcontroller should ideally be in C++ / embedded language. However, we will adopt a different approach. First, we need to develop a model using Simulink, which we utilize in our System Dynamics and Control course. Subsequently, with the aid of MATLAB, we can simply convert this model into C++ / embedded code. Finally, the processed data is transferred to the VESC control board, where the balance between the battery and the motor is regulated.

External Parts

Due to the critical features of the product, it must be mobile. Therefore, our literature review will focus only on products that meet this requirement. The desired design criteria (mobility) explain the working principle to us. In our project, which operates by driving the rear wheel, there are a few additional points that can be modified.

Swytch

Based on the literature review, there are two battery positioning options that meet mobility requirements. Similar products are listed below.

8.1.1 Swytch Kit

The batteries are designed to be separate from the system and can be fixed to the frame with an additional connection. This allows different battery sizes to be added without needing to modify the product frame. Products operating on this principle usually use a mounted motor and a mobile battery system. While our goal is not fully met by the easy plug-and-play feature of these products, they still provide inspiration. The reason we included the Swytch Kit in the comparison is that its working principle can be integrated with our system.



Figure 13 - Different Battery Placements

Livall Pikaboost

The Livall brand's operating system combines the entire system into a single product, rather than modeling individual faucets separately. This approach successfully meets all our criteria.



Figure 14 - Livall Pikaboost

Rubbee X

Like the Livall product, it comes with battery packs in a single housing. Unlike the Livall product, these products require a single, stable mount on the bike.



Figure 15 - Rubbee X

Battery

All devices used around the world have two main types of batteries: rechargeable and non-rechargeable batteries.

Primary (Non-Rechargeable) Batteries

Single-use batteries cannot be recharged after their energy has been depleted. Due to their disposable nature, they must be recycled in accordance with environmental conditions. Due to their long shelf life and portability, they are used in many industries, such as household appliances, medicine, and aerospace industries. There are various types, such as zinc-carbon, alkaline, and lithium.



Figure 16 - Dry Cell Battery

Secondary (Rechargeable) Batteries

These are energy storage systems that convert chemical energy into electrical energy and can perform this process multiple times in a reversible manner. Thanks to their rechargeability, they have a longer lifespan than primary batteries. Due to the high pollution they can cause to the environment, they must be recycled at waste collection facilities. They have a wide range of applications, including portable devices, EVs, and renewable energy etc.

Advantages:

- Frequent use thanks to the charging cycle
- Economical due to long service life
- More environmentally friendly than primary batteries
- High energy and power capacity
- High efficiency

For the reasons mentioned above, we chose to use a secondary battery in our device.

If we classify rechargeable batteries according to their chemical types, there are four main categories.

Nickel-Based Systems

- Nickel-Cadmium (Ni-Ca)
- Nickel-Metal Hydride (Ni-MH)

Lead-based Systems

- Lead-Acid (Pb-Ac)

Sodium-based Systems

- Sodium-Ion (Na-ion)

Lithium-based systems

- Lithium-ion (Li-ion)
- Lithium-polymer (Li-Po)
- Lithium iron phosphate (LiFePO₄)

In this project, the focus is on lithium-based secondary batteries due to their advantages such as lightness, cycle life, and energy density. While there are some different types of lithium-based systems (Li-ion, Li-Po, LiFePO₄, etc.), no specific type has been selected yet for this project. In later stages, the most suitable lithium-based battery type will be determined based on system requirements (e.g., power demand, safety, cost, weight).

Lithium-ion (Li-ion)

Among the most popular rechargeable battery types today, they have high energy densities and long cycle lives. Lithium-ion batteries are based on compounds containing lithium in both electrodes, typically graphite in the cathode and lithium in the anode. The charge-discharge process converts chemical energy into electrical energy.

Advantages:

- High energy density
- Low self-discharge effect
- High voltage per cell
- Long service life
- Low maintenance costs
- Fast charging support

Disadvantages:

- Safety risk in case of overcharging or over-discharging
- High cost compared to other batteries

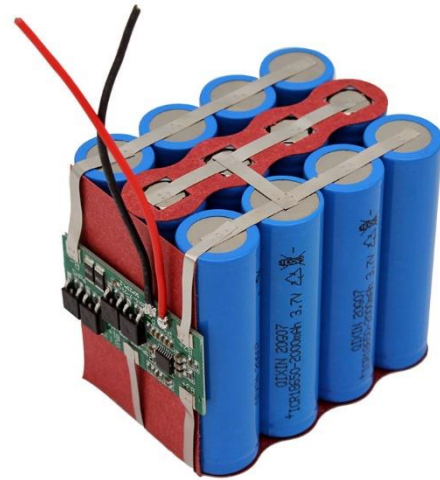


Figure 17 - Rechargeable Lithium Ion Battery

Lithium-polymer (Li-Po)

Li-Po batteries are a modification of lithium-ion batteries. Although liquid electrolytes take up more space than solid electrolytes, they can be used to make smaller batteries.

Advantages:

- Higher energy density than Li-ion
- Takes up less space
- Lightweight
- Flexible structure allows for production in different shapes



Figure 18 - Li-Po Battery

Disadvantages:

- High internal resistance
- Expensive due to the internal material of the electrolyte
- Sensitive to impact
- Shorter lifespan compared to Li-ion
- Low thermal stability.

Lithium Iron Phosphate (LiFePO₄)

LiFePO₄ batteries are a subtype of lithium-ion technology.

Advantages:

- High thermal stability, low risk of combustion and explosion
- Longer lifespan than Li-ion
- Wide operating temperature range
- Low internal resistance

Disadvantages:

- Low energy density
- High volume density
- High cost



Figure 19 - LiFePo4 Battery

How to read a Pugh Chart

When using Pugh Chart, we define one of the Concept as a baseline. This baseline scores “S” in every criteria. Other candidates compared with that baseline and gets different points.

If it’s better than our baseline, a “+” score is entered in the appropriate box

If its worse than our baseline, a “-“ score is entered in the appropriate box

If its same with our baseline, a “S” score is entered in the appropriate box

You can increase the amount of “+” or “-“ signs if necessary, but the maximum number of signs is usually 2. (“++” : much better, “--” : much worse)

For each candidate design option, the sum of the pros and cons is indicated. Generally in Pugh Charts, there isn’t any clear winner but a clear loser. Criteria can be weighted for more specific requests.

2. Pugh Charts and Comparison of Different Solutions

At this stage, the selected critical points are specified in detail and the criteria are color coded.

Color Code	Rating
Green	Good
Yellow	Moderate
Red	Inappropriate

Table 1 - Classification of colors

Microcontrollers

Microcontroller Feature	Arduino Uno	ESP32	STM32
Easy to Code	Easy	Middle -Arduino IDE Support	Middle
Software Support and Community	Extensive support	Extensive Support	Extensive Support (Technic)
Up to Date and Developable	Basic Customizability	Customizable	Basic Customizable
Remote Control	No	Wi-Fi and Bluetooth	No

Our primary objective when selecting a microcontroller is to prioritise features such as software support and up-to-dateness. Products must be codable. Generally, Arduino and its derivatives suffice for our needs. The second point to check is expandability. Some microcontrollers incorporate remote control technologies. Such microcontroller features are examined below.

Microcontroller Feature	Arduino Uno	ESP32	STM32
Easy to Code	S	S	S
Software Support and Community	S	+	S
Up to Date and Developable	S	+	S
Remote Control	S	+	S

TOTAL +	0	3	0
TOTAL -	0	0	0
TOTAL SCORE	0	3	0

Table 3 - Pugh Chart of Microcontrollers

VESC

We have now arrived at our most complex and valuable board. Let us assume we have a BLDC motor. Although there are cheaper and sufficient alternatives to control this motor, these boards are only suitable for basic speed control and cannot be modified due to being closed to external interference. For our project, VESC-type boards, which are open-source, modifiable, and more professional with high tolerance, are appropriate as the project develops over time. The VESC boards we select must have firmware installed; otherwise, professional settings cannot be performed. The VESC tool must be installed on a computer, and the

necessary configurations must be applied. The foremost setting is choosing the correct communication type with ESP32. We have decided to use UART communication because, according to the components and data in our project, UART is the fastest and easiest communication method.

Specification Board	xESC2 mini	xESC	xESC2	VESC 6	Basic Speed Controller
Max Current	10A / 40V	10A / 40V	10A / 40V	>80A / 60V	Variable
Interface	USB, UART, CAN, PWM, Analog	UART, PWM	USB, UART, CAN, PWM, Analog	USB, UART, CAN, PWM, Analog and more	Only PWM
Commutation	FOC	BLDC	FOC	FOC	BLDC
Control Mode	Current, Position, Speed	Current	Current, Position, Speed	Current, Position, Speed	With PWM
Sensor Compatibility	YES	YES	YES	YES	No Passive Control
Feedback Compatibility	YES	YES	YES	YES	No Feedback
Warming Problem	Slightly	No	No	No	No

Table 4 - Feature Table for Boards

Specification Board	xESC2 mini	xESC	xESC2	VESC 6
Max Current	S	S	S	S
Interface	S	S	S	S
Commutation	S	-	S	S
Control Mode	S	-	S	S
Sensor Compatibility	S	S	S	S
Feedback Compatibility	S	S	S	S
Warming Problem	S	S	S	S

TOTAL +	0	0	0	0
TOTAL -	0	2	0	0
TOTAL SCORE	0	-2	0	0

Table 5 - Pugh Chart for Boards

As shown in the table, we compared speed controllers and highlighted the most prominent technologies. Benjamin VEDDER (YODA), the father of the controller card technologies discussed here, has shared all of his work as open source and left it open for further development. The technologies shown are based on customizations made to this card design by YODA.

Our primary option is the xESC2 alternative. Compared to other options, the xESC2 is quite small, efficient, and sufficient. However, while its low cost creates positive feedback, it also raises security concerns because it's a customized, hand-built technology. We may need assistance with easy integration into the control unit. In the final report, we will conduct a cost analysis and explain in detail why we chose this product.

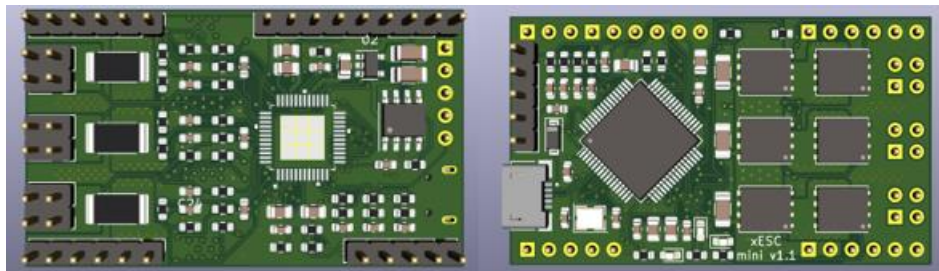


Figure 20 - xESC 2 Custom Design

Sensors

First, it's important to explain how we choose our sensors. Our options are briefly outlined below. However, our sensor selection should be driven by our criteria rather than brand/model.

Criteria:

- All sensor outputs should not exceed 3.3V, as microcontroller inputs generally have a tolerance of 3.3V. Our options here can be quite limited, but using a voltage divider can reduce the voltage to the desired range. A 5V supply voltage is not a problem.
- It should provide a linear output.

- (optional) An internal RC filter protects us from voltage fluctuations. Otherwise, filtering can be done through software.

Hall sensor

The Hall sensor motor is an integrated card, and the data from this provides information about the motor's speed. However, we must use this speed information unless a constant speed command is given in our control system, because setting an engine speed target isn't achieved by simply placing a throttle. We'll be comparing Hall sensors with our motors because the Hall sensor is integrated into the motor.

Throttle Lever

The incoming data provides information about the target. Here, we face a few issues. The main problems are as follows: the throttle may constantly output a small voltage even if no command is given (zero error). The responses from the throttle at the zero position should not be interpreted; otherwise, the product will behave as if it wants to decelerate as it accelerates (regenerative). Apart from this, speed stabilization mode and regenerative mode can be added later depending on progress. However, to avoid regulatory issues, this throttle sensor will be checked again.

e.g.

When the throttle is at position 0, the speed should be 0 / when the throttle is at position 1, 10 km/h is requested. (Incorrectly interpreted data)

When the throttle is at position 0, there should be no response / when the throttle is at position 1, an acceleration of 1 km is requested and full power is given. (Correctly interpreted data)

The mechanical movement is converted into an analog voltage between 0V–5V.

If connected with the midpoint at 2.5V (reference taken as 0V in the code):

- $>2.5V \rightarrow$ Positive demand (forward)
- $<2.5V \rightarrow$ Negative demand (regenerative braking)

This algorithm is created only for preliminary reference.

So how can we build this system?

We have two options: either buy ready-made products like a throttle or directly connect two potentiometers in reverse and calculate the voltage difference between them. Let us simply compare both options.



Figure 21 - Throttle Sensor

Throttle sensor is a bit expensive but it is not over our budget. (360 TL)

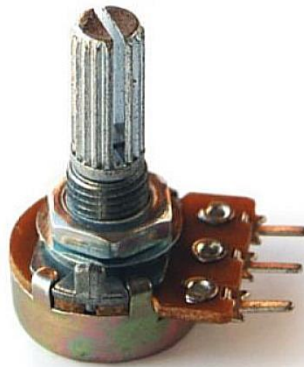


Figure 22 - Potentiometer

Potentiometer is quite cheap, around (20 TL)

Comparing these options falls under the cost analysis section, but let's briefly discuss them. Since we can easily install the two potentiometers by reversing them and calculating zero error, option 2 can be selected, but it's not our final decision. The output voltage ranges for the

products are not specified; if the outputs are at undesirable levels, a voltage divider should be used.

Temperature/Humidity, Gyro and current sensors

These should be interpreted within if blocks to provide protection. If the conditions are not met, the machine may reduce performance or shut down directly to avoid damage.

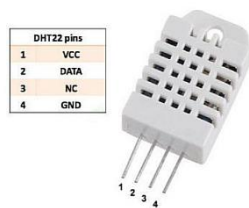


Figure 23 - Temperature and Humidity Sensor

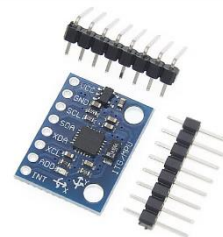


Figure 24 - Acceleration and Gyro Sensor

The reason we did not include current sensor types is that most of them come integrated into VESC card circuits.

At this stage, instead of comparing our sensors, we've identified key points to consider. Any sensor that meets these criteria, such as voltage and analog/digital separation, is suitable for us. Critical criteria are highlighted in green, while preferred criteria are highlighted in yellow.

Sensor	Type	Output Type	ESP32 Input
Throttle	Analog	0–3.3V	ADC
Temp/Humidity	Digital	DHT22	I ² C
Gyro/IMU	Digital	MPU6050 (I ² C)	I ² C
Current	Analog	ACS712	ADC

Table 6 - Sensors

Additional equipment

Level Shifter

Level shifters can be used in case we encounter problems with our voltage values. For example, if the VESC input we will soon examine is not 5V tolerant and a 5V electrical output from the ESP is transmitted to the VESC, it may be damaged (usually the opposite case is observed).

Voltage Divider

Level shifters are used for the purpose of reducing voltage. The board structures are quite different from each other, and as a result, voltage dividers are simpler and cheaper. While level shifters reduce voltage bidirectionally, voltage dividers operate unidirectionally. Therefore, if the current is unidirectional, voltage dividers are preferred due to their lower cost.

Products such as Dupont cables, headers, Faraday cages are standard and there is no difference in their working principles as long as quality is taken into account.

System Description

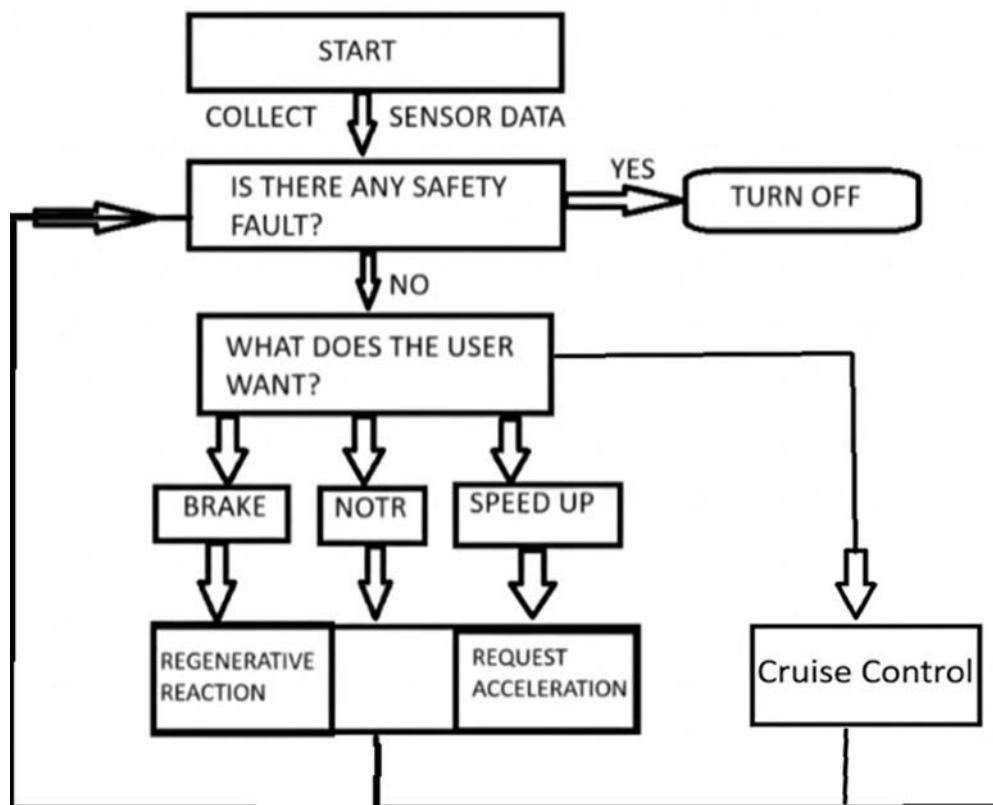


Figure 25 - Sensor Algorithm

Engine

As written in the first report, the types of engines we can use are limited due to the requirements of the system.

Why BLDC Brushless Motor?

Motors Criteria	DC Brushed	AC Induction	Stepper	Servo	DC Brushless (BLDC)
Efficiency	S	S	S	+	++
Regenerative Capability	S	S	+	S	++
Durability	S	+	S	+	+
Size/Compactness	S	--	--	--	S
Noise Level	S	S	++	++	++
Control Complexity	S	--	--	--	-
Cost	S	--	-	--	-
Precision	S	S	++	++	++
Suitability for Continuous Rotation	S	S	--	S	S
Integration with Arduino	S	-	-	-	+

TOTAL +	0	1	5	6	10
TOTAL -	0	-7	-8	-7	-2
TOTAL SCORE	0	-6	-3	-1	8

Table 7 - Pugh Chart for Motor Alternatives

We need to use a BLDC brushless and hub motor, but there are two types of motors that meet the requirements. The first is a direct-drive product without any chassis, and the other is a similar system that includes gears.

Criteria \ BLDC Types	Direct Drive Hub Motor	Geared Hub Motor
Torque	S	+
Speed	S	-
Efficiency	S	S
Regenerative Braking Capability	S	-
Noise Level	S	-
Weight	S	+
Size	S	+
Maintenance Requirement	S	-
Coasting	S	+
Cost	S	S
TOTAL +	0	4
TOTAL -	0	-4
TOTAL SCORE	0	0

Table 8 - Pugh Chart for BLDC Types

In fact, both working principles are suitable for us, but the fact that the direct drive motor type is generally average also ensures that it is trouble-free.

Materials Parameters	Aluminium Alloy	Carbon Fiber	ABS	Stainless Steel	PLA
Weight	S	+	S	--	S
Durability	S	+	S	+	--
Cost	S	-	+	S	++
Machinability	S	-	++	-	++
Heat Dissipation	S	-	-	S	--
Environmental Impact	S	--	-	--	-

TOTAL +	0	2	3	1	4
TOTAL -	0	5	2	5	5
TOTAL SCORE	0	-3	1	-4	-1

Table 9 - Casing Material Pugh Chart

Battery Type Parameters	Li-ion	Li-Po	LiFePO ₄
Energy Density	S	-	--
Cycle Life	S	S	+
Discharge Temperature	S	-	+
Safety	S	-	+
Cost	S	--	-
Weight	S	+	-
Charging Time	S	+	-
Physical Flexibility	S	+	-
Recyclability	S	-	+
High Current Performance	S	++	+

TOTAL +	0	5	5
TOTAL -	0	6	6
TOTAL SCORE	0	-1	-1

Table 10 - Batteries Pugh Chart

3. Design Approach and Process

In this section of our project, we are going to consider various design approaches and carry out detailed analysis to find the most appropriate solution to achieve our project's main goal of modularity, portability and ease of use. This design approach addresses some of the engineering challenges such as noise and contact efficiency and potential solutions to these problems.

Different placement and operating principles are being evaluated for the system that will form the basis of our project.

In this evaluation process, the advantages and disadvantages of each alternative are examined within the project's performance goals and realistic constraints.

Criteria for Mechanical Component Selection

Criteria for mechanical components and drive mechanism that will make the contact of our engine and the tire of the bike are listed below:

Material Properties: High coefficient of friction, wear resistance, light weight and durability.

Ease of Installation: A practical mechanism that will make our kit easily attachable and detachable, Adjustability and compatibility with the different tire types.

Protection: Protection of electronic components to the external factors, Prevention of liquid damage, Heat dissipation.

Pressure Application Mechanism: A reliable system that ensures the contact of our electrically assisted driving wheel and the tire.

Modularity and Portability

In design process, modularity and portability of the kit is considered as a fundamental necessity. Our goal is to bring all components together in a compact volume and make sure the user can easily carry it. The fact that the mounting mechanism can be quickly installed and removed without requiring any special tools is critical in minimizing the risk of theft.

Weight Distribution and Balance Criteria

The weight distribution of the kit on the bicycle should be optimized so as not to negatively affect riding balance and comfort. The location of heavy components such as the motor and battery will be aligned with the bicycle's centre of gravity.

Some of the fundamental engineering challenges we face are noise and contact efficiency. The general solution approaches to these challenges are as follows:

Noise Reduction Strategies

Material Selection: Researching and using materials with vibration and noise damping properties.

Mechanical Isolation: Using elements that absorb vibrations when placing the motor and drive mechanism inside the case.

Design Optimization: Optimizing surface geometry and contact pressure to reduce noise.

Contact Efficiency and Slip Prevention Strategies

High-Friction Surfaces: Developing surfaces with special textures or patterns that provide maximum friction with tire.

Adjustable Pressure Mechanisms: Integration of a manually adjustable pressure application mechanism that is adaptive to small differences in tire diameter and pressure.

Environmental Protection and Safety Measures

Water Resistance (IP Rating): All electronic components and connections are insulated against water and dust ingress in accordance with IP standards.

IEC Making electrotechnology work for you.

Ingress protection (IP) ratings guide

IP ratings are represented by combining the first and second digits of the below columns

1 st numeral - solid foreign objects	2 nd numeral - water
0 No protection	0 No protection
1 Protected against solid foreign objects of 50 mm Ø and greater	1 Protected against vertically falling water drops Vertically falling drops shall have no harmful effects
2 Protected against solid foreign objects of 12.5 mm Ø and greater	2 Protected against vertically falling water drops when enclosure tilted up to 15° Vertically falling drops shall have no harmful effects when the enclosure is tilted at any angle up to 15° on either side of the vertical
3 Protected against solid foreign objects of 2.5 mm Ø and greater	3 Protected against spraying water Water sprayed at an angle up to 60° on either side of the vertical shall have no harmful effects
4 Protected against solid foreign objects of 1.0 mm Ø and greater	4 Protected against splashing water Water splashed against the enclosure from any direction shall have no harmful effects
5 Dust-protected	5 Protected against water jets Water projected in jets against the enclosure from any directions shall have no harmful effects
6 Dust-tight	6 Protected against powerful water jets Water projected in powerful jets against the enclosure from any direction shall have no harmful effects
Example: IP 65 + Protected against water jets Dust-tight	7 Protected against the effects of temporary immersion in water Ingress of water in quantities causing harmful effects shall not be possible when the enclosure is temporarily immersed in water under standardized conditions of pressure and time
	8 Protected against the effects of continuous immersion in water Ingress of water in quantities causing harmful effects shall not be possible when the enclosure is continuously immersed in water under conditions which shall be agreed between manufacturer and user but which are more severe than for numeral 7
	9 Protected against high pressure and high temperature water jets Water projected at high pressure and high temperature against the enclosure from any direction shall not have harmful effects

Figure 26 - IP Ratings Guide

Electrical Insulation: Sufficient electrical insulation between internal circuits and external housing to eliminate potential electrical leakage risks.

Thermal Management: Designing an effective passive cooling system for heat-generating components such as motors and batteries.

Design Verification and Analysis Methods

Manual calculations, code-based analyses, and tools such as ANSYS (flow, vibration, harmonic analyses) and MATLAB Simulink (embedded software development and control algorithm simulations) will be used to verify the design. As we conduct these analyses, we will make the necessary changes to our design. These analyses will ensure that the design meets performance requirements (speed, range, carrying capacity) and is consistent with realistic design constraints (noise, efficiency, safety).

Manual Calculations

In the early stages of the project, manual calculations based on mechanical and electrical principles will be performed. These calculations will provide a quick and effective method for

the initial evaluation of design alternatives and preliminary determination of main component dimensions. Additionally, they will serve as a reference point for verifying input parameters and expected results before doing more complex simulations

Code Based Analyses

Special codes will be developed using MATLAB for specific engineering problems. These codes will be used in areas such as repetitive calculations, data processing, optimization algorithms, and mathematical modelling of control systems. For example, dynamic performance analyses such as battery discharge curves, motor efficiency maps, or energy consumption under different driving scenarios will be performed using this method. This approach will enable the effects of different scenarios to be examined by quickly changing design parameters.

ANSYS Analysis/Simulations

ANSYS software will be used to examine the mechanical integrity and performance of the design in detail.

Flow Analysis (CFD)

The air flow and heat distribution inside the kit's enclosure will be examined. The design of cooling channels and ventilation holes will be optimized through these analyses to ensure that heat-generating components such as the motor and battery remain at their optimal operating temperatures.

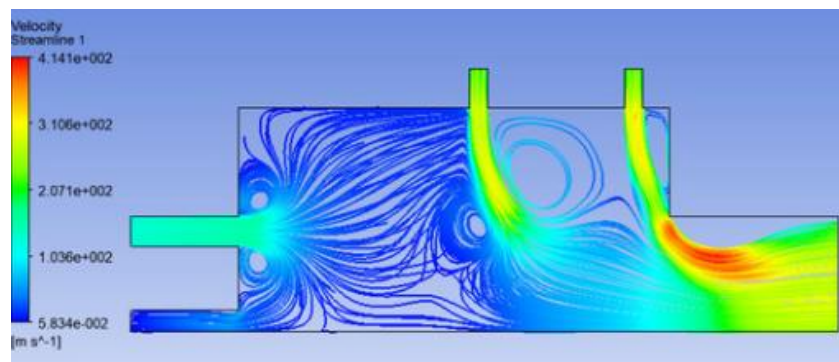


Figure 27 - Example Flow Analysis made on ANSYS

Vibration Analysis

The effects of road vibrations on the kit during bicycle operation and vibrations caused by motor operation will be analysed. These analyses are critical for increasing the fatigue life of mechanical components and reducing noise.

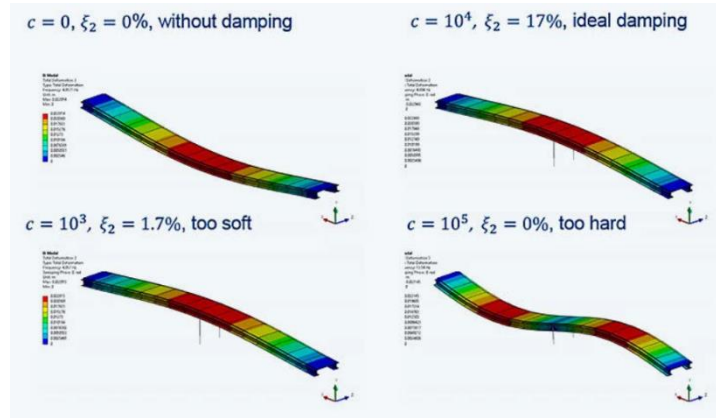


Figure 28 - Damped modal analysis for a bridge

Harmonic Analysis

Resonance conditions that may occur at the motor's operating frequencies will be investigated. Preventing resonance or minimizing its effects is vital for the stability and longevity of the system.

Structural Analysis

Stress, deformation, and safety factor analyses will be performed on critical components such as the kit's load-bearing structure, mounting points, and drive pulley under static and dynamic loads. This will ensure that the design safely meets the specified load capacity.

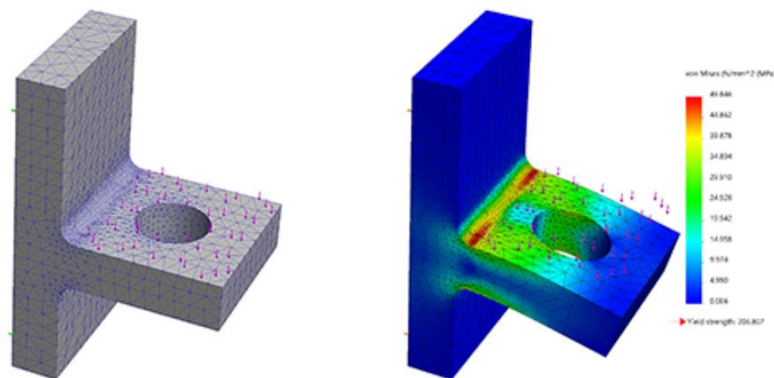


Figure 29 - Static Analysis of an object

Final Design Selection

1. Chosen Design

Design and Development

The primary objective of the design is to ensure user-friendly by enabling convenient charging at home due to its portability and facilitating easy installation through its modular structure. In line with this objective, the design process has focused on three main criteria: ease of attachment as well as detachment, a compact form, and aesthetic appeal.

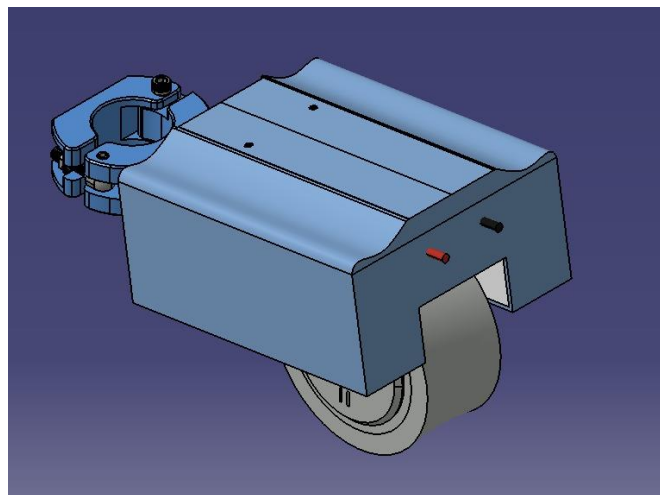


Figure 30 - 3D Model of the Product

Prior to the product design process, existing similar products on the market were examined. The evaluations revealed that these systems often suffer from several drawbacks, including excessive weight, lack of originality in design, and limited compatibility with specific bike types. These shortcomings have served as key reference points in defining the primary objectives of the new design.

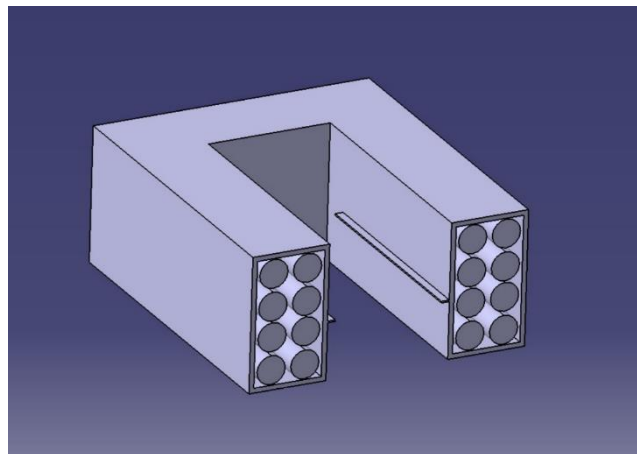


Figure 31 - 3D Model of the U-Shape Battery

All components were modelled in three dimensions using CATIA V5. The product consists of five main components: the clamping mount, battery, chassis, hub motor, and electronic

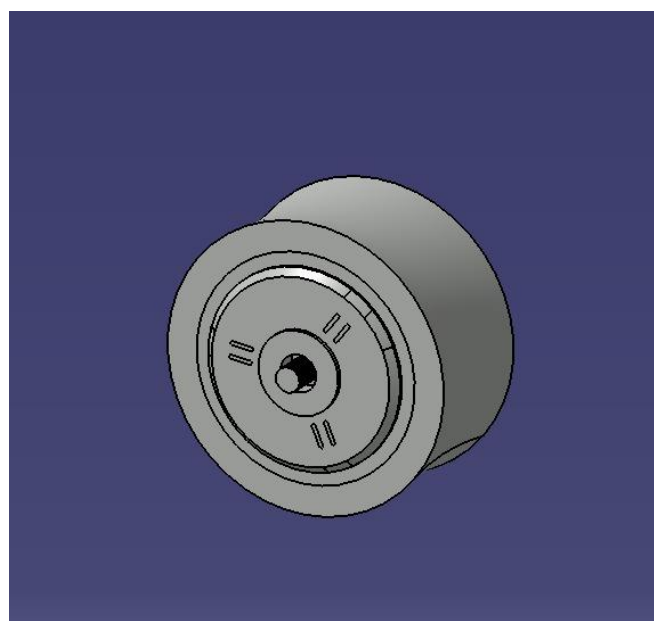


Figure 32- 3D Model of the Hub Motor

circuits. The clamping mount, which attaches to the seat tube, rigidly secures the chassis to the bike. Through adjustable clamping structure, it accommodates different tube diameters and achieves high retention force via tightening screws. On the surface of the clamping mount in contact with the chassis, there is a bolt hole integrated into the chassis. Through this connection, our product is attached to the saddle and thus to the bike. By using a suitably sized Allen key, both the fixation to the seat tube and the installation of our product onto the bike can be carried out with ease.

For the energy source of the system, a lithium-ion battery pack from the Xiaomi Mi Scooter Pro 2 was used. The battery module is positioned in a sealed enclosure at the top part of the chassis, protected from external factors. This arrangement not only permits balancing the system's center of gravity, but also allows for orderly routing of cable connections. Furthermore it ensures the battery is rigidly secured and protected against vibrations and impacts. A hub motor is used to transfer the obtained drive power to the bike. The 38.8 cm long battery module did not align with the overall aesthetics of the bike and had a negative effect on the balance of the center of gravity in the initial design trials. To improve this, the battery form was reshaped into a U-shape, thereby achieving both a compact appearance and a more balanced mass distribution. The motor is placed at the lower part of the chassis and contacts the rear wheel via a rubber-covered drive roller. This positioning ensures direct power transmission to the wheel while maintaining the compact structure of the system. The connection of the motor to the chassis is designed to be rigid, aiming to reduce vibrations. The hub motor is flanked by battery modules located laterally, and above it, protected by the chassis, are the ESP32 microcontroller and the VESC motor driver. In this way, the critical electronic circuits in our product are safeguarded against environmental effects such as external impact, vibrations, and liquid splashes.

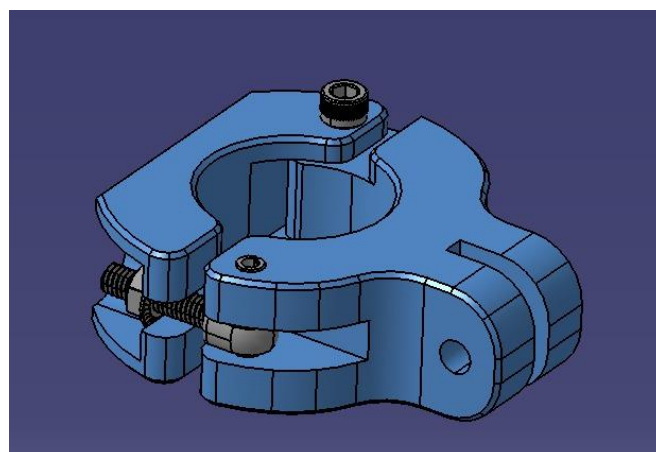


Figure 33- 3D Model of the Clamping Mount

Finally, mention is made of the chassis structure, which houses all components together and ensures the integrity of the system. The chassis simultaneously provides protection against

Figure 4. 3D Model of the Clamping Mount

environmental influences, thus enhancing system durability. One of the most original components in terms of design, the chassis has been shaped according to the geometry of the battery module used and the functional requirements of the system. Depending on the form of the battery, the chassis is directed upward at an angle of 9.163° in order to ensure appropriate placement of the hub motor and electronic circuits that can function properly. Additionally, the part of the chassis that attaches to the bike is directed downward at an angle of 2.15° , and the arms are designed with a length of 15 cm, so as to provide both rigid connection and aerodynamic advantage.

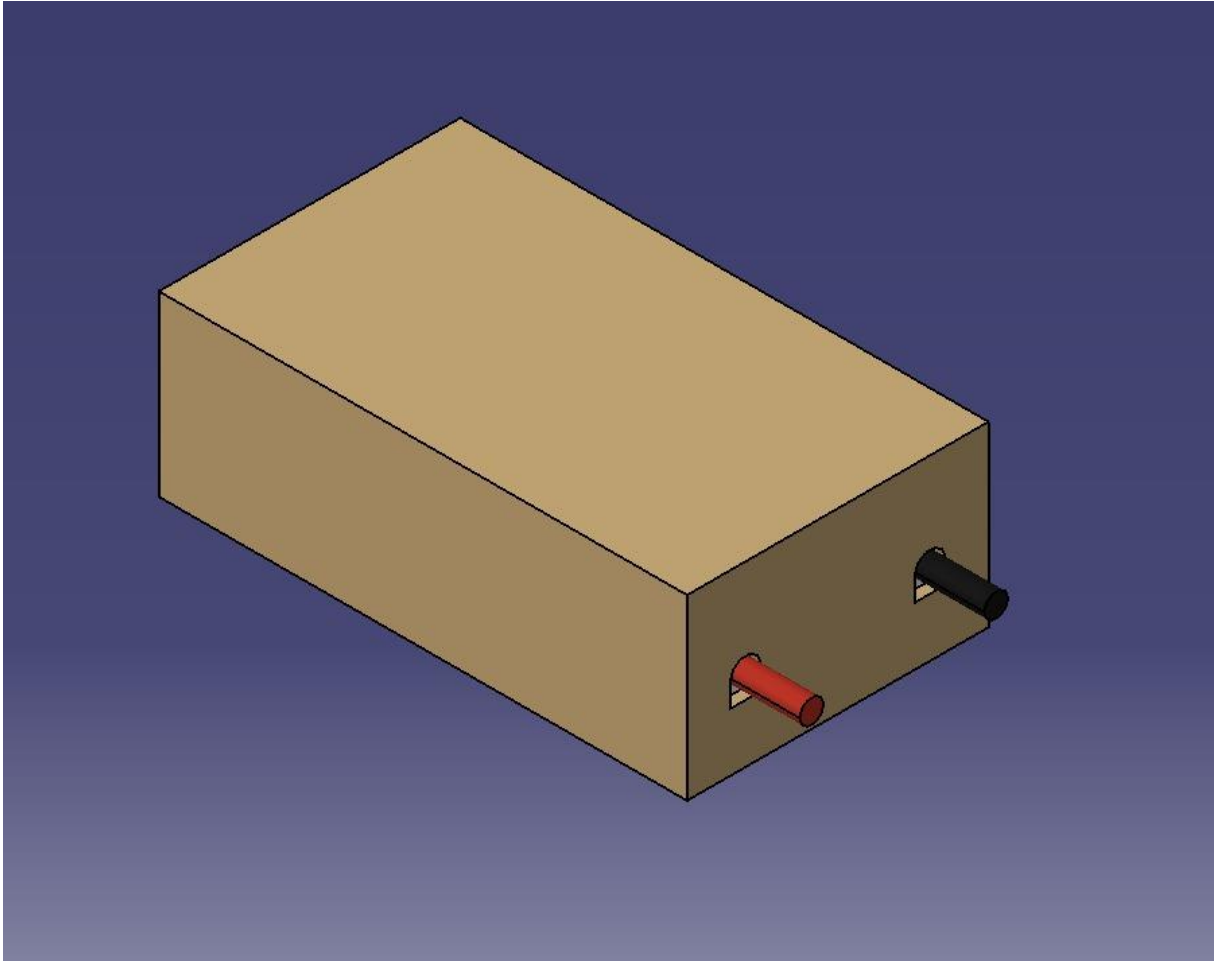


Figure 34- 3D Model of the VESC Motor Driver

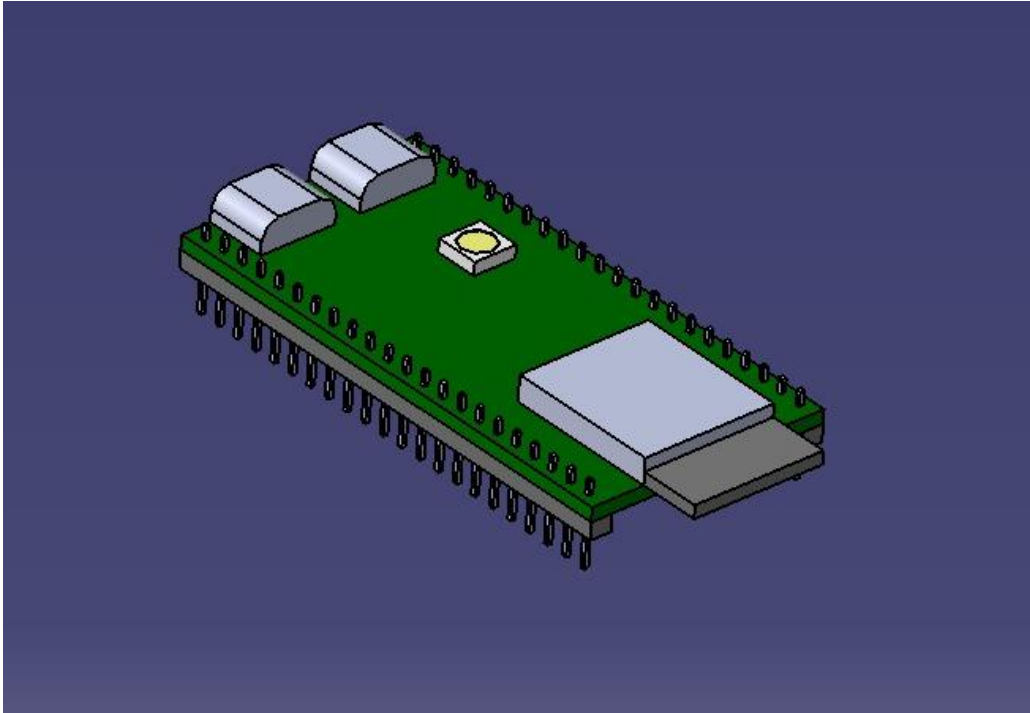


Figure 35- 3D Model of the ESP32 Microcontroller

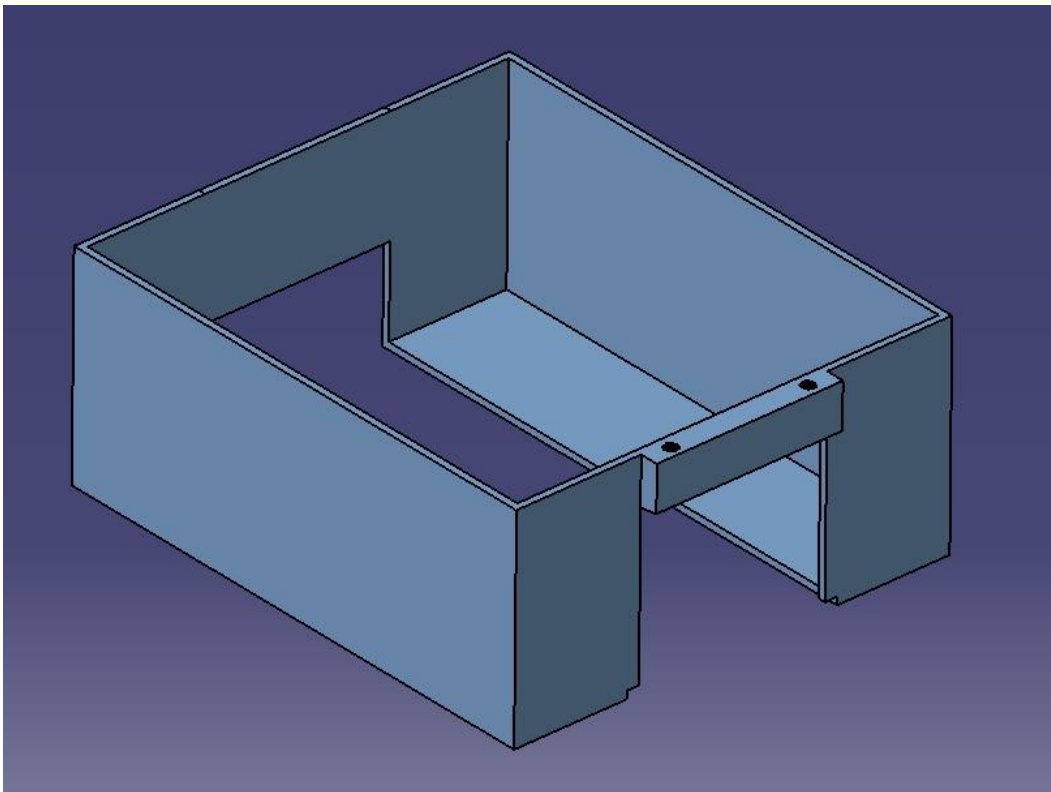


Figure 36- 3D Model of the Side Frame Chassis

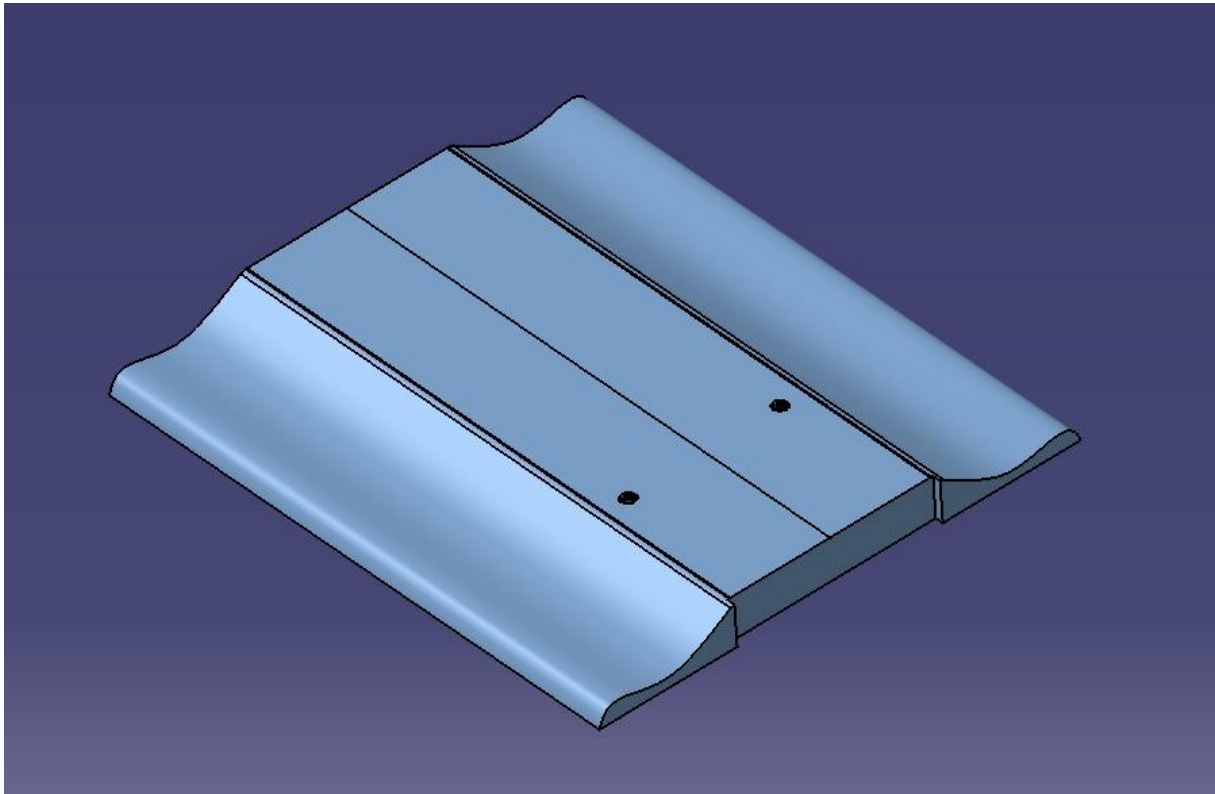


Figure 37 - 3D Model of the Upper Frame Chassis

Battery Design

Role of the Battery in the System

In electric transportation systems, the primary energy source is the battery. The battery provides the electrical energy required for the electronic components, lighting units, and the drive system with the highest power demand within the system, enabling the entire system to function.

The battery not only serves as an energy source but also plays a critical role that directly affects the overall performance of the system. The battery's capacity and voltage power determine the device's operating time, maximum power output, and driving continuity.

If the battery fails to perform its function, the entire system shuts down, and the vehicle cannot operate electrically. In terms of the functionality and continuity of the electric bicycle kit, the battery is considered one of the most critical subsystems.

Internal Structure of the Battery

The energy source used is the original lithium-ion battery module belonging to the Xiaomi Mi Scooter Pro 2 model. The battery pack is a series-parallel combination called 10S4P. 10S4P consists of a total of 40 18650 cylindrical cells, with 10 cells connected in series and each series group consisting of 4 cells connected in parallel. This configuration creates a battery pack with a capacity of ~12.4 Ah, operating in a voltage range of approximately 36–42 V (nominal ~37 V). Parallel groups are created by connecting the positive and negative terminals of four cells to each other using nickel strips in a solderless spot welding method, thereby increasing the battery capacity. To increase the voltage, groups of four cells are connected in series. Each series connection point is provided by nickel strips connecting the positive and negative poles of adjacent cell groups. This allows a single battery pack with the desired voltage and capacity to be obtained with a 10 series $\times$ 4 parallel structure.

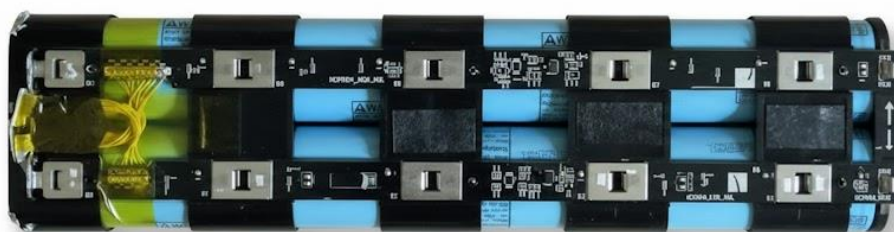


Figure XX. Internal layout of the 10S4P lithium-ion battery pack

Roles of the Battery Management System and Critical Aspects of Reuse

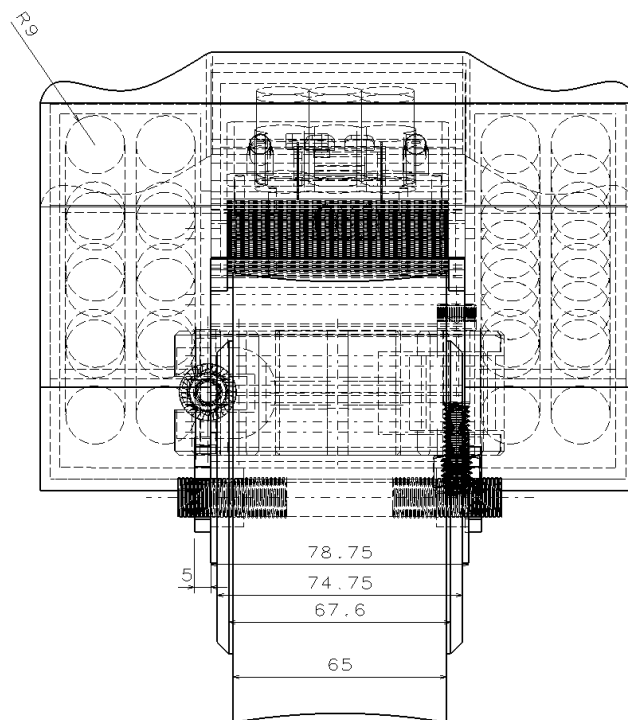
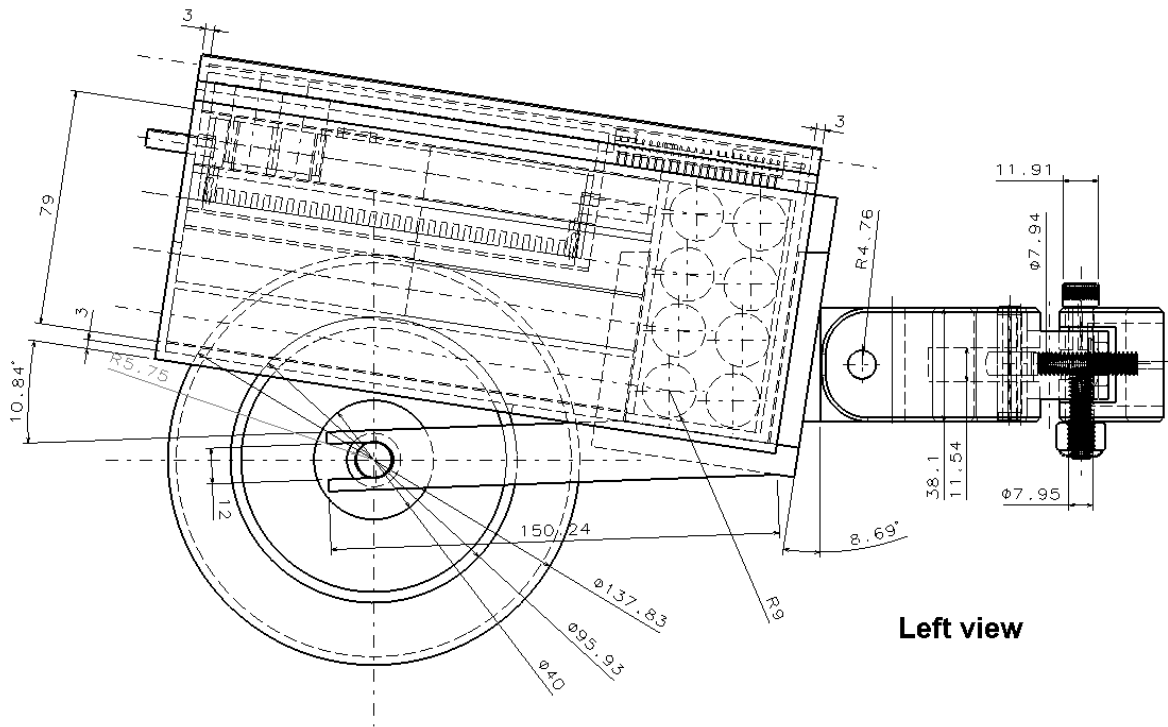
The BMS (Battery Management System) circuit in the original battery pack is the control unit responsible for the safe and efficient operation of the battery. The BMS circuit acts as a kind of “protective brain.” It performs charging, discharging, and balancing functions. It also continuously monitors the battery status to provide protection against extreme conditions. It extends battery life and ensures safe driving. For example, when the BMS circuit detects overcharging, over-discharging, high current, short circuit, and high/low temperature conditions, it cuts off the current to protect the cells. It ensures that all cell groups remain at the same voltage level by performing cell balancing. This balancing is performed, particularly in the final stage of charging, when some cell groups are ahead of others, by dissipating excess energy through resistors. The BMS monitors the voltage of each cell group in series separately.

Technical Drawing

Top view technical drawing of a mechanical component. The drawing shows a rectangular main body with a central circular feature and a complex, multi-ported bottom section. Key dimensions and labels include:

- Overall width: 52
- Overall height: 153.42
- Top edge dimensions: 19.77, 77.6, 6.2
- Central circular feature: R7.82
- Bottom section dimensions: 14.61, 27.94, 22.94, 15.5, 6.35, 66.24
- Bottom section radii: R50.8, R33.15, R3.98, R25.4
- Bottom section hole diameter: $\phi 7.94$
- Bottom section hole offset: 14.29
- Bottom section hole radius: R1.95
- Bottom section hole offset: 22.74
- Bottom section hole radius: R0.08

Figure 38- Technical Drawings of the Main Body – Top View



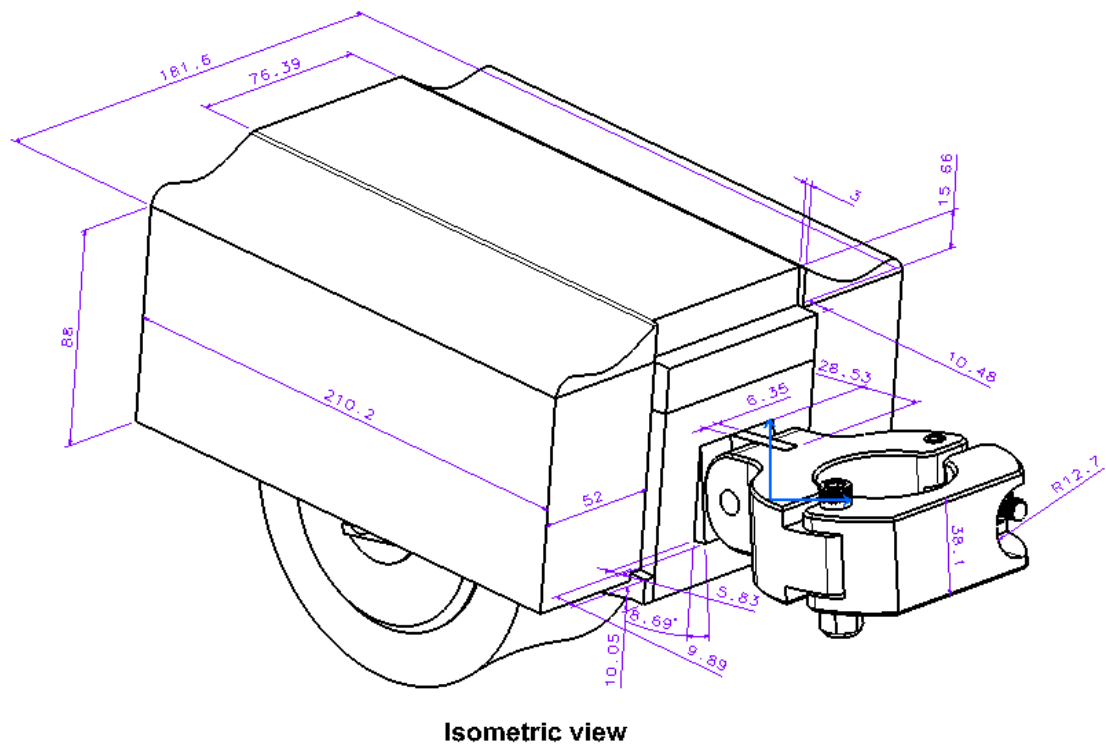


Figure 41 - Technical Drawings of the Main Body – Isometric View

2. Calculations & Simulations and Analysis

Calculations

Power & Torque Analysis

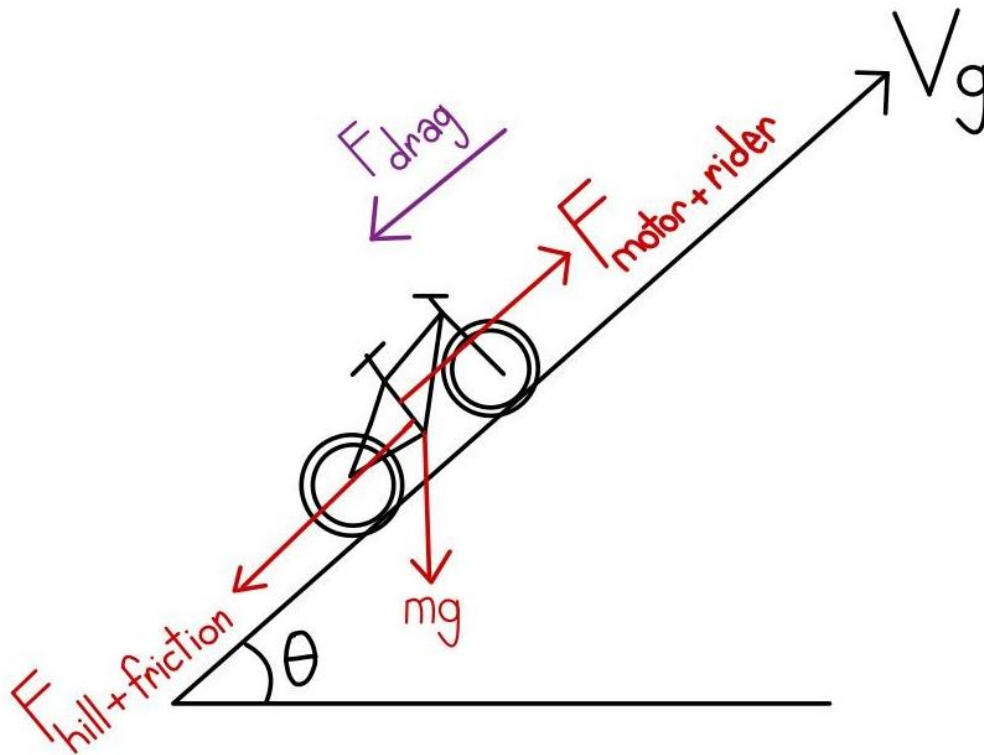


Figure 42- FBD of a bike going uphill

$$P_{\text{tot}} = P_{\text{drag}} + P_{\text{hill}} + P_{\text{friction}}$$

The key focus here should be on how energy is distributed. When we're trying to get work done, there are a few major obstacles that can get in the way.

Air resistance stands out as the most unpredictable element it can turn into our worst enemy at times, but in other situations, it might actually help us out. That's why it's crucial to check the air's relative speed.

Air density (1.225 kg/m^3) (ρ_{air})

Drag coefficient is 0.12 between 1.1 (C_d)

$$F_d = \frac{1}{2} C_d \rho_{air} A (v_g + v_{air})^2$$

So what is the power required to move a bicycle traveling at v_g ?

$$P_{drag} = \frac{1}{2} C_d \rho_{air} A (v_g + v_{air})^2 v_g$$

We also find the power required to carry the weight in a similar way by multiplying the speed by the opposite force exerted by the bicycle in the direction of travel.

$$P_{hill} = M g \sin(\theta) v_g$$

Here, we find the power generated by rolling resistance. Friction is the factor. Here, C_r is the rolling resistance coefficient that varies due to friction effects (between 0.002 and 0.012).

$$P_{fric} = M g C_r v_g$$

Here, we can simply calculate the force required. The greater the value, the greater the acceleration of the object.

$$P_{tot} \leq P_{pedal} + P_{motor}$$

While the sections we've covered so far have all been framed around force and power concepts, which stem from combining force and velocity parameters, in the second part, we'll shift our focus to examining these through torque and angular velocity, tackling the topic purely in terms of torque.

What should the total torque be?

The total resistive torque that the motor must meet is;

$$T_{load} = T_{drag} + T_{hill} + T_{friction}$$

Torque of Air Resistance

$$T_{drag} = \frac{1}{2} C_d \rho_{air} A (v_g)^2$$

Torque of Gradability

$$T_{hill} = M g \sin(\theta) v_g$$

Torque of Rolling Resistance

$$T_{fric} = C_r M g$$

PARAMETERS

$$m_{rider} = 65 \text{ kg}$$

$$m_{bike} = 3 \text{ kg}$$

$$m_{kit} = 5 \text{ kg}$$

$$M = 73 \text{ kg}$$

$$C_r = 0.002 \text{ kg}$$

$$r = 21 \text{ inch} = 0.267 \text{ m}$$

$$g = 9.81 \frac{\text{m}}{\text{s}^2}$$

$$C_d = 0.12$$

$$\rho_{air} = 0$$

$$A = 0.5 \text{ m}^2$$

$$\rho_{air} = 1.225 \text{ kg/m}^3$$

$$v_g = 25 \text{ km/h} = 6.94 \text{ m/s}$$

$$v_{air} = 20 \text{ km/h} = 5.56 \text{ m/s (Ters yönde)}$$

$$\theta = 0^\circ$$

Power of Air Resistance

$$P_{drag} = \frac{1}{2} * 0.12 * 1.225 * 0.5 * (6.94 - (-1.39))^2 * 6.94 = 17.70 \text{ W} =$$

Power of Gradability

$$\theta = 0^\circ \quad \text{so that} \quad P_{hill} = 0$$

Power of Rolling Resistance

$$P_{fric} = 73 * 0.002 * 9.81 * 6.94 = 9.93 \text{ W}$$

$$P_{tot} = P_{drag} + P_{hill} + P_{friction} = 27.6 \text{ W}$$

Air Resistance Torque

$$T_{drag} = \frac{1}{2} 0.12 * 1.225 * 0.5 * (6.96)^2 = 0.68 \text{ Nm}$$

Gradability Torque

$$T_{hill} = 73 * 9.81 * \sin(0) * 6.96 = 0$$

Rolling Resistance Torque

$$T_{fric} = 0.002 * 73 * 9.81 = 0.38 \text{ Nm}$$

$$T_{load} = T_{drag} + T_{hill} + T_{friction} = 1.06 \text{ Nm}$$

$$v_g = 10 \text{ km/h} = 2.78 \text{ m/s} \quad \& \quad v_{air} = 10 \text{ km/h} = 2.78 \text{ m/s} \quad \& \quad \theta = 18^\circ = 10^\circ$$

These values simulate the scenario that a cyclist climbing a hill at a 10° angle might experience under optimum system design and average environmental conditions.

Note: Weather sometimes works against us and sometimes helps us. That's why we eliminated the weather parameter.

Torque of Air Resistance

$$P_{drag} = \frac{1}{2} * 0.12 * 1.225 * 0.5 * (2.78 - (2.78))^2 * 6.94 = 0 \text{ W}$$

Power of Gradibilty

$$\theta = 10^\circ \quad \text{so that} \quad P_{hill} = 73 * 9.81 * \sin(10) * 2.78 = 345.53 \text{ W}$$

Power of Rolling Resistance

$$P_{fric} = 73 * 0.002 * 9.81 * 2.78 = 3.98 \text{ W}$$

$$P_{tot} = P_{drag} + P_{hill} + P_{friction} = 349.51 \text{ W}$$

Torque of Air Resistance

$$T_{drag} = \frac{1}{2} * 0.12 * 1.225 * 0.5 * (0)^2 = 0 \text{ Nm}$$

Torque of Gradility

$$T_{hill} = 73 * 9.81 * \sin(10) = 12.28 \text{ Nm}$$

Torque of Rolling Resistance

$$T_{fric} = 0.002 * 73 * 9.81 = 0.38 \text{ Nm}$$

Total Torque

$$T_{load} = T_{drag} + T_{hill} + T_{friction} = 33.57 \text{ Nm}$$

	Simulation 0 Requirements	Simulation 1 Requirements	Data Sheet
P	27.6 W	349.51 W	300W – 600W
T	1.06 Nm	33.57 Nm	16Nm(?)

Required Tangential Force

$$F_{hill} = 73 * 9.81 * \sin(10) = 124.29 \text{ N}$$

$$F_{friction} = 73 * 0.002 * 9.81$$

$$F_{hill} + F_{friction} = F_{total} = 125.72 \text{ Newton}$$

First of all, our requirements and available data are clear, but we can adjust these values. With our chosen motor and driver combination, we can increase the nominal power output of a 300W motor by 10% to 20%. However, beyond this point, we may need to develop new cooling.

The selected driver is quite powerful, allowing us to utilize the motor's full performance. Furthermore, the results of the thermal tests indicate that cooling can be improved with simple motor modifications.

For us, the $300W * 20\% = 360W$ value is the maximum safe zone that can be achieved by simply changing the driver and is sufficient.

Real Life Simulation

Parameter	Value	Rationale
Total Mass m	125kg	Assumption: 100kg rider + 25kg e-bike.
Gravity Acceleration g	9.81 m/s ²	Standard physical constant.
Rolling Coefficient C_{rr}	0.008	Typical tire/road coefficient for asphalt.

Air Density ρ	1.225 kg/m ³	Standard air density at sea level, 20°C.
Frontal Area A_f	0.6 m ²	Assumption for Samim Unlusoy. $0.7 \times width(0.5m) \times height(1.71m)$
Drag Coefficient C_D	0.75	Aerodynamic shape factor for a semi-recumbent position.
Wheel Radius r_w	0.35 m	Represents a 28-inch (700c) wheel size ($\approx 0.35m$ radius).
Wind Speed v_{air}	0 m/s	No einf. $v_g + v_{air} = v_g$
Assistance cons $K_{assistance}$	0.2	Based on tour mod
Rotary Mass Factor φ	1.1	Typically urban bike
Efficiency Constant n_m	0.88	Direct gear(Samim Unlusoy's note)
Wheelbase	1.15m	Bosch eBike Systems – Engineering Guidelines (2019)
Front–Rear Static Load Distribution	a=0.7 b=0.45	SAE & Bosch & Delft University Bicycle Dynamics
CG Height	0.6 m	SAE & Bosch & Delft University Bicycle Dynamics

BMS boundary acceptance	5/30 kph/W 200W	$\frac{Min}{Max}$ regen acceptance
-------------------------	-----------------------	------------------------------------

Table 11 - Simulation Setup Parameters

According to regulations the assistance constant determines how much power our engine should produce. While $P_{motor}(1 + K_{assistance}) = P_{b-out}$ represents the power delivered to the wheel, the expression $\frac{T_{mt}}{K_{assistance}} + T_m = T_w$ represents the torque going to the wheel. Similarly, the power coming from regeneration can be calculated. Below are the forces that prevent us.

Additionally, instead of a conventional mechanical braking mechanism, the braking system is built entirely on regenerative braking on the rear wheel (without emergency braking.). The main reason for this is that the motors used for e-bikes can generate sufficient counter force for most normal braking situations. The calculations presented in the following pages will show that even under the most optimal conditions (dry road...) what determines the braking distance is not the braking force itself but wheel slip, in other words the brake force distribution between the wheels. However, the calculations we will perform later will show that most of the energy that could be recovered from breaking is dissipated as heat through the mechanical brakes on the front wheel. Since creating a front-wheel-drive bicycle introduces disadvantages, the braking algorithm will follow Ehsani's rule in which braking demands up to 0.2 g are handled by regenerative braking on the rear wheel; in any other situation, we will distribute braking between the mechanical front brake and the regenerative rear brake using the defined k_{bf} constant. A detailed explanation is provided in the k_{bf} section.

The formulas shown below represent resistive forces for us, meaning that even if the user does not press the brake, the bicycle slows down due to these external effects. These effects have been taken into account in our calculations.

When vehicle dynamics are calculated, the calculations below are essentially performed, and the reason for the vehicle's acceleration or deceleration in the driving cycle clearly emerges. If the vehicle is decelerating, the reason for this is either that it is going uphill or it is performing regen. This means both formulas cannot take a positive value at the same time. The easiest way to apply these formulas is to look at the sign of the required Power.

$$P_{b-out} = \frac{V}{n_t n_m} [\overleftarrow{M\varphi \frac{dV}{dt}} + \overleftarrow{\frac{1}{2} C_d \rho_{air} A (v_g + v_{air})^2} + \overleftarrow{Mg \sin(\theta)} + \overleftarrow{Mg \times (f_r)}]$$

$$P_{b-in} = \alpha V \times n_t n_m [-M\varphi \frac{dV}{dt} - \frac{1}{2} C_d \rho_{air} A (v_g + v_{air})^2 + Mg \sin(\theta) - Mg \times 0.01 \times (1 + \frac{V}{160})]$$

To find the required max power, motor selection must be made by considering the $P_{motor} = P_{human} K_{assistance}$ calculation. For this, it is necessary to look at the largest positive value of the power-time graph; it must be equal to the $P_{motor}(1 + K_{assistance}) = P_{b-out}$ value. The resulting value must meet the nominal value of the motor to be selected. When calculating regen, our value symbolized by α gives us electrical efficiency. The α value is actually empirical and dynamic, therefore we must use the table in appendix 1.1 in the simulation.

This profile includes slow urban rides and intermediate accelerations, and there are actually two distinct riding styles. The first hour is generally flat and fast. Towards the end, there are power demands of 200-300W. The e-bike profile was taken with a total mass of 125 kg and parameters such as an aerodynamic drag coefficient of 0.75 (required parameters are specified above). Speeds below 5 km/h and power below 30W were considered dead zones. Battery charging power was limited to 200W, which is our BMS limit. In the ideal scenario, some peak values are observed. 200-watt values are common peaks for us. It can occasionally reach values like 300 watts, but these values are within acceptable limits for a nominal 300-watt and peak 600-watt electric motor. The primary reasons for these sudden spikes in the graph may be GPS data and demand for hills and potholes. The area above the X-axis of this data allows us to calculate the bike's total energy needs, while the area below it allows us to calculate the Regen energy.

Software In the Loop

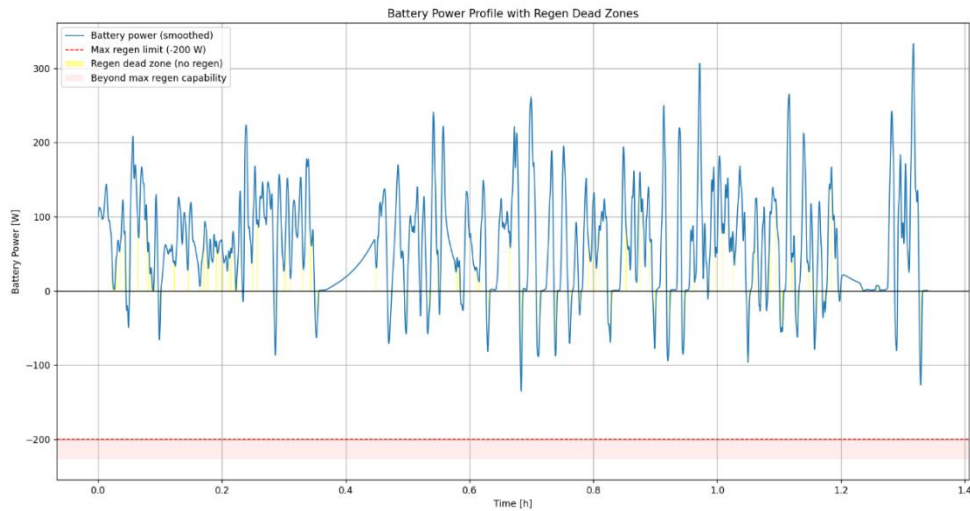


Figure 43- Battery Power Profile

The figure shows the kbf constant analyzed through dynamic evaluations and accompanying graphs. To achieve optimum balance in the regeneration analysis, the Brake Force Distribution is set to 0.60. If braking results in a deceleration below 0.2g, braking can be performed with 100% regeneration to maximize energy efficiency. In harder braking above 0.2g, 60% of the load is transferred to the front mechanical brake to prevent premature lock-up of the rear wheel. In the dynamic braking model, the battery charge acceptance limit remains constant at 200W. It's illogical to move the engine forward without also moving the bike's center of gravity forward, because just like there's a braking map, there's also a traction map. And the result is reduced performance.

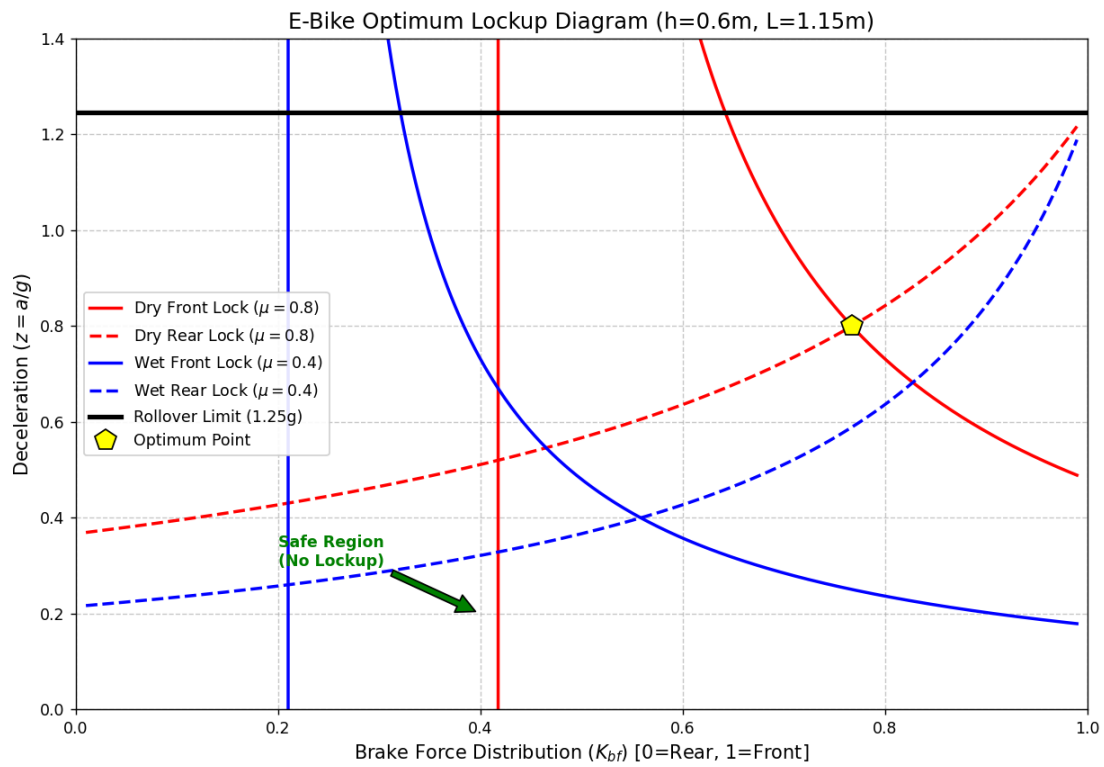


Figure 44- Optimum Lockup Diagram

This efficiency map is considered standard for small-scale electric motors in the IEEE and SAE literature. It was synthesized using loss modeling methods. The graph was not taken from a pre-existing source; the motor's copper losses, iron losses, and friction/wind losses were calculated separately for each area using physical equations, resulting in an efficiency map adapted to the Bafang 250W motor scale.

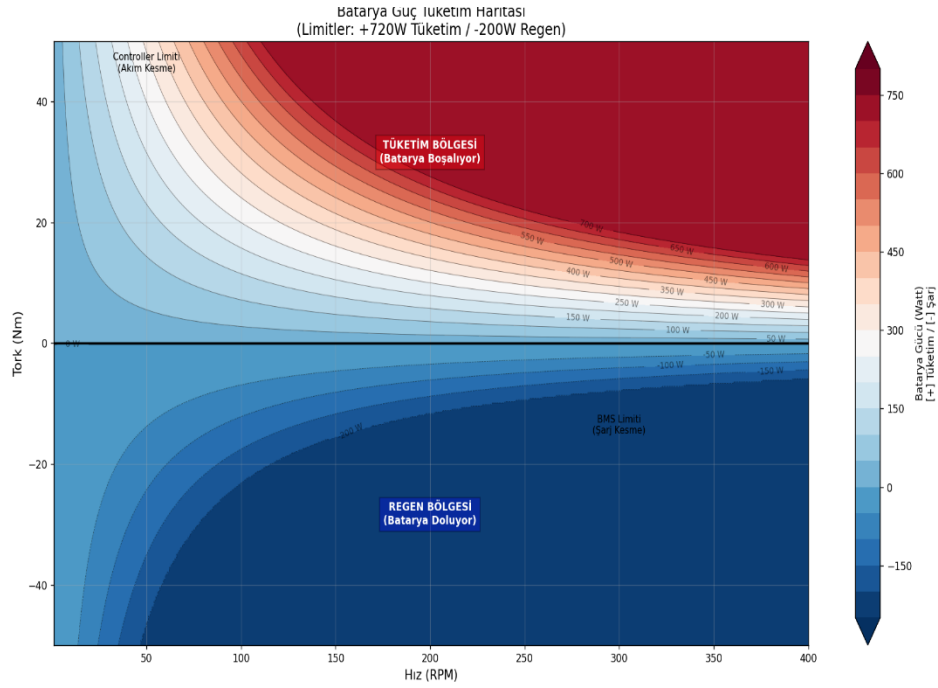


Figure 45 - Battery Power Consumption Map

It's impractical to estimate the charge status (SoC) of a charged battery by simply reading its voltage. A better indicator is to measure the open-circuit voltage after the battery has rested for a few hours. Temperature also affects all products that use batteries, but today we'll focus on its effect on the SoC. As this chart clearly shows, as the maximum voltage applied to the cell increases, the battery reaches higher capacity, but the charging time lengthens, and it becomes more difficult to reach full saturation. For example, a cell charged up to 4.20 V will reach full capacity, while lower voltages like 3.80 V will charge the battery faster but retain less capacity. This demonstrates a trade-off between battery life and performance.

```
def __init__(self):
    # --- A. ŞARJ VERİM HARİTASI (Coulombic Efficiency vs SoC) ---
    # Batarya doldukça iç direnç artar, iyon hareketi zorlaşır -> Verim düşer.
    self.soc_pts = np.array([0.0, 0.1, 0.5, 0.8, 0.9, 0.95, 1.0])
    self.eff_pts = np.array([0.88, 0.96, 0.96, 0.92, 0.80, 0.60, 0.10])
    self.eff_interp = interp1d(self.soc_pts, self.eff_pts, kind='linear', fill_value="extrapolate")

    # --- B. BMS AKIM LİMİTİ (CC-CV Charging Curve) ---
    # %80'e kadar tam akım, sonra voltaj sabitlenir akım düşer.
    self.limit_soc_pts = np.array([0.0, 0.75, 0.85, 0.95, 1.0])
    self.limit_factor_pts = np.array([1.0, 1.0, 0.60, 0.10, 0.0])
    self.limit_interp = interp1d(self.limit_soc_pts, self.limit_factor_pts, kind='linear', fill_value="extrapolate")
```

Figure 46 - Code for Battery Map

Mesafe	: 19.16 km
SoC Değişimi	: %60.0 -> %38.2

Bataryadan Çekilen	: 120.10 Wh
Geri Kazanılan	: 11.62 Wh (Regen)
Mekanik Fren Kaybı	: 1.76 Wh

NET TÜKETİM	: 108.48 Wh
VERİMLİLİK	: 5.66 Wh/km

Figure 47 - Efficiency Calculations

The validation of our simulation model was carried out using the industry-standard Grin Technologies Motor Simulator with Bafang G62 motor data. As seen in the reference graph, the motor's efficiency at its operating point is 84.3 percent, and these values are consistent with our reference standards of 85 percent peak efficiency and 99 percent accuracy. When looking at the consumption data in the simulator, do not be misled by the higher consumption values. the values we obtained represent urban and average fuel consumption.ith the developed Digital Twin model, dynamic BMS charge limits based on the battery state of charge (the CC-CV curve) were integrated and simulated to perform the calculations. Through this multilayer structure, mechanical brake pad losses, motor thermal inefficiencies, and battery internal resistance losses were separated, and in the real driving simulation of the system, the net energy consumption and regenerative recovery capacity were validated with high accuracy at approximately 4.55 Wh/km. As the recovered energy increases, the energy directed to the mechanical brake decreases because some data points fall below 200 W (the BMS maximum limit).

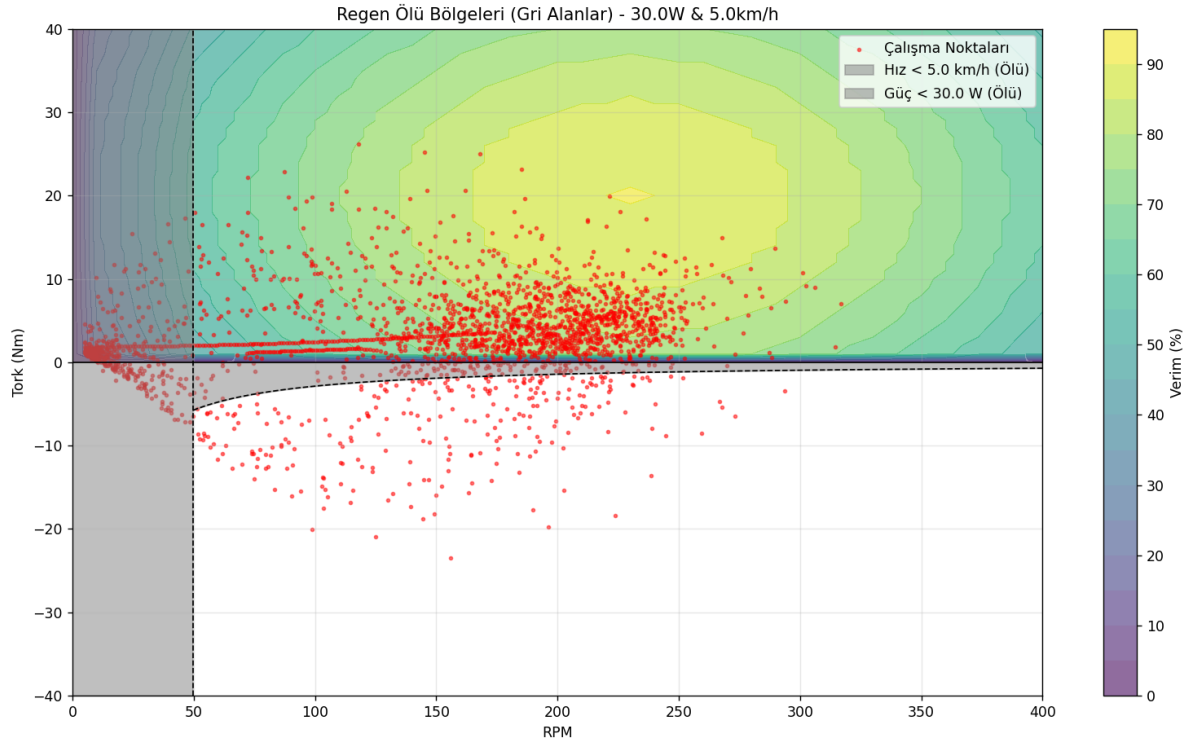


Figure 48 - Regen Dead Zones

Maximum Speed: 43.54 km/h

Gradability: %10.73

ACCELERATION TIME:28.10 seconds (40 km/h)

Component	Parameters	Simulation	Choosing parameters
Motor	Peak Power	530 W	600 W
Battery	Discharge Current	14.7 A (1.05C)	28.0 A (2Crate)
Battery	Regen	5.5 A (0.39C)	7.0 A (0.5Crate)
Controller	Max Current	14.7 A	20 A
Energy	Range	19.2 km	~80-100 km
Battery	18650 Lithium battery*40	105Wh	504Wh

Table 12 - Simulation Results

Roller Analysis

$$L_{10}(rev) = \left(\frac{C}{P}\right)^p * 10^6$$

$$p = \frac{10}{3} \quad (\text{for type of roller})$$

$$\frac{[(2\pi r) * rpm]}{60(hours)} = 25 \frac{km}{h} = 6.94 \frac{m}{s} \quad r = 6.61cm = 0.0661m$$

$$w = n = 1330rpm$$

$$\frac{L_{10}(rev)}{60n} = L_{10h} = \frac{\left(\frac{C}{P}\right)^p * 10^6}{60n}$$

$$F_r = MC_r g = 1.42 N$$

$$F_n = Mg \cos(\theta) = 716.3 N$$

$$\text{Factor of Safety} = 2$$

$$20,000 \text{ hours}$$

The main thing we need to pay attention to here is the load F_n value because our analysis is based on the load and the large value determines our dominant character.

When we substitute the given values, two C values emerge and we can understand the dominant force by looking at them.

$$C = P * (L_{10h} * 60 * n)^p * 6/p$$

$$C = 6.22N$$

$$C = 3.14kN$$

We can find the desired value by multiplying the larger value by the Factor of safety.

$$C_{all} = 6.28 \text{ kN}$$

However, these values are easily found in the catalog. We can use our calculations directly, because hours and C are directly proportional. We can calculate the final estimated lifespan.

Now, we will look up the necessary values in the Dynamic Load Rating Catalog and conduct our analysis.

We can find the estimated bearing life by dividing the C_{10} value of the bearing selected here by our value and multiplying by 10,000 hours.

Table 11-3

Dimensions and Basic Load Ratings for Cylindrical Roller Bearings

02-Series					03-Series			
Bore, mm	OD, mm	Width, mm	Load Rating, kN		OD, mm	Width, mm	Load Rating, kN	
			C ₁₀	C ₀			C ₁₀	C ₀
25	52	15	16.8	8.8	62	17	28.6	15.0
30	62	16	22.4	12.0	72	19	36.9	20.0
35	72	17	31.9	17.6	80	21	44.6	27.1
40	80	18	41.8	24.0	90	23	56.1	32.5
45	85	19	44.0	25.5	100	25	72.1	45.4
50	90	20	45.7	27.5	110	27	88.0	52.0
55	100	21	56.1	34.0	120	29	102	67.2
60	110	22	64.4	43.1	130	31	123	76.5
65	120	23	76.5	51.2	140	33	138	85.0
70	125	24	79.2	51.2	150	35	151	102
75	130	25	93.1	63.2	160	37	183	125
80	140	26	106	69.4	170	39	190	125
85	150	28	119	78.3	180	41	212	149
90	160	30	142	100	190	43	242	160
95	170	32	165	112	200	45	264	189
100	180	34	183	125	215	47	303	220
110	200	38	229	167	240	50	391	304
120	215	40	260	183	260	55	457	340
130	230	40	270	193	280	58	539	408
140	250	42	319	240	300	62	682	454
150	270	45	446	260	320	65	781	502

Figure 49 - Table of Dimension and Basic Load Ratings for Cylindrical Roller Bearings

As you can see, all the O-2 and O-3 series products we have meet our requirements.

The bearings were analyzed and no incompatibility was detected in the bearing integrated into the engine

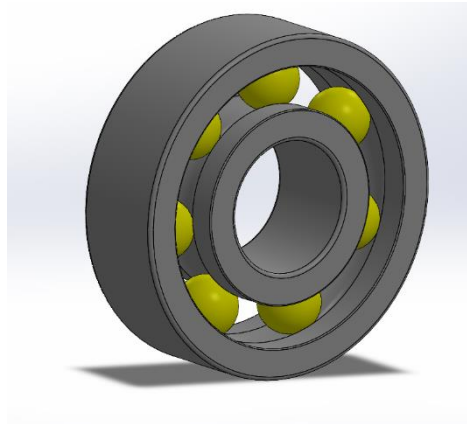


Figure 50 - Roller

Controller Design

To design a controller, we must have a detailed data sheet. Controllers, which can compensate for variables such as weight, tire size, and inclination, must have a thorough understanding of the system's dynamics. However, data sheets for the components used here are lacking. It's necessary to test and determine parameters such as current, resistance, and torque curves. This method also reveals the specific characteristics of the motor. Examining the control unit clearly reveals the need for a microcontroller. This microcontroller houses our sensory organs and sends commands to the internal controller based on changes in our environment.

We can get real-time data by placing the correct sensors on the pins on the microcontroller.

AS5600 VCC → 3.3V (ESP32 3.3V)
AS5600 GND → GND
AS5600 SDA → GPIO21, SCL → GPIO22
INA219 VCC → 3.3V, GND → GND, SDA/SCL → GPIO21/22 (same I2C line)
Throttle Vout → GPIO34 (ADC), Vcc throttle → Battery 5V/3.3V (according to the sensor), GND → GND
Brake switch one of the out GND, other GPIO26 (INPUT_PULLUP). When it activates, it rotates to the GND.
ESP32 TX2(GPIO17) → VESC RX, ESP32 RX2(GPIO16) ← VESC TX

- Battery + / - → VESC power input (VESkin).

Table 13 - Wiring Table

Controller Behaviour

There are codes on how to run such a system in the appendix.

We can make detailed adjustments to our internal controller using the VESC tool. We need this controller because it responds much faster than the external controller. It provides very clear feedback. Furthermore, the external controller lacks the ability to drive the motor, whereas the VESC is an advanced controller and can issue precise commands.

How should a mathematical model be derived? While we can't understand the physics of the engine based on the data we have, we can understand the physics of the bicycle, and the bicycle is our clearest connection to the environment. The physical movements of the bicycle directly affect the engine. You can access some of the necessary formulas above.

$$V = RI + L \frac{dI}{dt} + K_e w_m \quad \mathcal{L} \quad I(s) = \frac{V(s) - K_e \Omega_m(s)}{R + sL}$$

$$J \frac{dw}{dt} = I_e - B w_m - I_c \quad \mathcal{L} \quad I(s) = \frac{(sJ + B) \Omega_m(s)}{K_t}$$

$$G(s) = \frac{K_t}{s^2 L J + s(LB + R J) + (RB + K_t K_e)}$$

A parametric solution is presented here. Our motor is three-phase, with a 120-degree phase angle between each phase. The motor includes a Hall sensor, which will be our feedback. Parameters should be written in place after testing. Furthermore, the dominant characteristics of variable parameters in the controller design can be examined, and the K constants can be redesigned to achieve robustness.

Model In The Loop (Updated Version)(There is Last Iteration)

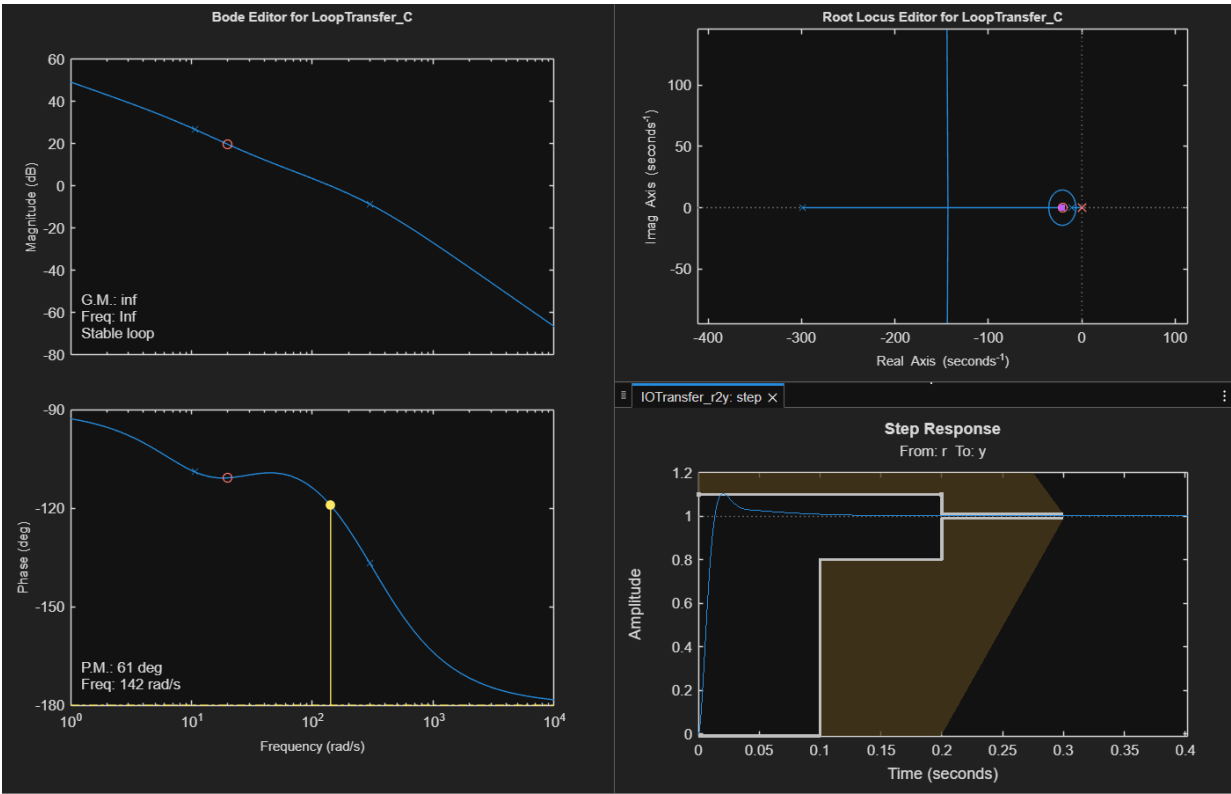


Figure 51 - Bode Editor

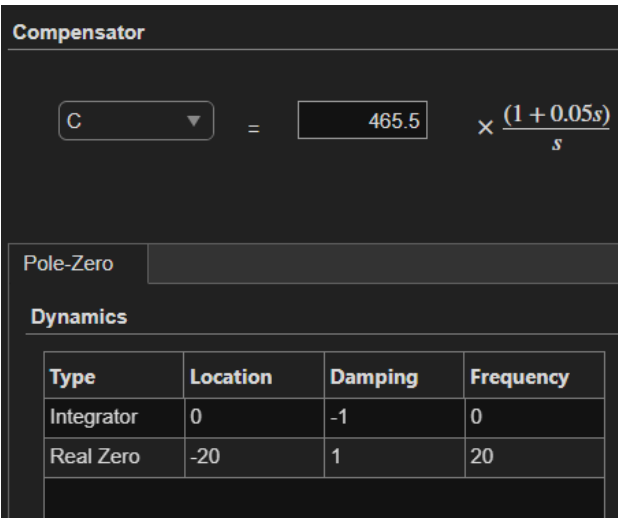


Figure 52 – Compensator

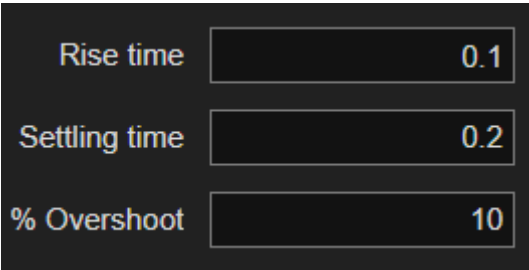


Figure 53 - Reaction Requirements

The steady-state error was eliminated, and the system's Phase Margin reached 61 degrees. This design, with the Step Response graph remaining within all design objectives, is a successful example of a PI controller that is both stable and high-performing.

Simulations

Advanced simulation software, primarily ANSYS Mechanical, is utilized to validate the mechanical integrity, thermal performance, and compliance of the electric bicycle conversion kit with critical design constraints (e.g., safety, noise, and efficiency). These simulations serve as an important verification step, complementing manual calculations and MATLAB-based code analyses.

1. Structural Analysis (Static)

A Static Structural analysis was performed to investigate the mechanical response of the final assembly, especially focusing on the clamping mechanism and overall chassis structure under maximum expected loads.

Objective: To determine stress distribution and total deformation of the components under static loading conditions, ensuring the design has an adequate Factor of Safety against material yielding.

Load Case: The 500N force applied to the middle axis that connects our drive rollers both ends on y-axis to simulate our bike tire's reaction force.

We placed the fixed supports on the inner top surface of the hinge that connects our kit to the bicycle frame. Our kit connects to the clamping mouth there, so we can accept that as a fixed support.

The solution log indicates that the model contains 103,519 total nodes and 99,581 total elements which is still in range for ANSYS Student version. Average meshing quality is 0.84823 and that value is okay for a mechanical analysis

Results: The results show that our max total deformation is 0.085658 mm and the average is 0.039528 mm. Equivalent (von-Mises) Stress is maximum 58 Mpa and average 0.36895. The factors of safety we obtain based on this maximum are as follows:

3.691 at its worst and average is 14.992

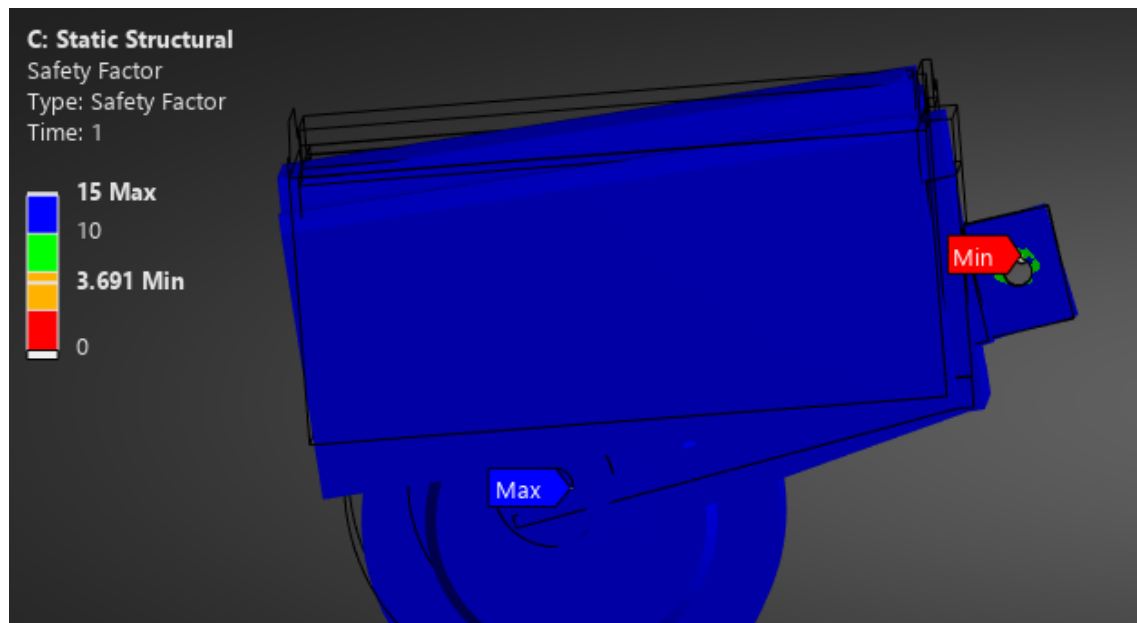
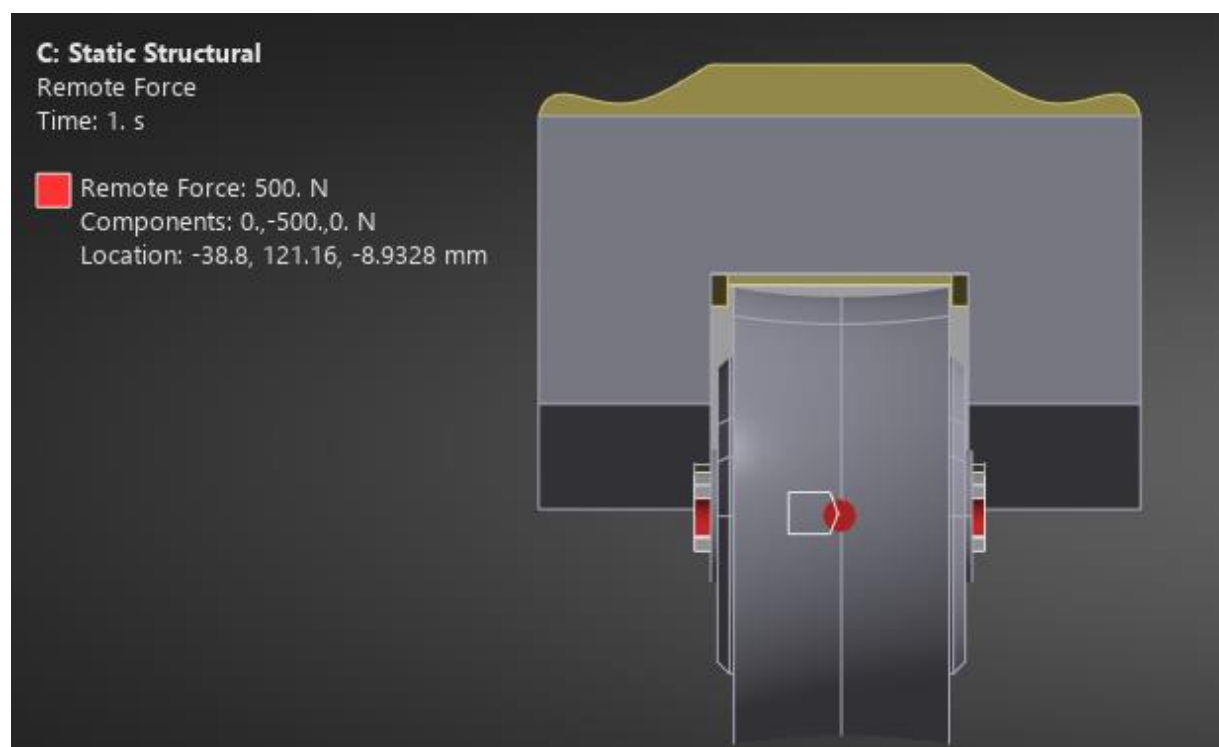


Figure 54 - Safety Factor of Static Structure



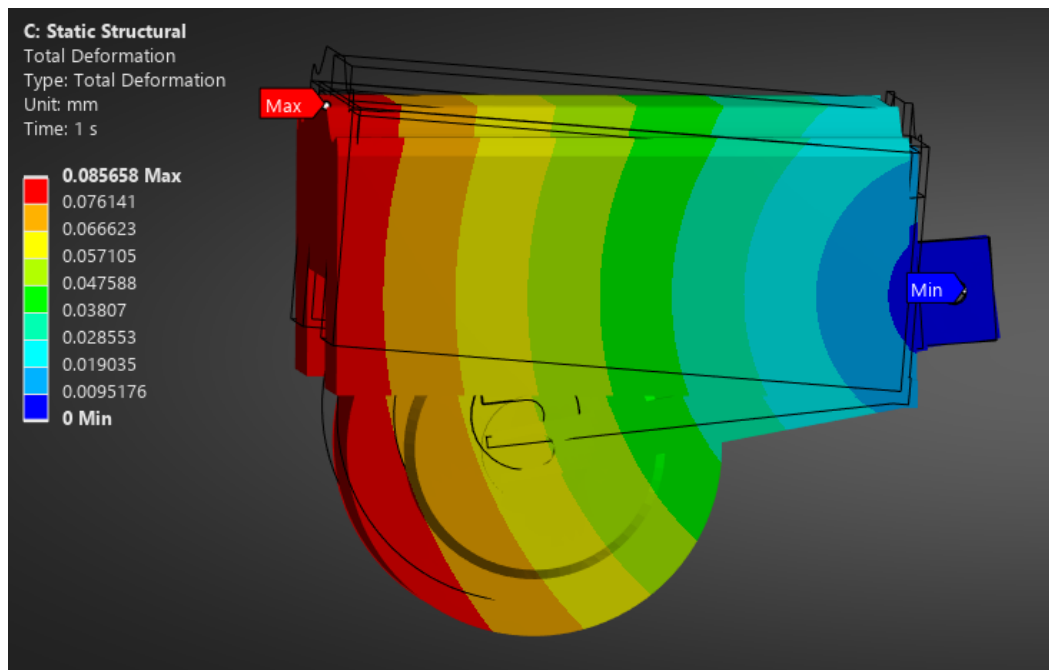


Figure 57 - Total Deformations on the Static Analysis

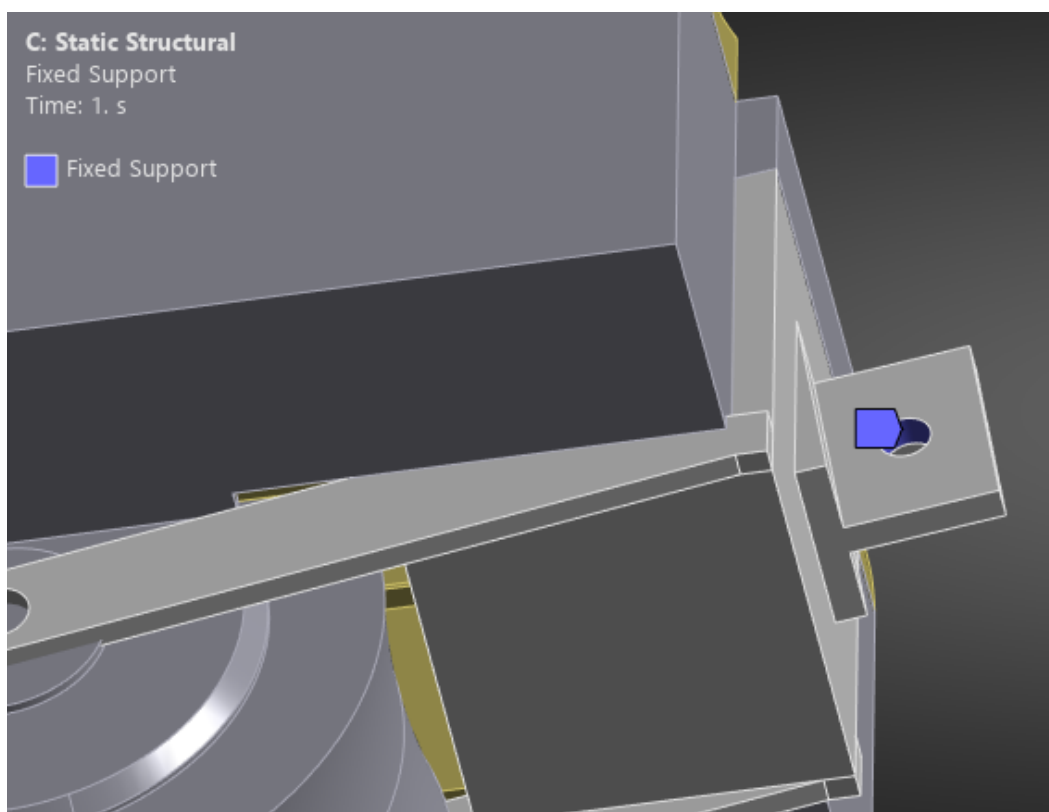


Figure 56 - Where the fixed support is

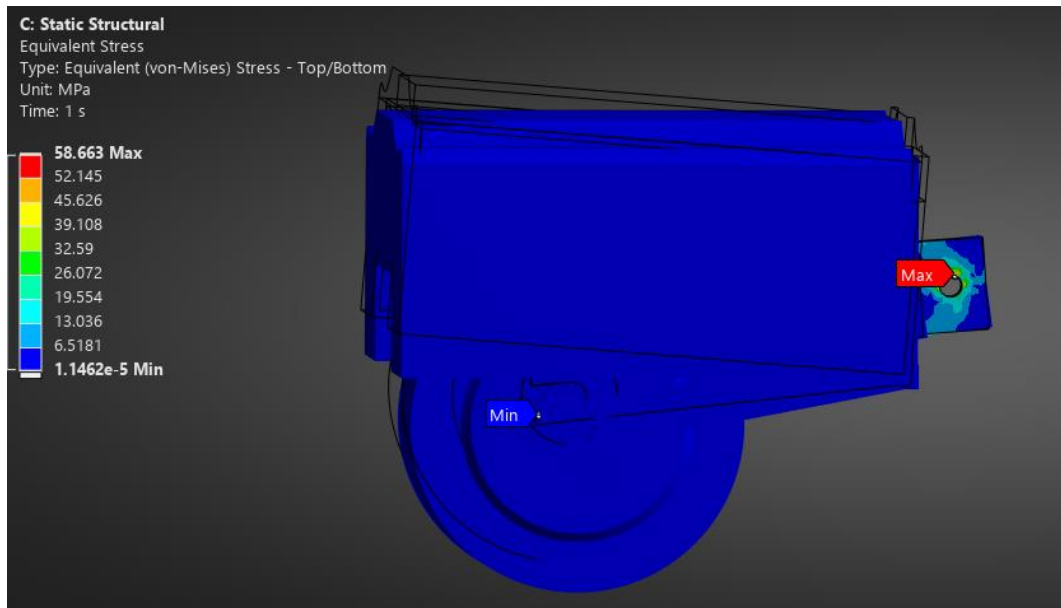


Figure 58 - Equivalent (von-Mises) Stress

3. Transient Thermal Analysis

Objective: A Transient Thermal analysis is essential for determining the maximum temperature reached by the internal electronic components, specifically the battery and VESC controller, over the expected operating time, ensuring the system operates within optimal thermal limits.

Heat Load: The primary heat source is Battery. We applied a 35 W heat flow to the battery protection's inner surfaces to simulate battery heat generation and we applied a 15 W heat flow to the VESC's inner bottom surface to simulate VESC chip's heat generation. ESP32 also generates heat but since it is too low, it can be neglected.

That simulation's total step time is 1800s. That means this setup goes on for 30 minutes. What we wanted to with that analysis is to learn about what is the worst outcome for a 30 minute driving cycle.

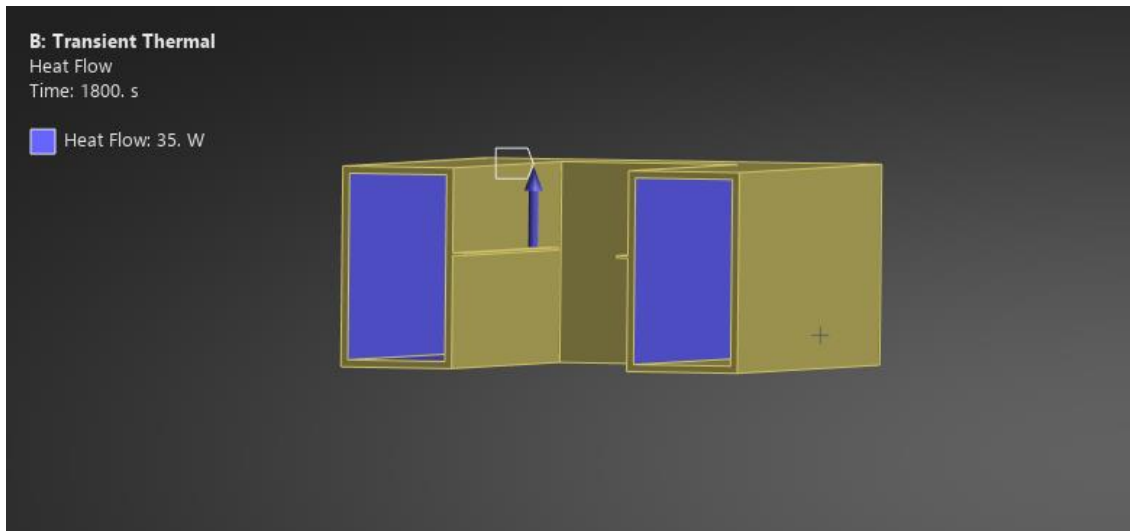


Figure 59 - 35 W Heat Flow

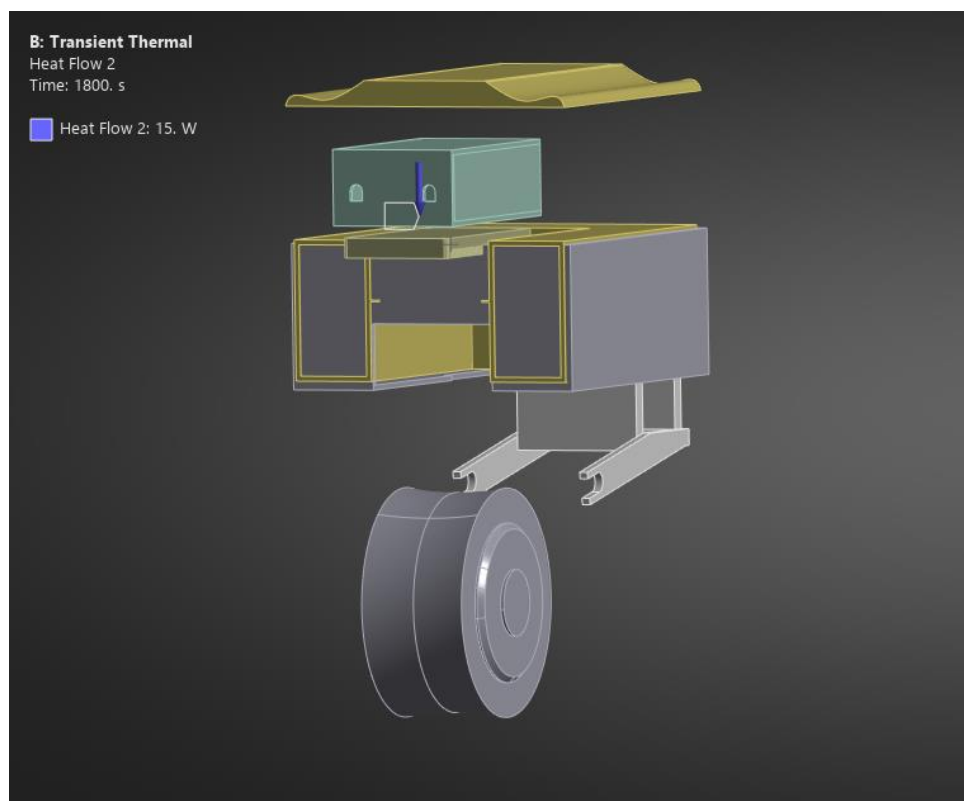


Figure 60 - VESC Heat Flow

Results: After 1800 seconds, the maximum temperature reaches approximately 57 degrees, while the average temperature is 27 degrees.

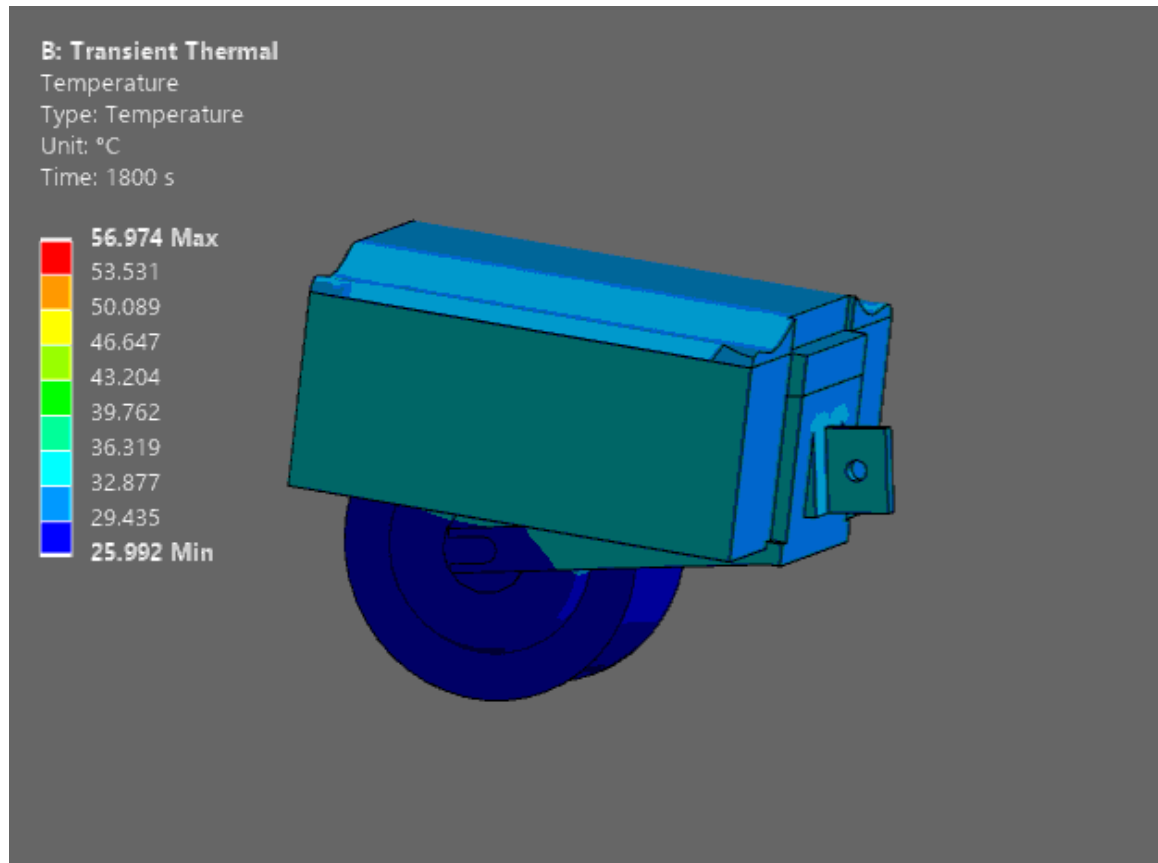


Figure 61 - Transient Thermal Analysis Result

3. Material Selection and Manufacturing Procedure

Material Selection

The choice of materials has a direct impact on the performance, durability, safety, and cost of our project. For each component, different materials examined from different aspects.

Casing

Durability and Impact Resistance

It must exhibit high resistance to impacts, drops, and wear that may occur during use. This is important both for protecting the components and increasing the product's lifespan.

Lightweight

Lightweight materials should be preferred to minimize the total weight of the kit and avoid negatively affecting the bicycle's riding dynamics.

Water and Dust Resistance

Since it will be used primarily in outdoor environments, the casing must provide full protection against water and dust ingress in accordance with IP standards. This will prevent electronic malfunctions and corrosion.

Thermal Conductivity and Heat Management

To ensure that heat generating components such as the motor and battery inside remain at optimal operating temperatures, the housing must have sufficient heat transfer capability.

Processability and Ease of Production

The selected material should be easily processable using the planned production methods.

Based on our project requirements, We selected ABS for the casing.

Why ABS?

ABS has near excellent energy absorption, making it suitable for portable and mobile equipment. ABS can easily be processed with injection molding and is also suitable for 3D Printed prototypes. ABS is also lighter than metal alternatives. That helps in keeping the total product weight low. It provides good electrical insulation and offers a good balance between performance and price.

Drive Roller

High Friction Coefficient

To prevent slippage in wet or dirty road conditions, it must provide a high and consistent friction coefficient with the bicycle tire, so the torque produced by the motor is efficiently transferred to the wheel.

Wear Resistance

Due to constant contact and friction with the bicycle tire, the material must be extremely resistant to abrasion. High abrasion resistance extends the wheel's lifespan and reduces maintenance requirements.

Vibration and Noise Reduction

The ability to absorb vibrations and noise is important. This increases riding comfort and contributes to the kit's quiet operation.

Temperature Stability

It must be able to maintain its mechanical properties despite heat generated by friction and changing ambient temperatures. For example, it should not soften at high temperatures.

Durability and Long Service Life

The wheel is expected to provide a long service life despite heavy use and environmental factors.

Processability and Formability

The material must be easily machinable to produce the wheel with the desired surface geometry and dimensions with precision.

After reviewing various perimeters, our selection is Polyurethane Resin.

Based on our needs Polyurethane Resin is the perfect option.

Why Polyurethane?

Polyurethane has excellent durability under continuous friction with tire surfaces. Provides good traction under wet conditions, improving torque transfer and start-up performance. The elastic nature of it, reduces vibration and noise and it also has good thermal stability over a wide range of temperature

Manufacturing Procedure

The key point throughout all stages of production is to ensure that components work together without any problems. Some changes will need to be made to the components selected as the basis for this.

Motor

As mentioned in the report, our motor is a brushless hub motor with a diameter of 1322mm. We will source this motor from the Xiaomi Mi 2 Pro model to keep costs down. Therefore, modifications to the motor's dimensions are required. After separating the motor from the rubber, we will need to separate the rotor and stator pair. Thanks to the two components we separate with the help of a magnet, we will have access to all parts of the motor. The stator part comes integrated into the outer core of the motor. After reducing the diameter of the outer shell with the help of a lathe, we will reassemble all the parts. This way, we will achieve our goal.

Battery

The battery section is quite complex and dangerous. Since ready-made battery kits are quite expensive, we will use the NED1004-K 37V - 48V 12.8Ah model that meets our requirements. This battery type is flat 'I' shaped, but the product we want should be 'U' shaped. The work to be done here generally involves the electrical field and should be done by a subcontractor firm. In short, the battery's 5 cell packs must be removed and redesigned in a 2+1+2 configuration, with a 90-degree rotation at each '+' sign. The empty plate on top of the battery packs must be removed and a new U-shaped plate must be integrated. Then, the cell, balance, and sensor cables must be assembled in the same order. Since the volume and surface area of the battery protection will not change, the battery cover must be reshaped. However, the joints created by the redesign of the battery protection must be reinforced.

Chasis

We need a small chassis to secure the motor shaft and battery. This chassis will be made of aluminum alloy. A lathe, vertical drill, file, and welding machine will be used in its shaping.

Outer Casing

To create the outer casing of our design, we will prioritize both ease of manufacturing and sustainability. In line with these objectives, we will produce the technical drawing developed in CATIA using ABS (Acrylonitrile Butadiene Styrene) plastic and a 3D printer. The primary reason for selecting this production method is the nature of Additive Manufacturing, which minimizes material waste and allows complex geometries to be achieved efficiently. The choice of ABS material specifically ensures the final component will possess the necessary mechanical strength, impact resistance, and thermal stability required for a functional casing. All individual parts of the design will be exported separately from the CATIA model, taking into account the optimal print orientation and support structure needed for successful printing. For pre-production preparation, we will upload these segregated parts to the Ultimaker CURA slicing software. Within the CURA application, we will define critical parameters such as filament flow rate, layer height, infill density, and print speed before sending the job to the printer. This stage is crucial for ensuring the precision and tight assembly tolerances required for mating and moving parts.

Drive Roller

We will use polyurethane resin for the drive roller. We will begin the drive roller manufacturing process by thoroughly cleaning the machine core (hub) and applying a specialized primer (bonding agent) to maximize adhesion with the polyurethane. Next, the two-component polyurethane resin will be accurately weighed and mixed, preferably under vacuum, to eliminate air bubbles. We will then slowly and carefully pour this prepared liquid mixture into a specialized mold where the core is centered. Following the casting, we will cure the roller (in an oven) at the temperature and for the duration required by the specific polyurethane type, ensuring it achieves its full mechanical strength. In the final stage, we will subject the demolded roller to turning and grinding operations to bring its outer diameter and surface finish to precise operating tolerances.

4. Cost Analysis

The cost analysis of our project is designed to support the economic feasibility of our design and its competitive pricing compared to existing market alternatives¹. The costs are structured under three main categories: direct material costs, labor/work package budgets, and indirect costs, using a "Bottom-Up" estimation approach. The maximum project budget is set at 30,000 Turkish Liras (TL).

Direct Material Costs

This table details the estimated cost of the core components, which are critical to the kit's production and align with our technical selections (e.g., BLDC motor for efficiency and performance , ESP32 for remote control , ABS for casing).

Component	Detail / Selection	Estimated Unit Cost (TL)	Total Cost (TL)
Motor	BLDC Hub Motor (Used)	2000	2000
Battery Pack	Lithium-ion Based (LiFePO4/Li-Po)	1500	1500
Motor Controller Board	VESC	3000	3000
Microcontroller	ESP32	300	300
Casing	ABS Material	700/kg	700
Drive Roller	Polyurethane Resin	500/kg	1000
Accelerometer and Gyro Sensor	MPU6050 6-Axis Sensor	120	120
Load Cell Amplifier	Voltage Dividers, Cables, Connectors, Mounting Mechanism	40	40
Raindrop Detection Sensor	5V Relay Module	100	100

Component	Detail / Selection	Estimated Unit Cost (TL)	Total Cost (TL)
Li-ion Battery Protection Board	3.7V 6A - 4M	50	50
Magnetic Hall Effect Sensor Board	KY-024	40	40
Plastic Black Potentiometer	WH0162-2-30T L45 B4.7K	40	40
Spring-Loaded Button	Number IC111	40	40
Flame Sensor	-	25	25
TOTAL MATERIAL COST (A)			8955

Table 14 - Cost Analysis

Labor and Work Package Budgets

This section details the pre-allocated budgets from the maximum project budget for specific phases, as defined in the work packages.

Work Package	Assigned To	Status	Budget (TL)
Electrical & Control System Development	Cihan Güldalı	Started	1000
Battery & Energy Management	Berke Arslan	Started	2000
Prototype Assembly	All Members	Not started	2000
Improvements on Design and Systems	Cihan Güldalı	In progress	1000
TOTAL WORK PACKAGE BUDGET (B)			6000

Table 15 - Work Package Costs

Total Cost Item	Result (TL)
TOTAL ESTIMATED COST	14955
MAXIMUM PROJECT BUDGET	30.000

Table 16 - Total Estimated Project Cost

Conclusion: The total estimated cost of the final design combined with the work package budget is targeted to be within the 30.000 TL maximum project budget.

Final Changes

Design and Development

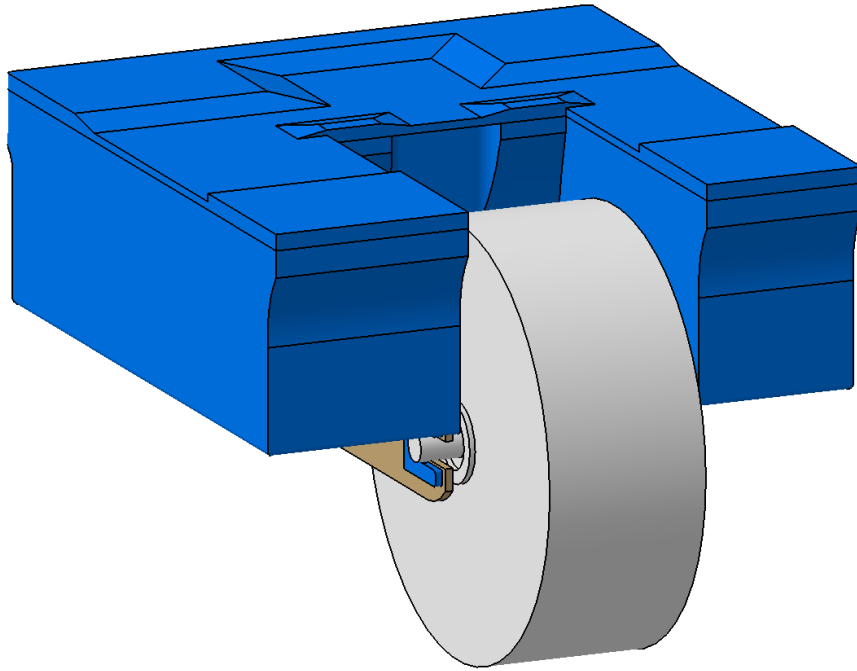


Figure 62 - 3D Model of the Product

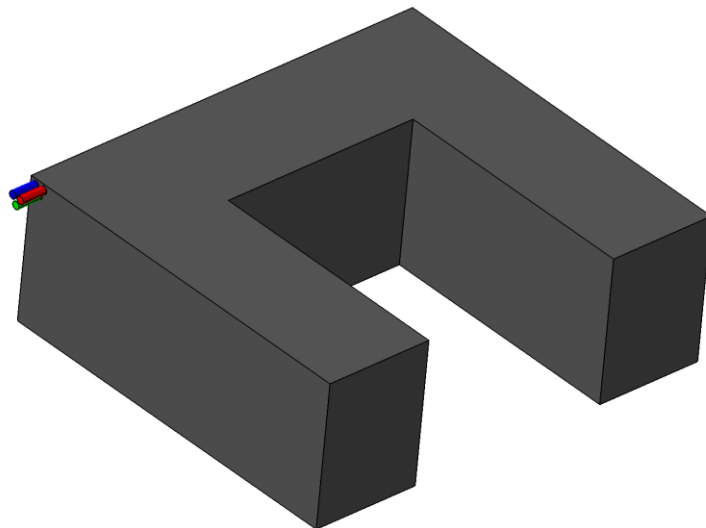


Figure 63 - 3D Model of the U-Shape Battery

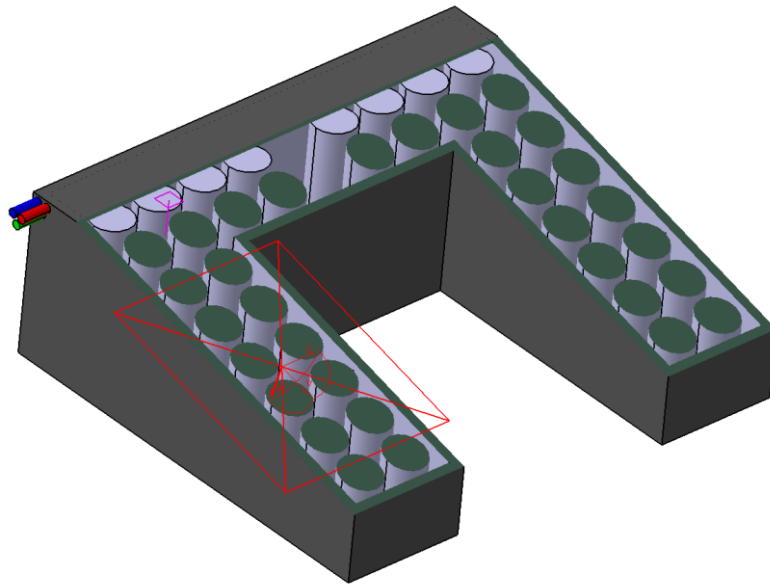


Figure 65 - Internal 3D model of the U-shaped battery pack



Figure 64 - Internal structure of the manufactured U-shaped battery pack

A custom-made lithium-ion battery pack with a 10S4P configuration was used as the energy source of the system. This battery pack consists of 40 Aspilsan A28 INR18650 2800 mAh 25 A rechargeable Li-ion cells. In the 10S4P configuration, 10 cell groups are connected in series, while each series group includes 4 cells connected in parallel. With this arrangement, the battery pack provides approximately 36 V nominal voltage, approximately 42 V when fully charged, and a total capacity of approximately 11.2 Ah. The parallel connection increases the current delivery capability of the cells, theoretically allowing a current potential of up to 100 A. However, for system safety, the operating current is limited by the Battery Management System.

The battery pack is divided into two side sections in accordance with the mechanical layout of the kit. Twenty cells are positioned on one side, while the remaining twenty cells are positioned on the other side. The central clearance was intentionally created to allow the hub motor to be placed in the middle section of the design. This clearance does not negatively affect the electrical connection layout or the operating performance of the battery. On the contrary, this arrangement allows the motor to be integrated into the system more effectively.

The overall external dimensions of the battery can be considered approximately 18.09×20.59 cm. The central clearance has a width of 9.97 cm and a depth of 12.14 cm, which allows the hub motor to fit easily into this region. In addition, the clearance is large enough to provide practical access during assembly. The two side arms of the battery have widths of approximately 5.34 cm and 5.28 cm. This U-shaped design provides the required mounting space for the hub motor while also allowing the battery to be placed compactly inside the casing.

A DALY BMS board is integrated into the battery pack. This BMS is used to ensure the safe and balanced operation of the battery. Its 40 A nominal and 60 A peak current capacity allows the system to supply energy safely under normal operating conditions and to meet short-term high current demands when required.

Together with all cells, the BMS board, and connection elements, the total weight of the battery pack is approximately 1.4 kg. This value is considered suitable for the portability target of the system and facilitates the positioning of the battery within the kit.

Roles of the Battery Management System and Critical Aspects of Reuse

The BMS (Battery Management System) circuit in the original battery pack is the control unit responsible for the safe and efficient operation of the battery. The BMS circuit acts as a kind of “protective brain.” It performs charging, discharging, and balancing functions. It also continuously monitors the battery status to provide protection against extreme conditions. It extends battery life and ensures safe driving. For example, when the BMS circuit detects overcharging, over-discharging, high current, short circuit, and high/low temperature conditions, it cuts off the current to protect the cells. It ensures that all cell groups remain at the same voltage level by performing cell balancing. This balancing is performed, particularly in the final stage of charging, when some cell groups are ahead of others, by dissipating excess energy through resistors. The BMS monitors the voltage of each cell group in series separately.

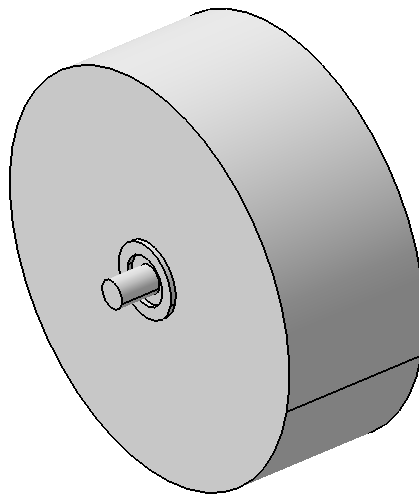


Figure 66 - 3D Model of the Hub Motor

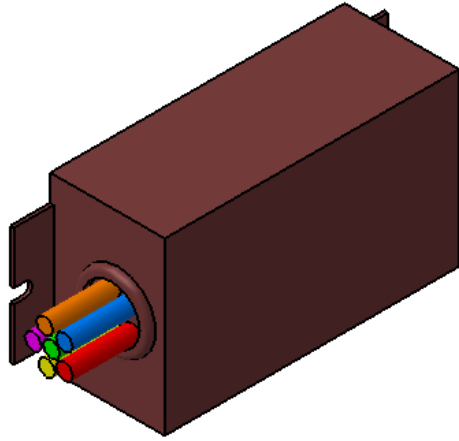


Figure 67 - 3D Model of the VESC Motor Driver

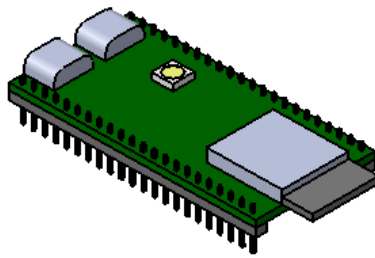


Figure 68 - 3D Model of the ESP32 Microcontroller

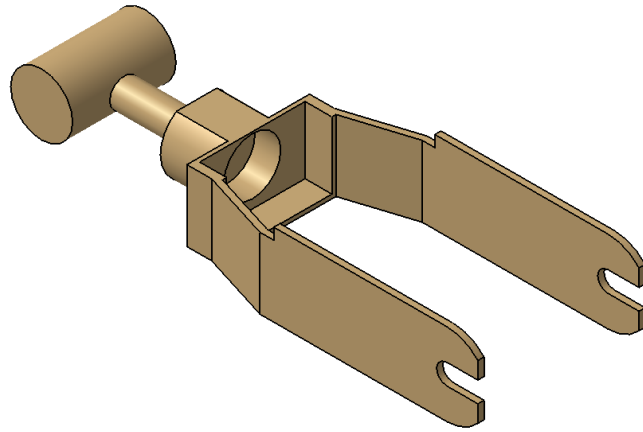


Figure 70 - 3D Model of the Chassis

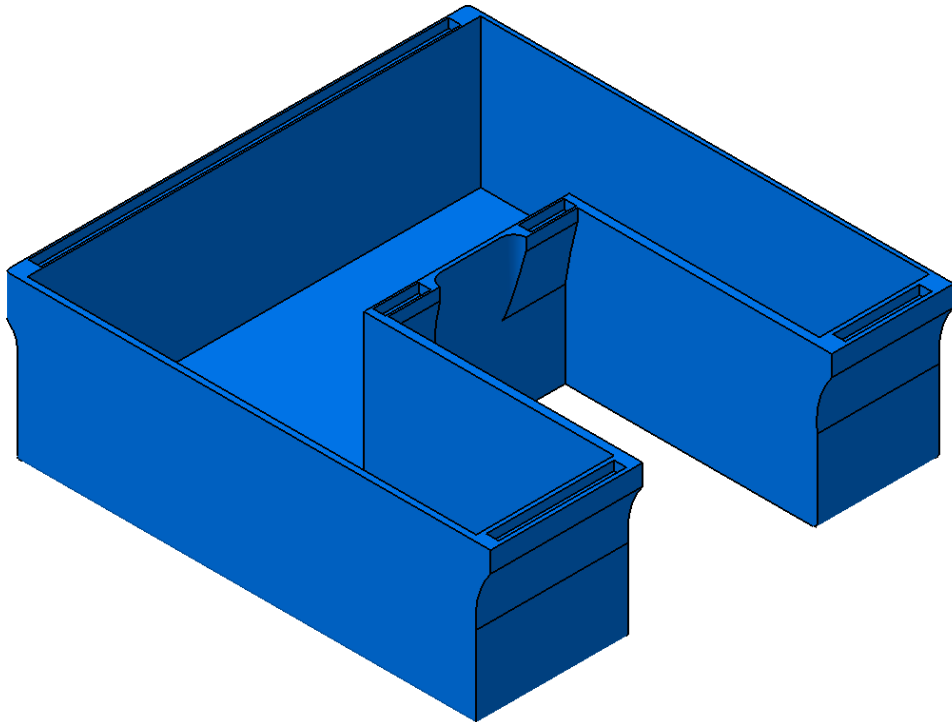


Figure 69 - 3D Model of the Side Frame Case

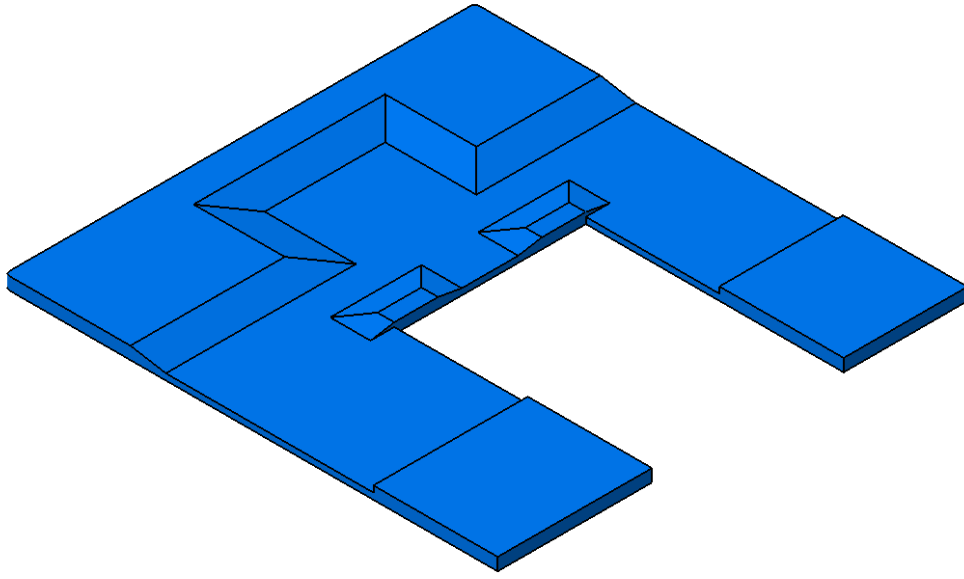


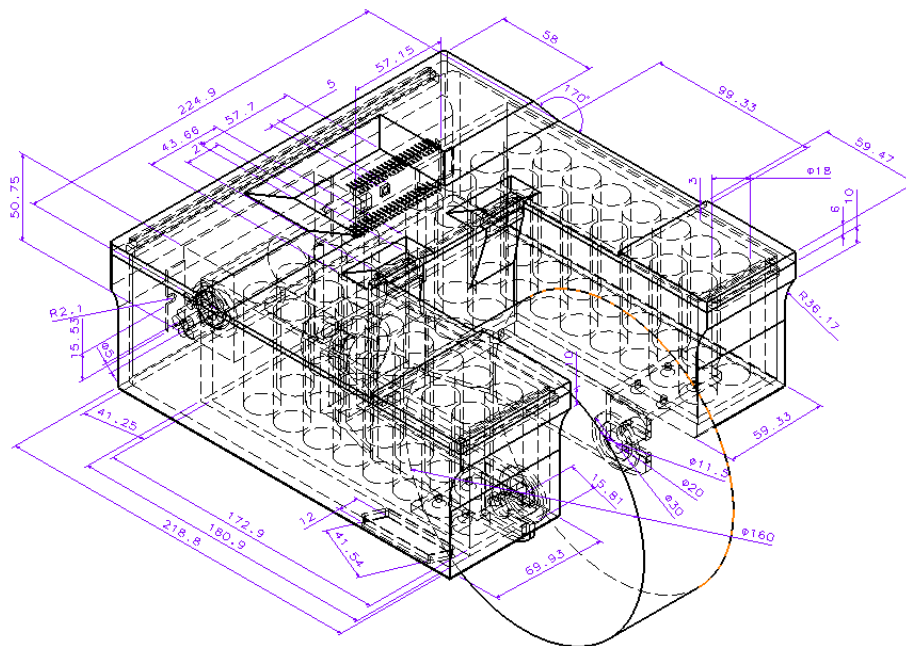
Figure 71 - 3D Model of the Upper Frame Case

The casing consists of two main parts: the body and the cover. Its primary function is to protect the battery, electronic components, and connection elements inside the kit from external effects. During the design process, lightweight structure, ease of manufacturing, and sufficient mechanical strength were prioritized; therefore, ABS Plus was selected as the casing material.

The body and cover are connected to each other using magnets. With this connection method, the need for screws is eliminated, making the assembly and disassembly processes more practical. The easily removable cover provides quick and convenient access to the internal components of the kit during maintenance, inspection, and wiring operations.

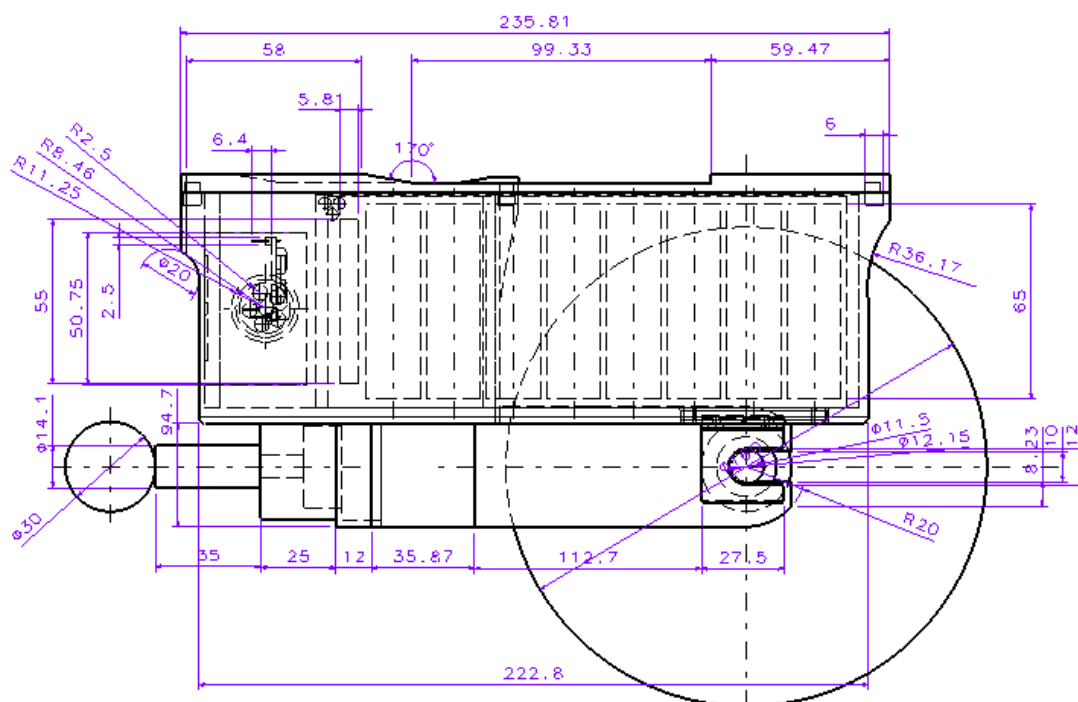
The total weight of the casing is approximately 692 grams. This value is compatible with the portability objective of the product and contributes to keeping the overall system weight as low as possible.

Final Technical Drawings



Isometric view

Figure 73 - Technical Drawings of the Main Body – Isometric View



Right view

Figure 72 - Technical Drawings of the Main Body – Right View

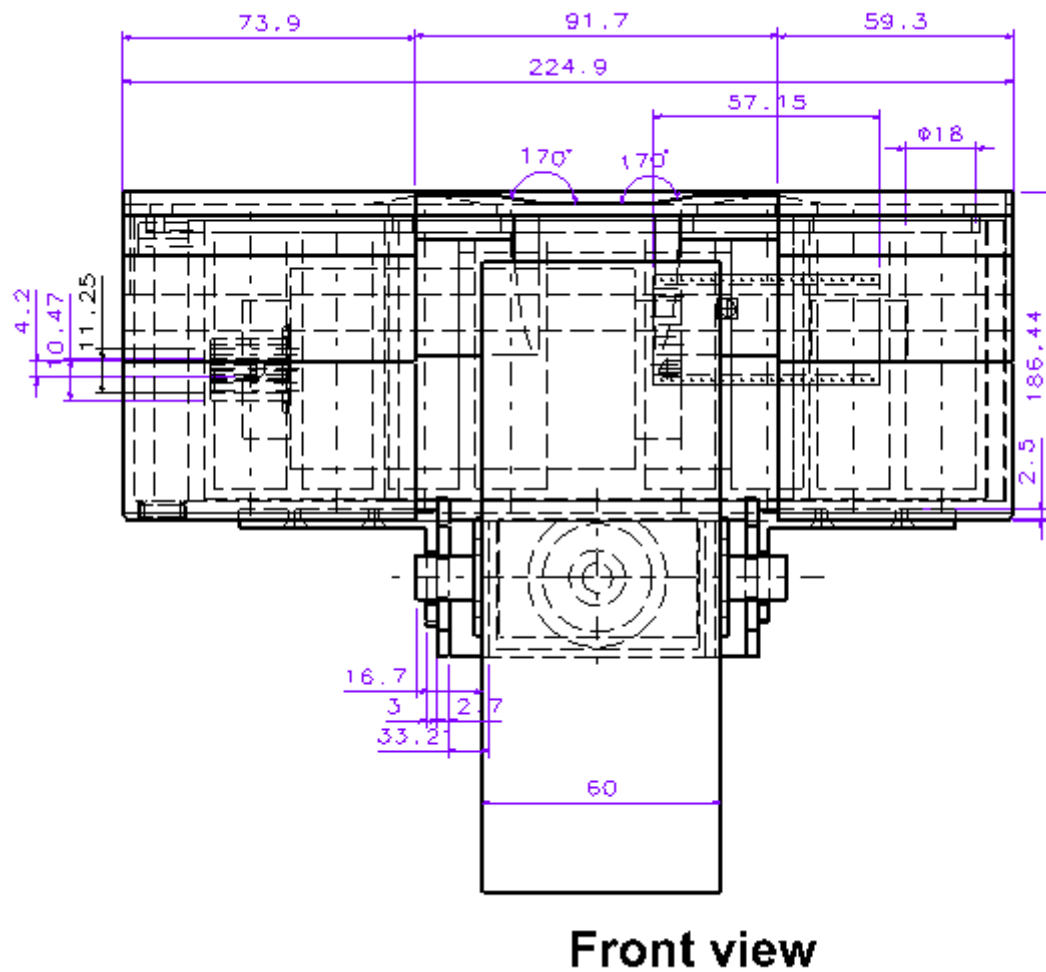


Figure 74 -Technical Drawings of the Main Body – Front View

Mechanical Integration and Assembly

The proposed kit consists of five main subsystems: the chassis, outer casing, electronic components, battery pack, and hub motor. These subsystems are integrated to ensure that the kit can be securely attached to the bicycle while maintaining the correct position of the drive unit with respect to the rear wheel.

The chassis functions as the primary load-bearing structure of the system. Its main role is to connect the kit to the bicycle seat tube and to maintain the hub motor in the required contact position with the rear wheel. Aluminium 6060-T4 was selected as the chassis material due to its favorable balance between low weight and mechanical strength. This material contributes to the portability of the product while providing sufficient structural capacity to carry the loads generated by the motor, battery pack, and outer casing.

The outer casing serves as the protective enclosure of the system. It is designed to protect the battery pack, VESC motor controller, ESP32 microcontroller, and internal wiring from external effects such as impacts, dust, and possible liquid splashes. Considering the requirements of lightweight construction, durability, and manufacturability, ABS Plus produced by 3D printing was selected as the casing material. In addition to protecting the internal components, the casing also provides partial side protection for the hub motor and contributes to the overall structural arrangement of the kit.

The electronic system mainly consists of a VESC motor controller and an ESP32 microcontroller. These components are positioned side by side inside the casing, close to the seat tube side of the bicycle. This layout provides easier access to the electronic components, enables a more organized cable arrangement, and supports a more balanced distribution of mass within the kit.

The battery pack is the component that has the greatest influence on the overall form of the casing. Due to its U-shaped geometry, it occupies the largest volume inside the casing and plays a determining role in the external appearance of the product. The battery is placed in the remaining internal volume after the positioning of the electronic components and is arranged to fit the inner geometry of the casing. As a result, the VESC, ESP32, battery pack, and the required electrical connections are integrated into a compact internal layout.

The hub motor is the main drive element responsible for transferring motion to the rear wheel. The fixed shaft of the motor is supported by both the fork-shaped structure of the chassis and the supporting fork structure of the casing. This double-sided support helps maintain the motor position, preserves the contact alignment with the rear wheel, and improves the mechanical integrity of the overall assembly.

Before the kit is mounted onto the bicycle, its internal assembly is completed. First, the U-shaped battery pack is placed inside the casing. Since the battery geometry is compatible with the internal form of the casing, it can be seated in position without requiring additional adhesive. Then, the VESC and ESP32 are fixed to the designated mounting points behind the battery, on the side closer to the seat tube. After these components are secured, the internal wiring between the battery pack and the electronic units is completed.

Following the internal assembly, the fixed shaft of the hub motor is connected first to the fork structure on the casing and then to the load-bearing fork structure on the chassis. This step completes the main mechanical structure of the kit. Finally, the chassis is attached to the bicycle seat tube using Allen-head screws. This fastening method allows the user to install and remove the kit when necessary, while maintaining sufficient rigidity during operation. Therefore, the final assembly provides a portable, compact, and practical electric bicycle conversion kit suitable for repeated attachment and detachment.

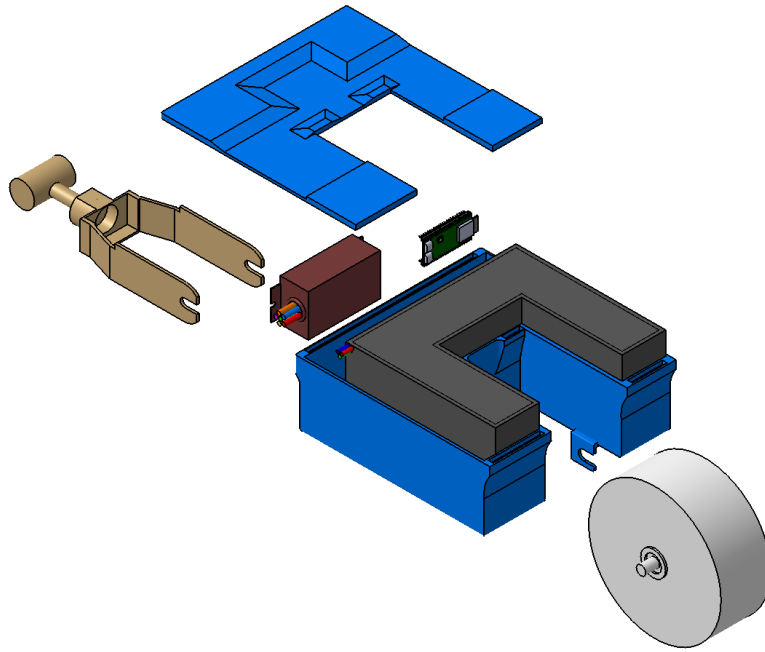


Figure 76 - Exploded 3D model showing the main components of the electric bicycle conversion kit



Figure 75 - Assembly showing the casing, battery pack, chassis, and hub motor

Tests Done on the Final Prototype and Test Results

Thermal Performance Tests

Main objective of thermal tests is to evaluate the thermal stability of the battery and VESC controller under simulated and actual maximum load conditions for an average ride. Our main heat sources were Battery and VESC. In the preliminary design part, we have conducted an ANSYS Transient Thermal analysis to evaluate the system's heat management. A 35 W heat load was applied to the battery and a 15 W heat load to the VESC over a 30-minute (1800s) period. The theoretical maximum temperatures obtained from these simulations were max 57 °C and the average of 27 °C for the whole system. With the real life tests, we have found out that temperature of 65 °C for maximum and average temperature of 34 °C are our real values. These temperature are slightly different than our simulation values.

The difference between these values are a direct result of analysis material selection and the real life material selection. Since the specifications of those materials are different, we got different values. The location of our parts are also changed. For a more compact design we merged our design in one case. This resulted to the higher temperatures.

Friction and Traction Tests

We had calculated that the physical interaction between the drive roller and the tire would be sufficient in our analysis but in real life conditions, when the ground has a real low value of friction coefficient, our drive roller slips a little bit. But in outside test conditions we did not experience any slip or contact inefficiency. The system was subjected to varying torque demands on asphalt and wet concrete. Only problem that we encountered was that the product had minor damages.

System Architecture and Technical Implementation

While urban micro-mobility solutions continue to gain momentum, one area that has yet to be overcome is the price of quality e-bikes when compared to the millions of usable conventional bicycles simply left to rust in hallway closets and garage sheds. In an effort to overcome this problem, we introduce a new approach for our friction-drive electric bicycle conversion using

a software-based approach for converting a conventional mechanical solution into an intelligent, firmware-powered solution based on the principles of Software-Defined Vehicles.

Our proposed solution makes use of an ESP32 as the primary control unit for the traction system using a distributed control loop with communication to the VESC over the high-speed UART interface. Maintaining control over a 60A powertrain during dynamic conditions means that the control of the phase currents must be precise. This can be done with the use of Field Oriented Control, ensuring that rotor angle feedback is maintained to keep the stator magnetic field at a perfect 90-degree angle relative to the permanent magnets. As a way to surpass the maximum RPMs due to the Back-EMF effect in high-RPM scenarios, a negative d-axis current injection can be achieved by way of Field Weakening control.

The early field tests brought out a clear difference between the IoT telemetry dashboard speed value and the ground truth speed obtained using GPS. This would require modifications on a deeper level than what was done previously. Since the kit operates directly with the tires, the linear velocity was measured from the driving surface area of the motor with a calibrated nominal radius of 0.141 meters. The speed conversion process would also depend on certain aspects of the internal motor structure – for instance, using a BLDC motor with 30 magnets required setting the poles per pair value to 15 – any other figure would mean an error. The revised formula used to convert the mechanical RPMs to speed is $\text{mechanical RPM}/60 * \pi * \text{diameter} * 3.6$.

From the mechanical point of view, operating the hub with a tire surface means there will be inevitable slippage, especially considering how much torque transfer takes place. To counter this issue, reverse engineering was conducted to obtain the necessary dimensions for a custom-made sleeve out of Polyamide, fitted with a custom tread pattern made out of Polyurethane for higher friction against the tire surface.

Operation of such a power-heavy vehicle in a vibration-filled urban setting presents challenges in software robustness. During testing in its early stages, impact on the potholes caused false positives in the form of crash interrupt signals in IMU polling loops that caused unwanted system reboots. As we rely solely on the chassis for structural safety, these IMU polling loops were software isolated for the exhibition setup to prevent a reboot cycle while maintaining processing power. In addition, regenerative braking lockout was added dynamically to protect the BMS from potential voltage fault issues. The voltage lockout function prevents the process of regenerative braking if it detects incoming voltage higher than 41.5V, which means that the battery is fully charged. Finally, the state machine monitor

constantly checks the BLE connection flag signal and automatically stops acceleration in case the BLE link between the smartphone control and TCU is disconnected. It also locks the throttle input after the connection to stop any possible voltage surge upon reconnecting.

```
496 // IMU polling loops software-isolated for the exhibition configuration
497 /*
498 if (now - t_imu >= IMU_POLL_MS) {
499     t_imu = now;
500     imuPoll();
501 }
502 */
503 (void)t_imu;
504 if (s1 < 0) {
505     // Coast motor if battery is fully charged to protect BMS
506     if (vesc.v_in > REGEN_BLOCK_VOLT) {
507         vescSetCurrent(0.0f);
508     } else {
509         vescSetCurrentBrake((-s1 / 100.0f) * regen_max);
510     }
511 }
512 // Halt motor immediately upon Bluetooth signal loss
513 if (RemoteXY.connect_flag == 0) {
514     vescSetCurrent(0.0f);
515     if (last_stop_reason != STOP_BLE_LOST) {
516         last_stop_reason = STOP_BLE_LOST;
517     }
518     return;
519 }
```

Figure 77 - BMS Protection and Signal Loss Codes

Range Tests

To verify the theoretical range of 60 km envisioned during the design and engineering phases, a comprehensive field driving test reflecting real-world conditions was conducted. The test took place within the Hacettepe University Beytepe Campus, known for its sloping terrain and stop-and-go dynamics.



Figure 79 - Test Area

Hacettepe Beytepe Kampüsü ve 1604. Cd. üzerinden

Trafiğin normal olduğu en hızlı rota

⚠ Bu güzergahta kullanım kısıtlaması veya özel yollar var.

[Ayrıntılar](#) [Önizle](#)

35 dk.

17,3 km

Figure 78 - Total Distance of the Test

Actual experimentation involved the use of a light urban vehicle weighing 120 kg and running for an hour on the ring route at the Hacettepe University Beytepe Campus and the 1604th Street, a track totaling 20 km of rugged terrain and urban conditions alike. After completion of the journey, the battery usage amounted to only 33% of its total capacity. Extending this data linearly implies that the full capacity of the battery would allow for a distance of 60 kilometers, precisely matching our predictions in the desktop study. This is even more convincing since this occurred under rugged terrain and at a weight of 120 kg.

Hill Climbing and Regenerative Braking

Real-world testing performed at the Hacettepe Beytepe Campus corroborated our predictions. For the range test, a 1 hour ride over an urban terrain with a total distance of 20 km utilized only 33% of the battery capacity, meaning that the range is about 60 km with the battery fully charged. For the hill-climb and regenerative brake tests, the car climbed to a height difference of 191 m, which meant climbing from 877 m to 1,068 m. The car kept climbing this hill until the battery was completely drained. The hill decent after this, where only regenerative braking was applied, made the motor able to recover enough energy to make the car climb up again 40% of the hill. It shows that the system is efficiently using energy in topographically demanding conditions.

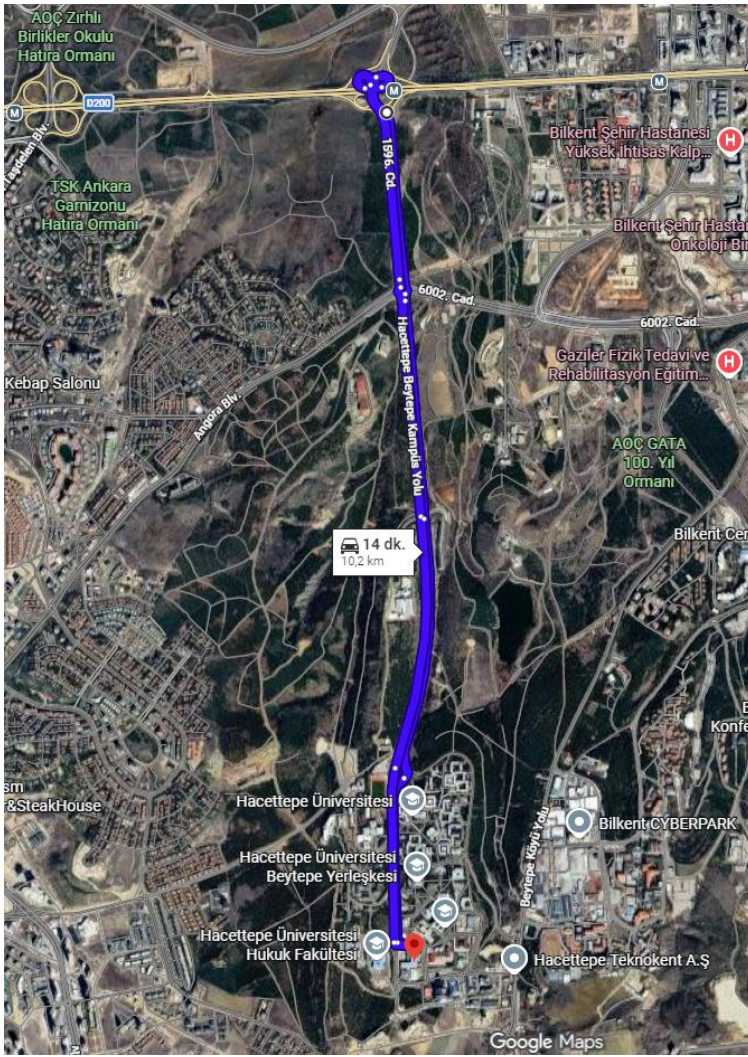


Figure 82 - Distance between Department and Station

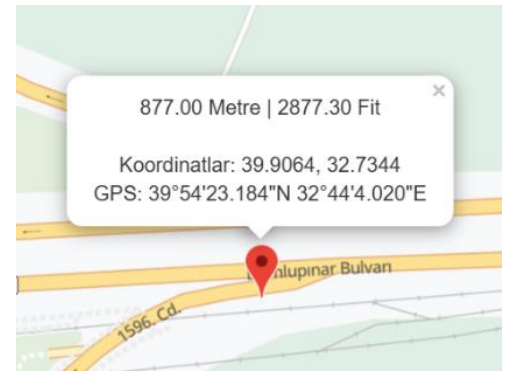


Figure 81 - Elevation of Beytepe Subway Station

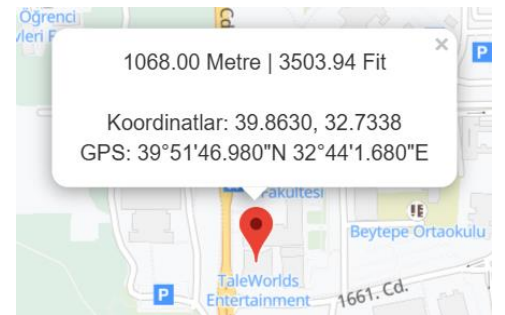


Figure 80 - Elevation of Department of Mechanical Engineering

Discussion of the Results

Since we changed the design, we got different temperatures than the analysis but the differences are not that high to take any action about it. This difference is a result of putting all components in the same 3D printed case.

Creating a friction drive conversion system that could actually work in real-world urban settings brought about issues that we were unable to predict in advance and solve only through practical application.

In addition to the problem associated with maintaining traction between the motor roller and the tire, another recurring issue was regenerative braking. In our initial testing sessions, the regenerative braking produced such an abrupt deceleration that it was uncomfortable to experience. While being acceptable in a safe testing environment, this kind of deceleration cannot be comfortable for daily commuting on real roads. We resolved this issue by adjusting the PID controllers in the VESC system and modifying the fail-safe logic.

Traction control proved to be a challenge too. The main difficulty here was that traction fails abruptly, and recovering from a slippage event halfway up the slope wastes valuable energy and can also pose safety hazards. To address this issue, we decided to rely on our own Traction Control Unit powered by ESP32, which constantly monitored for signs of slipping and reduced motor torque immediately upon detecting such signals. This ensured uninterrupted energy transmission up to a maximum of 19%.

Thirdly, thermal performance during continuous use became another issue. Temperature rises observed in the one hour/20 km city ride cycle showed potential issues with reliability that can be caused by overheating if not managed properly. In order to handle this problem, we decided to address it not with changes in hardware that might increase its mass and complexity, but with software solutions: thermal cut-off parameters were set in the motor controller and improved casing design was adopted to facilitate passive cooling.

For the final cost analysis, even though we projected a budget of 30.000 TL, our kit costed roughly 15.000 TL. The reason for this difference was that we had set our budget high at the start of the project as a precaution.

Conclusion

This report details a modular, portable electric bike conversion kit. Our goal is to develop an efficient kit that can electrify existing bikes. Our kit is built on reliability and efficiency, and features include speed control and regenerative braking.,

Each component has been individually examined within the report. For instance, alternative motors, control systems, battery technologies, designs, and simulations ensuring design reliability have been detailed. Among motors, BLDC motors have emerged as prominent in terms of weight and compactness. Regarding the control system architecture, ESP32 microcontrollers and VESC-based motor controllers provide professional-level support for development potential. External communication protocols have also been reviewed to ascertain their support.

As a result of our report, all systematic evaluations have been successfully completed. All factors have been analyzed by weighting them using a Pugh Chart; regenerative capabilities, efficiency, performance, and cost, among other factors, have been thoroughly examined. Sensor selections, along with their technical details, have been addressed in the report, and our critical limits have been specified. Regulations and directives that constrain us have been reviewed, and within this framework, we designed our requirements according to realistic design limitations. This presentation, which will serve as a guide in the future process, forms the foundation of our final results.

As a result of all analyses, it has been proven in all its details that the system can be manufactured and theoretically function. From this point on, the focus should be on production quality and analog measurement accuracy correction. Our research and development work is complete, and we will produce a product, which will then be transformed into an industrial product in accordance with product development standards and methods.

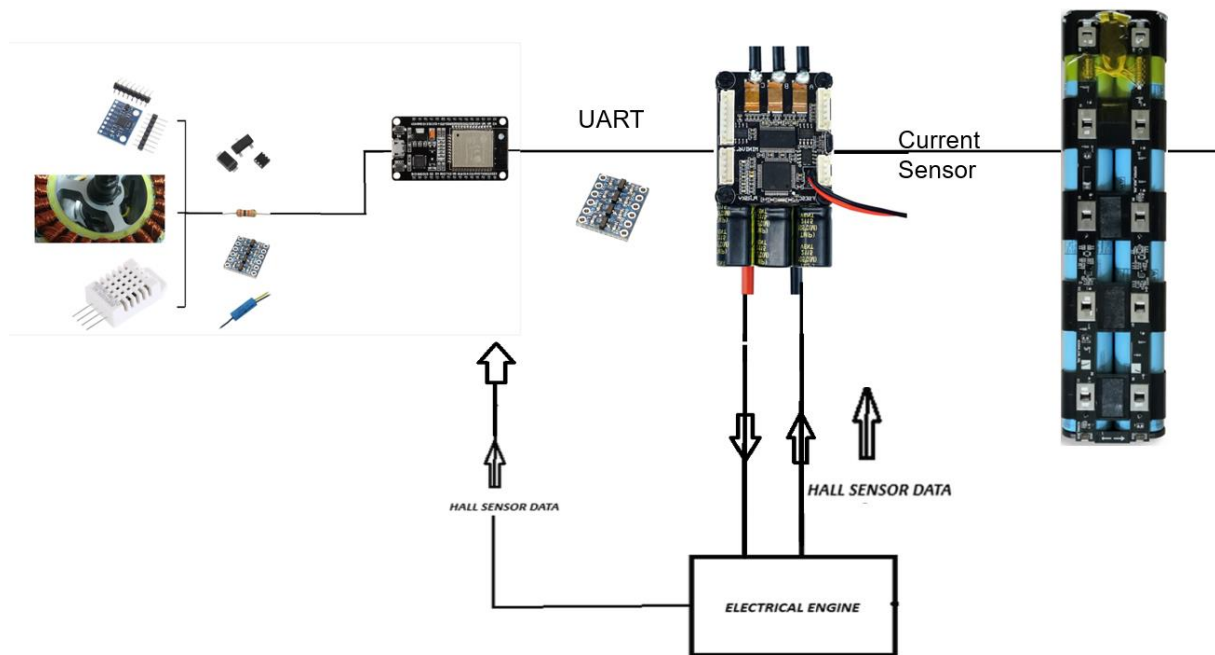


Figure 83 Overall Looking of System

As one may expect, the final prototype performs well to its design goals. Indeed, the verified 60-kilometer range and the 40% efficiency of regenerative braking confirmed with tests on a challenging and hilly urban-like campus course, where the combined weight of the test platform and the rider was 120 kilograms, prove that the prototype successfully reached our design specifications. For a first generation prototype, such results make us proud.

However, there are still some areas which could be improved upon for further work. Specifically, we identified wet weather traction as a problem which should be addressed in the future iterations of the prototype. Indeed, rain leads to a decrease in the coefficient of friction between the roller and the wheel, which is difficult to control through the software throttling alone. This will be solved through a redesign of the mechanical components of the system: the motor roller will feature knurled surfaces which provide increased friction, while a mechanical tensioning mechanism will increase the contact pressure automatically when moisture or slip is sensed.

Another aspect to be improved is the battery management system (BMS). While the current system works according to preset thermal cut-offs and works well for predictable riding patterns, its ability to work efficiently for varying patterns remains to be desired. Our plan includes implementation of an adaptive battery management system, based on the use of machine learning algorithms implemented in the microcontroller to learn the behavior of the rider and build a profile depending on such parameters as weight, topology of the terrain and velocity of the ride.

Future Works

While the final prototype successfully meets the core design objectives and performance criteria, several areas have been identified for future development to enhance the system's efficiency, reliability, and user experience. Future iterations of the conversion kit will focus on both hardware optimizations and advanced software integrations.

Hardware

Traction Control under Wet Conditions

To address the decrease in the friction coefficient during rainy weather, the mechanical components will be redesigned. The motor roller will incorporate knurled surfaces for better grip, and a mechanical tensioning mechanism will be introduced to automatically increase contact pressure when moisture or wheel slip is detected.

Drive Roller

Alternative manufacturing methods, such as utilizing direct rubber rollers or testing different polyurethane resin formulations, will be explored to maximize contact efficiency.

Battery

The system's range can be further extended beyond the validated 60 kilometers by integrating higher-quality lithium-ion cells into the custom U-shaped pack.

Motor

The lathe operations used to reduce the diameter of the salvaged BLDC hub motor will be further refined to ensure tighter mechanical tolerances.

Casing and Chassis Manufacturing

Future production phases will involve evaluating new 3D printing suppliers or alternative materials to improve the structural integrity and surface finish of the ABS casing.

Software

The current thermal cut-off logic will be upgraded to an adaptive BMS. By implementing machine learning algorithms on the microcontroller, the system will learn rider behavior and build custom efficiency profiles based on rider weight, terrain topology, and average velocity.

The existing ESP32 and VESC architecture will be expanded to process real-time data via Bluetooth and Wi-Fi. A Kalman filter will be implemented to increase the accuracy of position and velocity readings from the sensors.

Future software updates will feature predictive driving cycles and advanced State of Health (SoH) and Prognostic Estimation algorithms based on historical driving data.

The control system will undergo rigorous Hardware-in-the-Loop (HIL) testing to analyze system response times, which will be followed by real-world Software-in-the-Loop (SIL) validations to ensure the robustness of the algorithms.

Appendices

VESC Card Types

<https://github.com/ClemensElflein/xESC/tree/main>

VESC Tool

https://vesc-project.com/vesc_tool

BLDC Motor (Rotor Stator) Details

https://github.com/razmikTovmas/bldc_e_motor

```

File Edit Sketch Tools Help
[Icons] Select Board [Dropdown]

sketch_esp.ino
1  #include <Wire.h>
2  #include <AS5600.h>
3  #include <Adafruit_INA219.h>
4  #include "VescUart.h"
5
6  #define I2C_SDA_PIN 21
7  #define I2C_SCL_PIN 22
8
9  const int GAIN_PEDAL_PIN = 34;
10 const int STOP_SWITCH_PIN = 26;
11
12 const float ADC_SCALE_MAX = 4095.0;
13 const int AVG_SAMPLE_COUNT = 8;
14
15 HardwareSerial VESC_COMM_PORT(2);
16 VescUart driverLink(&VESC_COMM_PORT);
17
18 AS5600 rotarySensor;
19 Adafruit_INA219 powerMonitor;
20
21 void initializePeripherals() {
22     Serial.begin(115200);
23     Wire.begin(I2C_SDA_PIN, I2C_SCL_PIN);
24     delay(200);
25
26     if (!rotarySensor.begin()) {
27         Serial.println("Rotor sensor failure");
28     } else {
29         Serial.println("Rotor sensor online");
30     }
31
32     if (!powerMonitor.begin()) {
33         Serial.println("Power monitor failure");
34     } else {
35         Serial.println("Power monitor online");
36     }
37
38     pinMode(STOP_SWITCH_PIN, INPUT_PULLUP);
39
40     VESC_COMM_PORT.begin(115200, SERIAL_8N1, 16, 17);
41     driverLink.setTimeout(100);
42
43     delay(100);
44     Serial.println("System readiness check complete");
45 }
46 float measureGainInput() {
47     uint32_t totalReading = 0;
48     for (int i = 0; i < AVG_SAMPLE_COUNT; i++) {
49         totalReading += analogRead(GAIN_PEDAL_PIN);
50     }
51     float averageValue = totalReading / (float)AVG_SAMPLE_COUNT;
52
53     float normalizedInput = averageValue / ADC_SCALE_MAX;
54
55     if (normalizedInput < 0.03) normalizedInput = 0.0;
56     return constrain(normalizedInput, 0.0, 1.0);
57 }
58 void setup() {
59     initializePeripherals();
60 }
61
62 void loop() {
63     bool stopActive = (digitalRead(STOP_SWITCH_PIN) == LOW);
64
65     float pedalInput = measureGainInput();
66
67     float rotorAngle = 0.0;
68     if (rotarySensor.available()) {
69         rotorAngle = rotarySensor.getAngle();
70     }
71

```

ESP Communacition of Sensor (Arduinio) & Real Life Simulation (PyCharm)

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path
from dataclasses import dataclass
from scipy.interpolate import RegularGridInterpolator, interp1d
import os

# =====
# 0. YENİ "STANDART" HARİTA OLUŞTURUCU
# =====
def create_standard_map_file(file_path: Path): 1 usage
    # Eğer eski, kötü harita varsa sil, yenisini yapalım
    if file_path.exists():
        try:
            os.remove(file_path)
            print("[SİSTEM] Eski harita silindi, standart harita oluşturuluyor...")
        except:
            pass

    print(f"[SİSTEM] Standart Bafang/Bosch stili harita hazırlanıyor...")

    # Grid (Daha hassas)
    torques = np.linspace(start=0, stop=60, num=61)
    rpms = np.linspace(start=0, stop=450, num=46) # RPM aralığını genişlettik

    eff_data = []

```

```

def create_standard_map_file(file_path: Path): 1 usage
    # Merkez: 25 Nm @ 220 RPM -> %90 Verim
    opt_torque = 20.0
    opt_rpm = 230.0

    for t in torques:
        row = []
        for r in rpms:
            if t == 0 or r == 0:
                eff = 0.01
            else:
                # 1. Merkezde olan uzaklığı hesapla (Normalize edilmiş)
                r_dist = (r - opt_rpm) / 200.0 # RPM hassasiyeti
                t_dist = (t - opt_torque) / 35.0 # Tork hassasiyeti
                dist = np.sqrt(r_dist ** 2 + t_dist ** 2)

                # 2. Verim hesabı (Merkezden uzaklaştıkça düşer)
                # Max verim %91 kabul edelim
                base_eff = 0.91 - (0.4 * (dist ** 1.8))

                # 3. Fiziksel Düzeltmeler
                # Düşük RPM cezası (Bakır kaybı)
                if r < 50: base_eff *= (0.6 + 0.4 * (r / 50))

                # Aşırı Tork cezası (Doygunluk)
                if t > 45: base_eff *= 0.95

                # Sınırla
                eff = max(0.10, min(base_eff, 0.92))

            row.append(int(eff * 100))
        eff_data.append(row)

```

```

def create_standard_map_file(file_path: Path): 1 usage
    eff_data.append(row)

    cols = [f"{int(r)}_RPM" for r in rpms]
    df = pd.DataFrame(eff_data, columns=cols)
    df.insert(loc=0, column="Torque", torques)
    df.to_csv(file_path, index=False)
    print("[BAŞARILI] Yeni harita kaydedildi.\n")

# =====
# 1. AYARLAR (CONFIG)
# =====
@dataclass 2 usages
class BikeConfig:
    # Dosya Yolu (KONTROL ET)
    input_path: Path = Path(
        r"C:\Users\cihan\OneDrive\Desktop\ebike\470\data\E-Bike_Field_Study_Data_20251116 (3)_processed.csv")
    map_path: Path = Path("standard_motor_map.csv") # Yeni isim

    # FİZİK (21 JANT + I*ALPHA)
    mass_kg: float = 125.0
    g: float = 9.81
    c_rr: float = 0.004
    rho_air: float = 1.225
    area_frontal: float = 0.6
    drag_coeff: float = 0.75

    wheel_diameter_inch: float = 21.0
    wheel_mass_front_kg: float = 1.6
    wheel_mass_rear_kg: float = 5.6

```

```

class MotorMapReader: 1 usage
    def __init__(self, csv_path: Path):
        self.interpolator = None
        self.raw_df = None
        # Her zaman yeni standart haritayı oluştur
        create_standard_map_file(csv_path)
        self.load_map(csv_path)

    def load_map(self, path): 1 usage
        try:
            self.raw_df = pd.read_csv(path)
            torques = self.raw_df.iloc[:, 0].values
            rpms = np.array([float(c.split('_')[0]) for c in self.raw_df.columns[1:]])
            eff_grid = self.raw_df.iloc[:, 1:].values / 100.0
            self.interpolator = RegularGridInterpolator(points=(torques, rpms), eff_grid, bounds_error=False, fill_value=None)
        except Exception as e:
            print(f"Harita hatası: {e}")

    def get_efficiency(self, torque_nm, rpm): 1 usage
        if self.interpolator is None: return 0.85
        return max(float(self.interpolator((abs(torque_nm), abs(rpm))))), 0.01)

class BatteryManager: 1 usage
    def __init__(self):
        self.eff_interp = interp1d(x=[0, 0.5, 1.0], y=[0.95, 0.96, 0.90], fill_value="extrapolate")
        self.limit_interp = interp1d(x=[0, 0.8, 1.0], y=[1.0, 1.0, 0.0], fill_value="extrapolate")

```

```

class EBikeAnalyzer: 1 usage
    def __init__(self, config: BikeConfig):
        self.df = None
        self.motor_map = MotorMapReader(config.map_path)
        self.batt_mgr = BatteryManager()
        self.results = {}

    # --- A. SÜRÜŞ DÖNGÜSÜ SİMÜLASYONU ---
    def run_cycle_simulation(self): 1 usage
        if not self.cfg.input_path.exists(): raise FileNotFoundError("Veri dosyası yok!")
        print("Sürüş döngüsü simüle ediliyor...")

        self.df = pd.read_csv(self.cfg.input_path)
        self.df['date'] = pd.to_datetime(self.df['date'], utc=True)
        self.df = self.df.dropna(subset=['date', 'spd']).sort_values('date')

        t = (self.df['date'] - self.df['date'].iloc[0]).dt.total_seconds().to_numpy()
        v_ms = self.df['spd'].to_numpy() / 3.6

        # Fizik
        dt_avg = np.median(np.diff(t)) if len(t) > 1 else 1.0
        win = max(1, int(6.0 / dt_avg))
        a_smooth = pd.Series(np.gradient(v_ms, *varargs: t)).rolling(win, center=True, min_periods=1).mean().to_numpy()

        radius_m = (self.cfg.wheel_diameter_inch * 0.0254) / 2
        alpha = a_smooth / radius_m
        I_total = (self.cfg.wheel_mass_front_kg + self.cfg.wheel_mass_rear_kg) * (radius_m ** 2)

        F_total = (self.cfg.mass_kg * a_smooth) + (I_total * alpha / radius_m) + \
            (self.cfg.mass_kg * 9.81 * self.cfg.c_rr) + \
            (0.5 * 1.225 * 0.75 * 0.6 * v_ms ** 2)

        rpm_arr = (v_ms / (2 * np.pi * radius_m)) * 60

```

```

def run_cycle_simulation(self): 1 usage
    rpm_arr = (v_ms / (2 * np.pi * radius_m)) * 60
    torque_arr = F_total * radius_m
    P_req_wheel = F_total * v_ms

    curr_soc = self.cfg.initial_soc
    cap_j = self.cfg.battery_capacity_wh * 3600
    max_p = self.cfg.system_voltage * self.cfg.max_controller_current

    P_batt = np.zeros_like(P_req_wheel)
    P_mech_waste = np.zeros_like(P_req_wheel)
    soc_trace = np.zeros_like(P_req_wheel)

    for i in range(len(P_req_wheel)):
        p, trq, rpm = P_req_wheel[i], torque_arr[i], rpm_arr[i]
        dt = (t[i + 1] - t[i]) if i < len(t) - 1 else dt_avg
        eta = self.motor_map.get_efficiency(trq, rpm)

        if p >= 0: # Çekis
            p_draw = min((p / (eta * self.cfg.eta_chain)), max_p)
            P_batt[i] = p_draw
            curr_soc -= (p_draw * dt) / cap_j
        else: # Regen
            p_brake = -p
            g_val = -a_smooth[i] / 9.81
            if g_val <= self.cfg.regen_g_threshold:
                p_regen = p_brake
                p_waste = 0
            else:
                p_waste = p_brake * self.cfg.k_bf
                p_regen = p_brake * (1 - self.cfg.k_bf)

```

```

        soc_trace[i] = curr_soc

    self.df['P_batt_W'] = P_batt
    self.df['SoC'] = soc_trace
    self.df['Dist_km'] = np.cumsum(v_ms * dt_avg) / 1000
    self.df['RPM'] = rpm_arr;
    self.df['Torque'] = torque_arr

    # Sonuçlar
    P_pos = np.clip(P_batt, a_min: 0, a_max: None)
    P_neg = np.clip(P_batt, a_min: None, a_max: 0)
    self.results = {
        "dist": self.df['Dist_km'].iloc[-1],
        "E_batt_used": np.trapz(P_pos, t) / 3600,
        "E_regen_gained": np.trapz(P_neg, t) / 3600,
        "E_mech_waste": np.trapz(P_mech_waste, t) / 3600,
        "Net_Wh": (np.trapz(P_pos, t) + np.trapz(P_neg, t)) / 3600,
        "SoC_Start": self.cfg.initial_soc,
        "SoC_End": curr_soc
    }

# --- B. 0-40 KM/H HIZLANMA TESTİ (Dinamik) ---
def run_acceleration_test(self): 1usage
    print("0-40 km/h Hızlanma Testi Yapılıyor...")
    t, v = 0, 0
    dt = 0.1
    radius_m = (self.cfg.wheel_diameter_inch * 0.0254) / 2
    I_total = (self.cfg.wheel_mass_front_kg + self.cfg.wheel_mass_rear_kg) * (radius_m ** 2)
    P_elec_max = self.cfg.system_voltage * self.cfg.max_controller_current

```

```

times, vels = [0], [0]

while v * 3.6 < 50 and t < 60: # Max 50kmh veya 60sn
    # Motor Çekışı
    rpm = (v / (2 * np.pi * radius_m)) * 60
    # Basit tork limiti (Harita okumadan hızlı hesap)
    # Düşük hızda Tork limit, yüksek hızda Güç limit
    F_lim_torque = self.cfg.max_motor_torque_nm / radius_m
    F_lim_power = (P_elec_max * 0.85 * self.cfg.eta_chain) / max(v, 0.1)
    F_pull = min(F_lim_torque, F_lim_power)

    # Dirençler
    F_res = (self.cfg.mass_kg * 9.81 * self.cfg.c_rr) + \
        (0.5 * 1.225 * 0.75 * 0.6 * v ** 2)

    F_net = F_pull - F_res
    if F_net <= 0: break

    # I*Alpha'llı İvme: a = F_net / (m + I/r^2)
    eff_mass = self.cfg.mass_kg + (I_total / radius_m ** 2)
    a = F_net / eff_mass

    v += a * dt
    t += dt
    times.append(t)
    vels.append(v * 3.6)

# 0-40 Süresini bul
try:
    t_40 = next(tm for tm, vel in zip(times, vels) if vel >= 40.0)

```

```

        return None

def print_and_plot(self):
    usage
    if self.df is None: return
    r = self.results

    # Hızlanma verisini al
    t_40 = self.run_acceleration_test()

    # --- DETAYLI RAPOR ---
    print("\n" + "-" * 60)
    print(f"    FINAL ANALİZ (Standart Verim Haritası)")
    print("-" * 60)
    print(f"Mesafe           : {r['dist']:.2f} km")
    print(f"SoC Değişimi       : %{r['SoC_Start'] * 100:.1f} -> %{r['SoC_End'] * 100:.1f}")
    print("-" * 60)
    print(f"Bataradan Çekilen  : {r['E_batt_used']:.2f} Wh")
    print(f"Geri Kazanılan     : {abs(r['E_regen_gained']):.2f} Wh")
    print(f"Net Tüketim        : {r['Net_Wh']:.2f} Wh")
    print("-" * 60)
    if t_40:
        print(f"0-40 km/h Süresi   : {t_40:.2f} saniye")
    else:
        print(f"0-40 km/h Süresi   : Ulaşılamadı")
    print("-" * 60)

    # GRAFİKLER
    fig, (ax1, ax2) = plt.subplots(nrows=1, ncols=2, figsize=(16, 6))

    # 1. SoC Grafiği
    ax1.plot(self.df['Dist_km'], self.df['SoC'] * 100, color='#2ca02c', linewidth=2.5)

```

```

> if __name__ == "__main__":
    analyzer = EBikeAnalyzer(BikeConfig())
    try:
        analyzer.run_cycle_simulation()
        analyzer.print_and_plot()
        analyzer.df.to_csv("FINAL_RESULT_STD_MAP.csv", index=False)
        print("Sonuçlar kaydedildi.")
    except Exception as e:
        print(f"Hata: {e}")

```

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